

Orthography to Phoneme Conversion in Southern New England

Algonquian:

A Multi-Source Research Document

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Contents

1	Conversion Overview	2
1.1	Conversion Examples and Methodologies	2
1.2	Technique Comparison and Selection Rationale	4
1.3	Source Status and Expected KPIs	4
1.4	The Colonial Recorder Perception Problem	5
I	Source Workflows	6
2	Narragansett: Williams 1643	6
2.1	Source Description	6
2.1.1	Recorder Biography	6
2.1.2	Orthography System	7
2.1.3	Stress Marking	7
2.2	Vowel Inventory	7
2.2.1	Simple Vowels	7
2.2.2	Nasalized Vowels	7
2.2.3	Diphthong-Like Sequences	8
2.3	Consonant Inventory	8

2.3.1	Stops	8
2.3.2	Affricates	8
2.3.3	Nasals	8
2.3.4	Fricatives	8
2.3.5	Glides	8
2.3.6	The Silent /h/ — Preaspiration	8
2.4	Problems Encountered	9
2.4.1	Vowel Quality Ambiguity	9
2.4.2	Nasalization Implied but Not Marked	9
2.4.3	Stop Voicing Neutralized	9
2.4.4	English Spelling Conventions	9
2.4.5	Silent e	10
2.4.6	Preaspiration /h/ Invisible	10
2.4.7	<i>sh</i> Before Consonants	10
2.4.8	The <i>an/aun</i> Ambiguity for /ã/	10
2.5	Research Approach	10
2.5.1	Recommended Methodology	10
2.5.2	Annotation Plan	11
2.5.3	Cross-Linguistic Cognate Evidence	11
2.5.4	Open Questions for Revitalization	11
2.6	Conversion Workflow	11
2.7	Linguist Feedback	12
2.8	Related Work: Colonial Recorder Perception Problem	12
2.9	Bibliography	13
3	Wampanoag: Trumbull/Eliot 1903	14
3.1	Source Description	14
3.1.1	Recorder Biography: James Hammond Trumbull	14
3.1.2	The Lost Thesis Controversy	15
3.1.3	Posthumous Publication	15
3.1.4	The Eliot Orthography — Historical Origins	15
3.2	Vowel Inventory	16

3.2.1	Simple Vowels	16
3.2.2	Vowel Digraphs	16
3.3	Consonant Inventory	16
3.4	The ∞ Symbol — Historical Typography and Phonetics	16
3.4.1	The Misidentification	16
3.4.2	What ∞ Actually Represents	17
3.4.3	The ∞→oozzz Pipeline Mapping	17
3.5	Entry Structure — Detailed Parsing Guide	17
3.5.1	Standard Entry Template	18
3.5.2	Entry Variants	18
3.5.3	Source Abbreviations	18
3.5.4	Grammatical Labels	18
3.6	Problems Encountered	18
3.6.1	The ∞ () Symbol Misidentification	18
3.6.2	Preaspiration Gap	19
3.6.3	Apostrophe Ambiguity	19
3.6.4	Dialect Stratification	19
3.6.5	Vowel Quality	19
3.6.6	c Ambiguity	19
3.6.7	Boundary Markers	19
3.6.8	The Devil Problem: <i>achm∞wonk</i> vs. <i>hobbamock</i>	19
3.7	Research Approach	20
3.7.1	Recommended Methodology	20
3.7.2	Proto-Algonquian Correspondences	21
3.7.3	Confirmed Cognate Mappings for Pipeline Testing	21
3.8	Conversion Workflow	21
3.9	Linguist Feedback	21
3.10	Related Work: Colonial Recorder Perception Problem	21
3.11	Open Questions for Linguist Review	22
3.12	Bibliography	23

4 Mohegan-Pequot: Fielding 2013 23

4.1	Source Description	23
4.1.1	Fidelia A. H. Fielding (1827–1908)	23
4.1.2	The Fielding Diaries	24
4.1.3	Corpus Structure	24
4.1.4	Comparative Significance	25
4.1.5	Stephanie Fielding’s Grammatical Analysis	25
4.1.6	Fielding’s Methodology	25
4.2	Vowel Inventory	26
4.2.1	Vowel System	26
4.2.2	Diphthongs	26
4.3	Consonant Inventory	26
4.3.1	Allophonic Variation (from Granberry 2003)	26
4.4	Historical Phonology: Proto-Algonquian Reflexes	27
4.4.1	PA Vowel System	27
4.4.2	The Y-Dialect Classification	27
4.4.3	Consonant Reflexes	27
4.4.4	Cluster Reflexes	27
4.5	Comparison with Prince & Speck (1904)	27
4.5.1	The Prince-Speck Glossary	27
4.5.2	Orthographic Differences	27
4.5.3	The Brothertown Material	27
4.5.4	Convergence and Divergence	28
4.6	Current Orthographic Conventions	28
4.6.1	The Modern Practical Orthography	28
4.6.2	Comparison with Granberry’s Orthography	29
4.7	Problems Encountered	29
4.7.1	Small Corpus	29
4.7.2	Alternate Spellings	29
4.7.3	Circularity Risk	29
4.7.4	The Pequot Extinction and the Treaty of Hartford (1638)	29
4.8	Research Approach	30
4.8.1	Recommended Methodology	30

4.9	Conversion Workflow	30
4.10	Linguist Feedback	31
4.11	Related Work: Colonial Recorder Perception Problem	31
4.12	Mohegan Revitalization	32
4.12.1	The Mohegan Language Project	32
4.12.2	Current Revitalization Efforts (as of 2024)	32
4.12.3	Collaboration with Jessie Little Doe Baird	32
4.12.4	The Samson Occom Papers	32
4.12.5	Terminology: Reclamation vs. Revitalization	33
4.13	Bibliography	33
5	Wampanoag: Multi-Source (Wood, Winslow, Shepard, Eliot)	34
5.1	Source Description	34
5.2	Orthography Systems	35
5.2.1	William Wood (1634)	35
5.2.2	Edward Winslow (1624)	36
5.2.3	Thomas Shepard (1647)	36
5.2.4	John Eliot (1666)	37
5.3	Vowel and Consonant Tables	37
5.4	Source-Specific Orthographic Patterns	37
5.5	Problems Encountered	38
5.6	Research Approach	39
5.6.1	Source-Adaptive Parsing Strategy	39
5.6.2	Resolving Ambiguity	39
5.7	Conversion Workflow	39
5.8	Linguist Feedback Template	40
5.9	Related Work: Colonial Recorder Perception Problem	40
5.10	Key References	41
6	Wampanoag: Winslow 1624	42
6.1	Source Description	42
6.2	Recorder Biography	43
6.3	Linguistic Intermediaries	43

6.3.1	Tisquantum (Squanto, c. 1585–1622)	43
6.3.2	Hobbamock (Hobomok, d. 1642)	44
6.4	Complete Vocabulary	44
6.5	Syntactic Complexity	45
6.6	Orthography System	45
6.6.1	Vowel Grapheme-to-Phoneme Mapping	45
6.6.2	Consonant Grapheme-to-Phoneme Mapping	45
6.6.3	Preaspiration /h/ in Winslow	45
6.7	Pidginization Analysis (Goddard 1977)	46
6.7.1	Key Evidence	46
6.7.2	The Massachusetts Pidgin Hypothesis	46
6.8	Cross-Source Cognate Table	46
6.9	Dialect Affiliation: Pokanoket Wampanoag	46
6.10	Problems Encountered	47
6.11	Research Approach	48
6.11.1	Step 1: Corroboration-Only	48
6.11.2	Step 2: Key Cognate Mappings	48
6.11.3	Step 3: h-Transcription as Secondary Evidence	48
6.11.4	Step 4: Segment k- Prefix	48
6.11.5	Step 5: Verify Against Trumbull	48
6.12	Conversion Workflow	49
6.13	Linguist Feedback Template	49
6.14	Related Work: Colonial Recorder Perception Problem	50
6.15	Key References	51
7	Wampanoag: Wood 1634	51
7.1	Source Description	51
7.2	Recorder Biography	52
7.3	Orthography System	52
7.4	Vowel Tables	53
7.5	Consonant Tables	53
7.6	Key Orthographic Observations	53

7.6.1	b vs. p Alternation	53
7.6.2	Geminate Consonants	53
7.6.3	qu Digraph	53
7.6.4	Word-Initial a- Prefix	53
7.6.5	Preaspiration /h/	53
7.7	Number System	53
7.8	Comparison: Wood vs. Eliot Orthography	54
7.9	Triple-Spelling Deep Dive: “Devil”	54
7.10	A-Prefix Analysis	54
7.11	Pidginization Theory	55
7.11.1	Goddard’s Analysis (1977, 1981)	55
7.11.2	Evidence in Wood’s List	55
7.11.3	Counter-Argument	55
7.12	Problems Encountered	56
7.13	Research Approach	56
7.13.1	Conversion Pipeline Recommendations	56
7.14	Conversion Workflow	57
7.15	Linguist Feedback Template	58
7.16	Related Work: Colonial Recorder Perception Problem	58
7.17	Key References	59
8	Mohegan-Pequot: Prince-Speck 1904	60
8.1	Source Description	60
8.1.1	Publication History and Citation Clarification	60
8.1.2	J. Dyneley Prince (1868–1945): The Transcriber	61
8.1.3	Frank G. Speck (1881–1950): The Fieldworker	61
8.1.4	The Speck–Prince Handoff	62
8.1.5	Fidelia Fielding’s Orthography and Its Influence	62
8.1.6	Orthography System	63
8.2	Phonemic Reconstruction	63
8.2.1	Consonant Inventory (Reconstructed)	63
8.2.2	Vowel Inventory (Reconstructed)	63

8.2.3	Preaspiration Phonology	63
8.2.4	Nasalization	63
8.3	German Orthographic Influence	64
8.3.1	Affricate Representations	64
8.3.2	Sibilant Representations	64
8.3.3	Vowel Length Marking	64
8.3.4	Consonant Fortition	64
8.3.5	Umlaut-Like Notation	64
8.4	Preaspiration Evidence	65
8.4.1	Explicit /h/ in Prince-Speck	65
8.4.2	Preaspiration (/hk/, /hp/, /ht/)	65
8.4.3	Why Prince-Speck Catches Preaspiration	65
8.4.4	Comparison with Other Sources	65
8.5	English Loanword Typology	65
8.5.1	Loanword Categories	66
8.5.2	Identifying Loanwords	66
8.5.3	Handling in Conversion Pipeline	66
8.6	Comparison with Other SNEA Recording Traditions	67
8.6.1	Williams 1643 (Narragansett)	67
8.6.2	Eliot 1660s (Wampanoag Mass Bible)	67
8.6.3	Stiles 1760s–90s (Mohegan-Pequot)	67
8.6.4	Comparison with Stephanie Fielding (2013) Modern Mohegan	68
8.7	Problems Encountered	68
8.8	Research Approach	69
8.9	Conversion Workflow	69
8.10	Linguist Feedback	70
8.11	Related Work: Colonial Recorder Perception Problem	70
8.12	References	71
9	Mahican: Edwards 1787	72
9.1	Source Description	72
9.1.1	Jonathan Edwards Jr. — Biographical Context	72

9.1.2	The 1788 Publication	72
9.1.3	Orthography System	73
9.2	Phonemic Reconstruction	73
9.2.1	Key Phonological Correspondences (PA → Mahican)	73
9.2.2	The /i / vs. /e / Issue	73
9.3	The <i>gh</i> Grapheme Mystery	74
9.4	Phonological Processes Affecting Edwards' Orthography	74
9.4.1	Vowel Reduction and <i>uh</i>	74
9.4.2	Nasalization of *a	74
9.4.3	Word-Final Vowel Deletion	75
9.4.4	Schwa Lowering Before /h, x/	75
9.4.5	Rounding Assimilation	75
9.5	Morphological Patterns in Edwards' Data	75
9.5.1	Person Prefixes	75
9.5.2	Verbal Suffixes	75
9.5.3	Noun Inflection	75
9.6	Comparative Sources for Orthography Interpretation	76
9.6.1	The Schmick Manuscript (Primary Internal Source)	76
9.6.2	Zeisberger's Delaware Materials	76
9.6.3	Later Mahican Sources	77
9.7	Problems Encountered	77
9.8	Research Approach	78
9.8.1	Reference Cognate Set for Validation	78
9.8.2	Software Architecture	78
9.9	Conversion Workflow	79
9.10	Linguist Feedback	79
9.11	Related Work: Colonial Recorder Perception Problem	79
9.12	References	80

II Thematic Synthesis 81

10 Preaspiration Across Sources 81

10.1	Cross-Source Comparison of Preaspiration Reporting	81
10.1.1	Edwards (1787) — The Gold Standard	81
10.1.2	Prince-Speck (1904) — The Trained Linguist	82
10.1.3	Eliot (1666) — Partial and Inconsistent	82
10.1.4	Winslow (1624) — Partial but Notable	82
10.1.5	Williams (1643) — The Invisible /h/	82
10.2	Cross-Linguistic Evidence for Preaspiration	83
10.3	Cowan (1973): Indirect Evidence for Preaspiration	84
10.4	Implications for Conversion Workflows	84
11	Colonial English Phonology as a Filter	85
11.1	Early Modern English Phonology Baseline (c. 1600–1620)	85
11.2	Per-Recorder Linguistic Background	85
11.2.1	Winslow (1624) — The Honest Amateur	86
11.2.2	Wood (1634) — The Pre-Orthographic Recorder	86
11.2.3	Williams (1643) — The Linguistically Gifted Amateur	86
11.2.4	Eliot (1666) — The Systematic Missionary	86
11.2.5	Edwards (1787) — The Fluent L2 Speaker	87
11.2.6	Prince-Speck (1904) — The Two-Stage Chain	87
11.3	The Kopris (2014) Framework	87
12	Vowel Ambiguity Patterns	88
12.1	Per-Source Vowel Ambiguity	88
12.1.1	Williams’ Vowel System in Detail	89
12.1.2	Eliot’s Vowel System	89
12.1.3	Wood’s Vowel Instability	90
12.1.4	Edwards’ Vowel System — The /i / Correction	90
12.2	Systematic vs. Random Variation	90
12.2.1	Random Variation: Wood’s Triple Spelling	90
12.2.2	Systematic Variation: Williams vs. Winslow	91
12.2.3	Diagnostic Criteria	91
12.3	Schwa Reduction Patterns	91
12.4	Nasalization Ambiguity	92

12.5 Cross-Linguistic Vowel System Comparison	93
13 Cross-Source Calibration Methodology	93
13.1 The Calibration Pipeline	94
13.2 Cross-Source Disambiguation: Specific Examples	94
13.2.1 Massachusetts Disambiguates Narragansett Vowels	94
13.2.2 Edwards Disambiguates Preaspiration	95
13.2.3 Prince-Speck Confirms Fielding	95
13.2.4 Wood and Winslow Corroborate Each Other	95
13.3 Sister-Language Dictionaries as Calibration Anchors	95
13.3.1 Eastern Algonquian (Closest Genetic Relatives)	96
13.3.2 Central Algonquian (Most Conservative Phonologically)	96
13.3.3 Plains Algonquian (Divergent, Boundary Conditions)	97
13.4 The Cognate Set Approach	97
13.5 How PA Reconstructions Anchor Vowel Quality	97
13.6 Calibration Confidence Levels	98
14 Dialect Classification Evidence	98
14.1 The Four-Way Dialect Classification	99
14.2 Costa (2007) Isoglosses	99
14.3 Mohegan-Pequot as a Y-Dialect: The Evidence	99
14.4 Central Algonquian Reference Points	100
14.5 Mahican as Transitional	101
14.6 Cross-Linguistic Cognate Evidence Table	101
14.7 Implications for the Conversion Workflow	102
14.8 Cross-Linguistic Reassessment of Dialect Classification	102
III External Resources and Open Questions	103
15 Non-SNEA Algonquian Resources	104
15.1 Proto-Algonquian Reconstructions	104
15.2 Eastern Algonquian (Closest Genetic Relatives)	104
15.3 Central Algonquian (Most Conservative Phonologically)	105

15.4 Plains Algonquian (Highly Divergent)	106
15.5 Comparative SNEA Dialectology	107
16 Open Problems	107
16.1 Phonological	107
16.2 Methodological	107
16.3 Corpus-Level	108
17 Aggregated Linguist Feedback	108
17.1 Feedback Categories	108
17.2 Cross-Cutting Patterns	108
18 Future Work	109
18.1 Immediate Priorities	109
18.2 Longer-Term Directions	109

1 Conversion Overview

1.1 Conversion Examples and Methodologies

This section presents concrete orthography-to-phoneme conversions across the eight source languages. For sources where implementation is complete, real conversion results are shown. For sources where only research has been completed, [PROTO] examples are given — these represent the current best hypothesis grounded in the research evidence, with documented uncertainty.

Narragansett (Williams 1643) — [PROTO]

Input: wétu (house)

Output: /wi:tə w/

Method: Comparative calibration via PA *wi:kiwa:mi, Massachusett cognate *wetu* (Eliot). The initial *wé-* maps to /wi:/ via Williams' circumflex-as-length convention; final *-tu* maps to /tə w/ via PA reflex.

Uncertainty: Vowel length in the first syllable is inferred from the circumflex accent, which Williams uses for primary stress — whether this marks phonemic length or stress prominence is debated.

Input: Aaunchemókaw (Record 3834)

Output: /ã:čimo:hka:w/ [PROTO]

Method: FSM with preaspiration inference. Williams writes *k* but the only phonetic record (\ph field) shows /hk/. The preaspiration gap — English-speaking recorders routinely fail to hear /h/ before stops — is the single most important finding for this source.

Uncertainty: The nasalization of the initial vowel is inferred from the *Aaun-* spelling, which may represent /ã:/ or /aun/.

Wampanoag (Trumbull/Eliot 1903) — [PROTO]

Input: matta (no, not)

Output: /mata/

Method: Direct Eliot orthography mapping. Eliot's system is the most regularized colonial orthography in the corpus. Confirmed via PA *mati-.

Input: achmōwonk (devil)

Output: /čomu:waŋk/ [PROTO]

Method: 4-stage FSM pipeline: (1) normalize ō (Eliot’s oo-ligature) to /u:/, (2) map *ach-* to /č/ via Eliot’s c-before-a rule, (3) resolve the *-wonk* suffix via PA *-a:kani, (4) comparative fallback against Pokanoket *hobbamock* (Winslow) to check for dialect divergence.

Uncertainty: The ō symbol (Eliot’s oo-ligature, U+A74F) was historically misidentified as an infinity symbol. Its phonetic value is long /u:/, but the precise quality (tense vs. lax) is debated.

Mohegan-Pequot (Fielding 2013) — [PROTO]

Input: wetu (house)

Output: /wetu/

Method: Direct 1:1 phoneme mapping. Fielding’s modern practical orthography is essentially phonemic. Confirmed via PA *wi:kiwa:mi and cognate with Narragansett *wétu* and Massachusetts *wetu*.

Wampanoag (Wood 1634) — [PROTO]

Input: abbona (five)

Output: /apana/ [PROTO]

Method: Normalize *b* to /p/ (Wood writes *b* where Eliot/Trumbull use *p* — direct evidence of voicing neutralization). Segment the initial *a-* as a demonstrative prefix.

Uncertainty: Wood’s triple spelling for ‘devil’ (*abbomocholabomocholabamacho*) is the strongest evidence in the corpus for colonial vowel instability. The medial vowel could be /a/, /ə/, or /o/.

Mahican (Edwards 1787) — [PROTO]

Input: ghusooh (eight)

Output: /xə sə w/ [PROTO]

Method: The *gh* grapheme is the most unusual across all eight sources. Candidates include /x/ (voiceless velar fricative), /h/, or /k/. PA *nešwa:šika and Wampanoag cognate *shwoshuk* suggest /x/ is most likely — a Mahican phoneme not present in SNEA languages.

Uncertainty: The *gh* value cannot be confirmed without cross-linguistic cognate evidence from Munsee Delaware or other Mahican-adjacent languages.

Mohegan-Pequot (Prince-Speck 1904) — [PROTO]

Input: bahkeder

Output: /pahketə r/ [PROTO]

Method: German orthography normalizer: *b*- initial = /p/ (German final devoicing rule applied to perception), *-hk-* = /hk/ (preaspiration preserved — Prince-Speck is one of the few sources that writes preaspiration explicitly), *-der* = /tə r/ (German *d* = /t/ in final position).

Uncertainty: The *-er* suffix may represent /ə r/ or a reduced syllable. Loanword detection needed — ~20–30% of the 446 entries are English loanwords.

1.2 Technique Comparison and Selection Rationale

The technique selection for each source is itself a research output, not a predetermined input. Each source’s unique characteristics — corpus size, orthographic regularity, ambiguity profile, available comparative evidence, recorder linguistic background — determine which approach is appropriate. A one-size-fits-all approach would fail because the sources span 263 years of colonial documentation with radically different orthographic systems.

Table 1 compares the approaches considered across all sources, with expected accuracy and rationale for selection or rejection. The per-source rationale is documented in detail in Part I.

1.3 Source Status and Expected KPIs

Table 2 shows the current status of each source. Research is complete for all eight sources. Implementation is pending for all sources. Proto-KPI targets are research-grounded predictions based on the known ambiguity profile of each source.

Note: Expected accuracy is lower for sources with extreme vowel ambiguity (Williams, Wood) or tiny corpora (Winslow). Mohegan-Pequot (Fielding) is expected to have the highest accuracy because its modern practical orthography is already phonemic. These targets will be updated as implementation validates or refutes the research predictions.

1.4 The Colonial Recorder Perception Problem

Every source in this study shares a fundamental challenge: English-speaking recorders with different linguistic backgrounds hearing Algonquian phonemes through their native phonological filter. This is not unique to SNEA — it is a well-documented phenomenon across colonial documentation of North American languages.

Dr. Keith Cunningham — Nanticoke

Dr. Keith Cunningham (Linguist, Nanticoke Indian Tribe) completed his doctoral dissertation *A Phonological Analysis of Nanticoke With Practical Applications for Language Revitalization* at Georgetown University (2024). The dissertation analyzes three 18th-century Nanticoke word lists (Heckewelder 1785) using the same methodology employed here: historical phonology from colonial recorders, with the same challenges of multilingual informants and orthographic complexity. His 2024 Algonquian Conference presentation *A Revised Phonological Analysis of the Heckewelder Vocabulary of Nanticoke* directly parallels the analyses in this document. Cunningham has also co-authored a Nanticoke language textbook, bridging directly into practical revitalization — the same goal as the SNEA orthography-to-phoneme conversion pipeline.

Cunningham’s work demonstrates that the colonial recorder perception problem extends beyond SNEA into the broader Eastern Algonquian family (Nanticoke, Maryland Algonquian). His methodology — PA comparative calibration, recorder-bias identification, cross-source verification — is directly transferable to the SNEA sources.

Dr. Craig Kopris — Wyandot

Dr. Craig Kopris (Independent scholar) has nearly three decades of experience in language description and endangered language revitalization, with emphasis on Iroquoian languages. His research note *Les sons wyandots perçus par des oreilles étrangères* (Wyandot Sounds through Foreign Ears, *Recherches amérindiennes au Québec*, 2014) is the exact framing of the core problem addressed in this document: how European recorders systematically misperceived and mis-transcribed indigenous phonemes. Kopris demonstrates that knowing the sound system of the transcriber’s first language is the key to recovering the phonetic details of the target language.

His paper *Wyandot Phonology: Recovering the Sound System of an Extinct Language* (Proceedings of the Second Annual High Desert Linguistics Society Conference, 1999) applies the same recovery

methodology to Wyandot, a Northern Iroquoian language unrelated to Algonquian. This demonstrates that the foreign-ear perception problem is universal across colonial documentation of North American languages, not limited to any single language family. His Dictionary of Wyandot project (NEH award FN-255576-17) shows the long-term application of these methods to language revitalization.

Relevance to This Project

The work of Cunningham and Kopriv establishes that the methodological framework for addressing the colonial recorder perception problem is well established. The SNEA orthography-to-phoneme project extends this framework to eight sources across four languages (Narragansett, Wampanoag, Mohegan-Pequot, Mahican) spanning 263 years of colonial documentation (1624–1904). The techniques developed here — FSM with weighted transitions, comparative calibration via PA, colonial English phonology baselines, preaspiration detection — contribute to an ongoing methodological conversation about recovering phonology from colonial sources across Eastern North America.

Part I

Source Workflows

2 Narragansett: Williams 1643

2.1 Source Description

Recorder: Roger Williams (1603–1683), founder of Rhode Island Colony. **Source:** *A Key into the Language of America* (1643), London: Gregory Dexter. **Language:** Narragansett [xnt] (Southern New England Algonquian). **Corpus:** 2,134 records. **Orthography:** English-based ad-hoc spelling, seventeenth-century Early Modern English conventions. Williams marks stress with diacritics (á primary, à low/falling, â rising-falling) — a major advantage over other colonial sources such as Eliot’s Massachusetts Bible (1663), which uses no stress marking.

2.1.1 Recorder Biography

Roger Williams (1603–1683) was a Puritan minister, theologian, and founder of the Colony of Rhode Island and Providence Plantations. A graduate of Pembroke College, Cambridge (B.A. 1627), Williams was a skilled linguist who reportedly mastered Latin, Greek, Hebrew, Dutch, French, and several Al-

gonquian languages. He arrived in Massachusetts Bay in 1631 and began intensive work with the Narragansett people after his banishment from the Massachusetts Bay Colony in 1635. His *A Key into the Language of America* (1643) was the first extended study of a Native American language published in English. The work takes the form of a phrase book organized by topic (greetings, family, hunting, etc.), with each Narragansett phrase accompanied by an English translation and occasional commentary.

2.1.2 Orthography System

Williams used an English-based ad-hoc spelling system with no standardized orthography. He applied English sound-spelling conventions (seventeenth-century Early Modern English) to Narragansett, an Algonquian language with a radically different phoneme inventory. The system was designed to be usable by English colonists, not by Algonquianists — it systematically collapses phonemic distinctions that English lacks.

2.1.3 Stress Marking

Williams' orthography is **stress-marked**, unlike Eliot's Massachusetts — this is a major advantage. Eliot's Massachusetts Bible (1663) uses no accent marks, making stress reconstruction for Massachusetts far more difficult. Williams' accents provide critical data about Narragansett prosody.

Aubin's proposed stress rule (from personal correspondence) posits that all long vowels are stressed and all alternate even-numbered short vowels are stressed. However, this rule requires correct vowel length classification, which has not been achieved for extinct SNEA languages. The problem is circular: vowel length is needed to assign stress, but stress is one of the cues for vowel length. Costa (2007) discusses the prosodic structure of SNEA languages and notes that Narragansett stress patterns appear similar to Loup A and Massachusetts, with a preference for stress on long syllables and alternating rhythm on short syllables — but the documentary record is too thin to confirm this definitively.

2.2 Vowel Inventory

2.2.1 Simple Vowels

2.2.2 Nasalized Vowels

A critical feature of SNEA phonology: Proto-Algonquian *a and *o developed nasalized reflexes in Narragansett before certain consonants. Williams does not mark nasality directly; he uses nasal coda consonant digraphs to imply it.

The nasalization is phonemic: *n now* (yes) vs. *núnów* (no) — only nasality distinguishes them, and Williams collapses this distinction orthographically in related forms.

2.2.3 Diphthong-Like Sequences

2.3 Consonant Inventory

SNEA languages neutralized the PA voicing distinction in stops. Williams' English ear interpreted this as a gradient between voiced and voiceless, writing both *p* and *b* (though *b* is rare), *t* and *d*, *k* and *g* (nearly absent). In almost all cases, *p*, *t*, *k* represent an unaspirated stop — between English /b/ and /p/, /d/ and /t/, / / and /k/.

2.3.1 Stops

Cowan (1973) demonstrated that PA *-k- (simple velar) and *-hk- (preaspirated velar) merged phonetically in Narragansett as /hk/ medially, but can sometimes be distinguished through the behavior of preceding vowels. PA *-k- shortens a preceding long vowel; PA *-hk- does not. This provides an indirect method for reconstructing preaspiration from vowel length patterns — even though Williams never writes the /h/.

2.3.2 Affricates

2.3.3 Nasals

2.3.4 Fricatives

Goddard & Bragdon (1988) note that in Massachusetts, the cognate system shows */s/ leniting to /h/ in intervocallic position — a sound change that may also operate in Narragansett but is invisible in Williams' orthography (since he never writes /h/).

2.3.5 Glides

2.3.6 The Silent /h/ — Preaspiration

English speakers routinely fail to hear preaspirated /h/ before stops. This is the single most important finding for this source. Williams almost never wrote /h/ where modern reconstructions put preaspiration:

- In Narragansett (and other SNEA languages), /h/ commonly precedes /p/, /t/, /k/ in word-medial position (e.g., /hka /, /hp/, /ht/)

- English speakers perceive these as single stops /k/, /p/, /t/ — preaspiration is not a phonemic distinction in English
- Williams wrote *k* where the actual pronunciation was /hk/ (preaspirated k)
- **Confirmed by \ph records:** The 3 records with \ph fields show missing /h/ in Williams

All three show the same pattern: Williams writes *k* where the reconstruction has /hk/. The /h/ is invisible to English ears.

2.4 Problems Encountered

2.4.1 Vowel Quality Ambiguity

Williams' vowels *a*, *e*, *i*, *o*, *u* each map to 2–4 possible phonemes. The vowel *a* alone can represent /ə/, / /, or /a /. This extreme ambiguity means that any single Williams spelling is consistent with multiple phonological interpretations.

2.4.2 Nasalization Implied but Not Marked

Sequences like *an*, *aun*, *om* may indicate nasal vowels /ã/, /õ/ or oral vowel + consonantal *n*. Disambiguating requires:

1. Checking the cognate in Massachusetts (Eliot marks nasal /ô/ explicitly)
2. Checking PA reconstruction (do expected reflexes predict nasalization in this environment?)
3. Checking morphological structure (is the *n* part of a suffix, or a nasalization marker?)

2.4.3 Stop Voicing Neutralized

p, *t*, *k* represent unaspirated stops that English speakers perceived variably as voiced or voiceless. Williams writes both *p* and *b*, *t* and *d*, *k* and *g* in free variation.

2.4.4 English Spelling Conventions

English digraphs (*ea*, *oo*, *ch*, *sh*) follow English values, but inconsistently due to 17th-century English pronunciation being different from modern English. For example, *ea* in 1643 was / /, not modern /i /.

2.4.5 Silent e

Word-final *e* is often silent, but not always. This affects at least 15–20% of entries. Comparison with Massachusett cognates (Eliot) is the primary disambiguation strategy.

2.4.6 Preaspiration /h/ Invisible

Only 3 of 2,134 records have phonetic (\ph) fields. All three show the same pattern: Williams writes *k* where the reconstruction has /hk/. If this pattern is representative, >50% of Williams' stops may be missing a preceding /h/.

2.4.7 *sh* Before Consonants

Williams writes *sh* before consonants where Massachusett cognates suggest plain /s/: Williams *shp*- corresponds to Massachusett *sp*-. Costa (2007) notes that this palatalization before consonants is also attested in Loup A records, suggesting it is a genuine SNEA dialect feature, not a Williams idiosyncrasy.

2.4.8 The *an/aun* Ambiguity for /ã/

The most consequential ambiguity: Williams writes both *an* and *aun* for /ã/, but also writes *an* for plain /an/ (oral a + n). Disambiguating requires comparative evidence from Massachusett (Eliot marks nasal /ô/ explicitly), PA reconstruction, and morphological analysis.

2.5 Research Approach

2.5.1 Recommended Methodology

DO NOT USE REGEX — the ambiguities are fundamentally context-dependent. Recommended approach:

1. **Finite State Machine (FSM):** Encode known spelling-to-phoneme rules as weighted state transitions, with fallback probabilities for ambiguous mappings.
2. **ML classifier:** Train on the 3 \ph seed examples (plus indirect evidence from comparative Algonquian) to predict preaspiration placement and vowel quality.
3. **Morphological analysis:** Use known Narragansett morphology (prefixes, stems, suffixes from Aubin/Hagenau) to segment Williams' spellings before applying conversion.

4. **Comparative evidence:** Use Proto-Algonquian reconstructions and sister languages (Massachusett, Mohegan-Pequot) as priors for ambiguous mappings.

2.5.2 Annotation Plan

The 3 \ph entries alone are insufficient for ML training. Recommended annotation plan:

1. Hand-annotate 50 representative entries with conversions
2. Use the annotations as training data
3. Evaluate on the remaining entries with spot-check verification
4. Cross-validate against Massachusett cognates (Goddard & Bragdon 1988) for at least 200 entries to calibrate the Narragansett-specific isoglosses

2.5.3 Cross-Linguistic Cognate Evidence

Massachusett, the closest sister language to Narragansett, was documented by John Eliot about 20 years after Williams. Eliot used a different orthographic convention — less English-phonetic, more systematic (he devised it for Bible translation).

2.5.4 Open Questions for Revitalization

1. **Vowel length:** Was Narragansett vowel length phonemic (as in Massachusett)? Williams' orthography suggests yes (áh vs. a for long vs. short /a/), but the marking is inconsistent.
2. **Preaspiration:** How pervasive was /h/ before stops? The 3 \ph entries suggest it was common.
3. **Nasalization phonemicity:** Was nasalization phonemic (as in Massachusett) or allophonic? Costa (2007) suggests phonemic in at least some SNEA dialects.
4. **Final vowel status:** Did Narragansett drop word-final vowels (like Mahican) or retain them (like Massachusett)? Williams' variable practice suggests either retention with devoicing or transition state.

2.6 Conversion Workflow

[Pending implementation --- see Issue #1362]

2.7 Linguist Feedback

[Template --- content pending review]

2.8 Related Work: Colonial Recorder Perception Problem

The core challenge of this research card — English-speaking recorders misperceiving and mis-transcribing Algonquian phonemes — is a known phenomenon documented across Eastern North America. Two researchers' work is directly relevant:

Dr. Keith Cunningham (Linguist, Nanticoke Indian Tribe). His doctoral dissertation *A Phonological Analysis of Nanticoke With Practical Applications for Language Revitalization* (Georgetown University) analyzes three 18th-century Nanticoke word lists (Heckewelder 1785) using the same methodology: historical phonology from colonial recorders, with the same challenges of multilingual informants and orthographic complexity. His 2024 Algonquian Conference presentation *A Revised Phonological Analysis of the Heckewelder Vocabulary of Nanticoke* directly parallels the Williams analysis — both involve English-speaking recorders working without standardized orthography, both require PA comparative calibration, and both feed into practical language revitalization. Cunningham's work demonstrates that the colonial recorder perception problem extends beyond SNEA into the broader Eastern Algonquian family (Nanticoke, Maryland Algonquian).

Dr. Craig Koprís (Independent scholar). His paper *Les sons wyandots perçus par des oreilles étrangères* (Wyandot sounds perceived by foreign ears, *Recherches amérindiennes au Québec*, 2014) is the exact framing of the Williams problem: how European recorders systematically misperceived and mis-transcribed indigenous phonemes. His *Wyandot Phonology: Recovering the Sound System of an Extinct Language* applies the same recovery methodology to an unrelated language family (Iroquoian), demonstrating that the foreign-ear perception problem is universal across colonial documentation of North American languages. His Dictionary of Wyandot project (NEH award FN-255576-17) shows the long-term application of these methods to language revitalization.

The Williams Narragansett corpus (2,134 records, only 3 with phonetic transcription) represents one of the most extreme cases of this problem — but the methodological framework for addressing it is established in the parallel work of Cunningham (Algonquian) and Koprís (Iroquoian).

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3 Wampanoag: Trumbull/Eliot 1903

3.1 Source Description

Recorder: James Hammond Trumbull (1821–1897), drawing on John Eliot’s 17th-century Massachusetts writing system. **Source:** *Natick Dictionary*, Bureau of American Ethnology Bulletin 25 (1903, posthumous). **Language:** Wampanoag (Massachusetts) [wam]. **Corpus:** 4,491 records (largest in the SNEA collection). **Orthography:** Eliot orthography — the first standardized writing system for an Algonquian language, developed by John Eliot and Massachusetts collaborators (Job Nesutan, John Sassamon) in the 1650s–1660s.

3.1.1 Recorder Biography: James Hammond Trumbull

James Hammond Trumbull (1821–1897) was an American philologist, historian, and bibliographer. He served as Connecticut State Librarian (1854–1897) and was a founding member of the American Philological Association. His work on the Algonquian languages of southern New England spanned four decades, beginning with his early comparative study *On the Best Method of Studying the North American Languages* (1859).

Trumbull’s major Algonquianist works include:

- *The Composition of Indian Geographical Names* (1870) — Connecticut Historical Society
- *Indian Names of Places in Connecticut* (1881) — systematic geographic toponymy
- *Natick Dictionary* (published posthumously 1903) — his magnum opus

3.1.2 The Lost Thesis Controversy

Trumbull delivered a landmark paper before the American Philological Association titled *The Method of the Indian Languages as Exemplified by the Massachusetts Dialect* — based on a “lost thesis” he had written at Yale in the 1840s. The paper argued for the polysynthetic structure of Algonquian languages and established Trumbull’s reputation as the leading American philologist of Native languages before the Boasian revolution.

Trumbull’s methodology combined:

1. **Text-based extraction:** Exhaustively reading through all of Eliot’s Massachusetts translations (Bible, primers, catechisms) and manually extracting every lexical item onto slips
2. **Cross-referencing with cognates:** Comparing Massachusetts forms with related languages (Naragansett via Williams 1643, Mohegan-Pequot via field notes, Abenaki, Micmac, etc.)
3. **Morphemic analysis:** Breaking down polysynthetic forms into constituent morphemes — a radical approach for its time
4. **Etymological comparison:** Identifying cognate sets across the Eastern Algonquian languages

3.1.3 Posthumous Publication

Trumbull died in 1897 with the dictionary manuscript substantially complete but not yet in press. Edward Everett Hale (the Unitarian minister and author) supervised the final editing at the Bureau of American Ethnology. The preface acknowledges that some inconsistencies in cross-references and source abbreviations may reflect the incomplete final revision by Trumbull himself. The published dictionary includes 4,491 main entries.

3.1.4 The Eliot Orthography — Historical Origins

John Eliot (1604–1690) arrived in Massachusetts Bay in 1631 and began preaching to the Massachusetts people in the 1640s. His language teacher was **Job Nesutan**, a Massachusetts man who served as Eliot’s primary linguistic informant and translator. Eliot also worked with **John Sassamon** (a Massachusetts convert educated at Harvard) who later became a crucial intermediary.

Eliot’s orthography was developed in the 1650s as he prepared his translation of the Bible. It was:

- **Phonemically oriented** for its time — Eliot attempted to represent the sounds of Massachusetts systematically, not just through English ear-approximations

- **English-based** — using English letter values as the starting point, then extending them
- **Regularized** — unlike Wood (1634) or Winslow (1624), Eliot chose consistent spellings for the same morpheme across occurrences
- **Influenced by Hebrew and Greek grammar** — Eliot’s *Indian Grammar Begun* (1666) explicitly draws parallels between Massachusett structures and Biblical Hebrew

The *Mamusse Wunneetupanatamwe Up-Biblum God* (1,180 pages) was printed by Samuel Green and Marmaduke Johnson in Cambridge, Massachusetts. It was the first complete Bible printed in the Americas in a Native language. Production required a specially cast type font (some characters unique to Massachusett), translators and typesetters literate in the orthography, and Massachusett readers to proofread.

3.2 Vowel Inventory

3.2.1 Simple Vowels

The core challenge of the Eliot orthography is that 5 vowel letters must represent a larger Massachusett vowel inventory (likely 8–10 vowel phonemes including length distinctions).

3.2.2 Vowel Digraphs

3.3 Consonant Inventory

3.4 The ∞ Symbol — Historical Typography and Phonetics

3.4.1 The Misidentification

The ∞ symbol (Unicode INFINITY, U+221E) is one of the most distinctive features of the database’s Trumbull entries. However, this is almost certainly a **typographic misidentification** in the digitization process. The original Trumbull dictionary uses a different symbol:

The actual symbol: Eliot’s Bible and Trumbull’s dictionary use a ligature of two o letters joined — what typographers call an “oo ligature” — Unicode (U+A74F, LATIN SMALL LETTER OO). In the early BAE Bulletin 25 printed type, this appeared as a figure-8-like symbol because the two o’s were cast as a single piece of type:

Original Eliot/Trumbull: (oo ligature)

What it looks like at small sizes: ∞ (infinity-like)

What DB has: ∞ (Unicode INFINITY)

The digitizer apparently did not recognize the typographic convention and encoded it as the visually-similar infinity symbol.

3.4.2 What Actually Represents

In Eliot's orthography, ∞ represents **long /u/** — specifically, it is Eliot's way of writing the long u sound, contrasting with oo which may represent a different vowel quality or length. Eliot's own description in *The Indian Grammar Begun* (1666):

“oo, and ∞, are two differing vowels, or rather the same vowel differently sounded; oo is short and ∞ is long.”

This suggests oo = /u/ (short) and ∞ = /u/ (long). However, database evidence showing 654 occurrences of the symbol (now encoded as ∞) suggests a more complex distribution. In many entries, ∞ appears where a preaspirated/breathy vowel sequence would be expected phonologically.

3.4.3 The ∞→oozzz Pipeline Mapping

Per the lessons-learned documentation, the mapping ∞ → oozzz was established empirically based on:

1. **Distributional patterns:** ∞ appears in contexts where /u/ + /h/ + /z/ would be expected from PA reconstruction
2. **Morphophonemic alternation:** Where cognate forms in other SNEA languages show /u/ + preaspiration + sibilant sequences
3. **Goddard & Bragdon (1988) transcription conventions:** Their modern phonemic transcription suggests underlying forms with /u h/ sequences

The oozzz string is therefore a **working normalization** — it captures the phonetic complexity (length + breathiness + sibilant quality) that the original ∞ was designed to encode, even if the exact phonetics remain debated.

3.5 Entry Structure — Detailed Parsing Guide

Trumbull entries have a complex multi-part structure parsed into a single 1x field. Understanding this structure is essential for FSM-based parsing.

3.5.1 Standard Entry Template

*|achmøwonk|, vbl. n. `news' C.

Field breakdown:

- * — Cross-reference marker
- |...| — Citation form delimiters (morphemic boundaries)
- , — Field separator
- vbl. n. — Grammatical label (verbal noun)
- `news' — Gloss in backtick quotes
- C. — Source abbreviation

3.5.2 Entry Variants

Root/compound form (no gloss):

|-adchaubuk|, in comp. words, root, or roots

Cross-reference entry:

|asqhuttøche| `whilst' C. = |asq-uttøche|.

Multiple glosses:

|wame|, adj. `all, whole, every' C.

3.5.3 Source Abbreviations

Trumbull used a system of abbreviations to tag each entry's source:

3.5.4 Grammatical Labels

3.6 Problems Encountered

3.6.1 The ∞ () Symbol Misidentification

The symbol was historically misidentified as an infinity symbol (∞). It is Eliot's oo ligature (U+A74F), representing long /u/. The digitization error encoding it as Unicode INFINITY (U+221E) has propagated through the database, requiring normalization in the conversion pipeline.

3.6.2 Preaspiration Gap

Eliot writes *hk/hp/ht* only 15% of expected positions. Edwards (1787 Mahican) is the only colonial source that consistently marks preaspiration. For Trumbull/Eliot entries, preaspiration /h/ must be inferred from comparative evidence (Edwards' Mahican data, PA reconstructions, morphotactic position) rather than read directly from the orthography.

3.6.3 Apostrophe Ambiguity

The apostrophe may represent /h/, / /, or vowel elision — no consistent rule. The apostrophe function is ambiguous without comparative evidence and cannot be resolved solely from within the Trumbull entry.

3.6.4 Dialect Stratification

Variation between Natick (Eliot) and Pokanoket (Winslow/Wood) may be orthographic, phonological, or dialectal. The *achm[∞]wonk* vs. *hobbamock* divergence (the “devil problem”) suggests real dialect differentiation between the “praying town” dialect of Natick and the Plymouth-area Pokanoket dialect.

3.6.5 Vowel Quality

The five-letter vowel system (a, e, i, o, u) must represent 8–10 phonemes. Disambiguation requires distributional analysis, PA correspondence, stress position, and morphological position.

3.6.6 *c* Ambiguity

Before back vowels (a, o, u) → /k/; before front vowels (e, i) → /s/. Exception: some irregular forms where *c* = /k/ before e. Word-final *c* → /k/.

3.6.7 Boundary Markers

The | . . . | pipe convention is near-universal (4,488 of 4,491 entries) but embedded glosses inside pipes are rare, cross-reference syntax (=) may span pipe boundaries, and hyphens within pipes indicate morpheme boundaries, not word boundaries.

3.6.8 The Devil Problem: *achm[∞]wonk* vs. *hobbamock*

One of the most revealing lexical puzzles in the corpus involves the word for ‘devil’:

This shows that Trumbull/Eliot may be drawing from a different lexical stratum (the “praying town” dialect of Natick) than Winslow/Wood (Plymouth-area Pokanoket). The lexical difference is not just orthographic — it may reflect real dialect divergence.

3.7 Research Approach

3.7.1 Recommended Methodology

DO NOT USE REGEX — the ∞ symbol, pipe boundaries, embedded glosses, and cross-reference syntax require context-sensitive parsing.

Stage 1: Preprocessing FSM. Tokenize Trumbull entries by:

1. Recognizing | . . . | citation boundaries and stripping pipes
2. Identifying ∞ and replacing with ooxxx (first normalization)
3. Extracting backtick-delimited glosses
4. Isolating grammatical abbreviations (vbl. n., C., etc.)
5. Handling * prefix and = cross-reference syntax
6. Separating source abbreviations (C., B., etc.) for provenance tracking

Stage 2: Phonemic Conversion FSM. Apply Eliot-to-phoneme mapping rules:

1. Map Eliot vowel digraphs (ee, oo, aa, etc.) to phonemes
2. Handle $c \rightarrow k/s$ disambiguation based on following vowel
3. Infer preaspiration /h/ from written *hk*, *hp*, *ht* clusters (direct evidence), morphotactic position (coda of stressed syllable, before /p t k/), comparative evidence from Edwards (1787) Mahican data, and PA reconstructions showing */h/ in corresponding position
4. Resolve $\infty \rightarrow$ ooxxx (already mapped in Stage 1)
5. Resolve apostrophe to /h/ or / / based on context and comparative evidence

Stage 3: Comparative Fallback. For ambiguous vowel quality (the 5-vowel $\rightarrow$ 8–10-phoneme mapping problem):

1. PA comparison: Use Aubin (1975) reconstructions as primary disambiguation

2. Sister-language evidence: Williams (Narragansett), Edwards (Mahican), Fielding/Prince-Speck (Mohegan-Pequot)
3. Known cognate patterns: Eliot orthography → modern Wampanoag revitalization (Weeden, Baird)
4. Goddard & Bragdon (1988) transcriptions: Modern phonemic analyses of Massachusetts texts

Stage 4: Statistical/Distributional Analysis. The large corpus (4,491 entries) enables statistical approaches not viable for smaller sources:

1. Vowel quality prediction model based on surrounding consonant environment and stress position
2. Preaspiration probability model based on morphotactic position and PA correspondence
3. ∞ behavior model: Distributional patterns of ∞ across phonological environments

3.7.2 Proto-Algonquian Correspondences

3.7.3 Confirmed Cognate Mappings for Pipeline Testing

3.8 Conversion Workflow

[Pending implementation --- see Issue #1363]

3.9 Linguist Feedback

[Template --- content pending review]

3.10 Related Work: Colonial Recorder Perception Problem

The core challenge of this research card — English-speaking recorders misperceiving and mis-transcribing Algonquian phonemes — is a known phenomenon documented across Eastern North America. Two researchers' work is directly relevant:

Dr. Keith Cunningham (Linguist, Nanticoke Indian Tribe). His doctoral dissertation *A Phonological Analysis of Nanticoke With Practical Applications for Language Revitalization* (Georgetown University) analyzes three 18th-century Nanticoke word lists (Heckewelder 1785) using the same methodology: historical phonology from colonial recorders, with the same challenges of multilingual informants and orthographic complexity. His 2024 Algonquian Conference presentation *A Revised Phonological Analysis of the Heckewelder Vocabulary of Nanticoke* directly parallels the Trumbull/Eliot analysis — both involve working with standardized missionary orthographies (Eliot for Massachusetts, Heckewelder for

Nanticoke) and both require PA comparative calibration. Cunningham’s work demonstrates that the colonial recorder perception problem extends beyond SNEA into the broader Eastern Algonquian family.

Dr. Craig Kopris (Independent scholar). His paper *Les sons wyandots perçus par des oreilles étrangères* (Wyandot sounds perceived by foreign ears, *Recherches amérindiennes au Québec*, 2014) is the exact framing of the Trumbull/Eliot problem: how European recorders systematically misperceived and mis-transcribed indigenous phonemes. His *Wyandot Phonology: Recovering the Sound System of an Extinct Language* applies the same recovery methodology to an unrelated language family (Iroquoian), demonstrating that the foreign-ear perception problem is universal across colonial documentation of North American languages. His Dictionary of Wyandot project (NEH award FN-255576-17) shows the long-term application of these methods to language revitalization.

The Trumbull corpus (4,491 records, the largest in the SNEA collection) benefits from Eliot’s relatively systematic orthography — but the preaspiration gap and vowel ambiguity problems documented in this section are the same foreign-ear perception issues that Cunningham and Kopris address in their respective language families.

3.11 Open Questions for Linguist Review

1. **∞() phonetics:** Is the oozzz mapping phonetically accurate, or should it be revised to uu with a diacritic for preaspiration? The zzz component seems ad-hoc.
2. **Preaspiration inference rules:** Can a reliable set of morphotactic rules predict when preaspiration /h/ was present but unwritten in Eliot’s orthography? Or does this require per-entry comparative verification?
3. **Dialect stratification:** How much of the variation between Trumbull/Eliot (Natick) and Winslow/Wood (Pokanoket) is orthographic vs. phonological vs. dialectal? The *achm∞wonk* vs. *hobbamock* divergence suggests real dialect differentiation.
4. **Apostrophe resolution:** Is there a predictable pattern for when ' = /h/ vs. / / vs. simple vowel elision? Edwards (Mahican) apostrophe usage may provide clues.
5. **Eliot’s c before e:** Are there identifiable exceptions to the c before front vowels → /s/ rule?
6. **Edwards cross-over:** How much does Edwards’ (1787) Mahican preaspiration system overlap with Massachusetts? If the systems are parallel, Edwards’ data can directly inform Trumbull /h/ inference.

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4 Mohegan-Pequot: Fielding 2013

4.1 Source Description

Recorder: Stephanie Fielding (Cits Pátunáhshô), Mohegan tribal member and linguist, with David Costa (Myaamia Center, Miami University of Ohio). **Source:** *Modern Mohegan Dictionary* (2006, revised 2013). **Language:** Mohegan-Pequot [xpq] (Y-dialect of SNEA). **Corpus:** 145 records (FF-marked entries from Fidelia Fielding’s diaries). **Orthography:** Modern practical orthography — 12 consonant + 6 vowel symbols, 1:1 phoneme-grapheme correspondence, circumflex for length.

4.1.1 Fidelia A. H. Fielding (1827–1908)

Fidelia A. Hoscott Fielding (Mohegan name: **Jits Bodunaxa** “Flying Bird”) was the last fluent speaker of the Mohegan-Pequot language. Born in 1827 at Mohegan, near Norwich, Connecticut, she learned Mohegan as a child from her grandmother, **Martha Uncas Shantup** (1761–1859), a direct descendant

of Sachem Uncas. As the number of fluent speakers dwindled through the late 19th century, Fielding maintained her language by speaking with her sister and, later, by talking to herself (Prince & Speck 1904:18–19).

4.1.2 The Fielding Diaries

Between 1902 and 1904, Fielding kept a series of diaries written in the Mohegan language. These are the only known surviving documents produced by a native speaker of Mohegan-Pequot. The corpus consists of:

- **Three diaries** and a copy of the Lord’s Prayer in Mohegan (Cornell University Library, acquired 2004 from the Huntington Free Library)
- A fourth diary was lost in a fire at Columbia University around 1906 (Granberry 2003:10)
- A sermon text published by Prince and Speck (1903) in *American Anthropologist*, vol. 5, pp. 193–212
- A long narrative tale recorded in phonetic notation by Frank Speck (Speck 1904)

The diaries were written in Fielding’s own orthography — an English-based ad-hoc spelling system reflecting her pronunciation. After her death in 1908, the diaries passed to Frank Speck at the University of Pennsylvania, and later to Cornell University. In November 2020, Cornell repatriated the diaries to the Mohegan Tribe in a formal ceremony, where they now reside in the Mohegan Archives at Uncasville, Connecticut.

4.1.3 Corpus Structure

The Brothertown corpus under analysis contains **145 records** from the Fielding dictionary. This is a subset of the full dictionary (which numbers several thousand entries, many reconstructed via comparative Algonquian methods by David Costa). Records marked **FF** in the dictionary indicate words attested directly in Fidelia Fielding’s diaries. The 145 records constitute our working corpus because they have explicit Mohegan-to-phoneme field mappings, whereas the larger set has modern orthography forms requiring parsing.

4.1.4 Comparative Significance

Prince and Speck (1904:18) explicitly compare Fielding to **Dorothy Pentreath**, the last speaker of Cornish, noting the “strangely pathetic” nature of a dying language spoken only by a single elderly woman who kept it alive by talking to herself. Despite the “last stages of decay” they observed, they found that “enough of the original phonetics and grammatical phenomena” remained to judge the language’s structure.

4.1.5 Stephanie Fielding’s Grammatical Analysis

Stephanie Fielding (Cits Pátunáhshô), a full-blooded Mohegan tribal member and descendant of Fidelia Fielding by marriage (her great-great-grandfather’s aunt), served as the Mohegan Tribe’s linguist and authored the *Modern Mohegan Dictionary* (2006, revised 2013). The dictionary was reviewed by David Costa who researched the lexicon and grammar, developed the Mohegan alphabet, translated scripts and wrote sample sentences, and proofed the grammar paradigms.

The dictionary includes:

1. **Pronunciation Guide** — the practical orthography with phoneme mappings
2. **Mohegan Grammar Paradigms** — a systematic grammatical sketch covering pronouns, nouns, verbs, and particles
3. **Mohegan to English Dictionary** — stems with part-of-speech codes and inflections
4. **English to Mohegan Word Finder** — English lookup with Mohegan stems

Fielding’s grammar covers the major Algonquian grammatical categories: pronouns (prefixes and suffixes attached to verb stems, pro-drop/polysynthetic), nouns (two genders: Animate vs. Inanimate, with plural, obviative, locative, and absentative inflections), verbs (four orders: Independent, Conjunct, Imperative, Participial, with transitivity marking VAI, VII, VTA, VTI), and derivation (person-marking prefixes *nu-* 1st person, *ku-* 2nd person).

4.1.6 Fielding’s Methodology

Stephanie Fielding worked extensively from texts, particularly the **Eliot Bible** (John Eliot’s 1663 Massachusetts translation), using a concordance to find Wôpanâak cognates and applying Mohegan-specific sound changes to reconstruct Mohegan forms. This comparative method is standard for Algonquian language reclamation. She also drew from Fidelia Fielding’s diaries (primary source, marked FF), sister

languages (Wôpanâak, Narragansett, Munsee Delaware), Proto-Algonquian reconstructions with sound-law application, and David Costa’s grammatical charts and comparative research.

Stephanie Fielding spent the 2017–2018 academic year as a presidential visiting fellow at Yale University, teaching a course comparing the Mohegan and Delaware languages. She has also worked with the Unkechaug people of Long Island to help reclaim their nearly identical language (the primary difference being the Mohegan /y/ vs. Unkechaug /ɾ/ correspondence).

4.2 Vowel Inventory

4.2.1 Vowel System

The Mohegan vowel system in the Fielding practical orthography consists of six plain vowels, four long vowels, and three diphthongs:

4.2.2 Diphthongs

Granberry (2003:14) also reconstructs /oi/ / j/ as marginal diphthongs (as in English “boy”) from certain sources.

4.3 Consonant Inventory

4.3.1 Allophonic Variation (from Granberry 2003)

Granberry’s phonological reconstruction (using the “reconstituted phonetic units” methodology of Mary Haas 1954) identifies allophonic patterns:

- Stops /p, t, k/ are **lenis-fortis pairs** overlapping in distribution — the practical orthography uses voiceless symbols but actual realization varies
- /kw/ word-finally often reduced to [k] before consonants
- Vowels show positional length variation: pre-stress short, stressed medium-long, post-stress reduced
- The phoneme /e/ (plain *e*) ranges from [i] to [e] depending on environment, likely reflecting the historical merger of PA *i and *e in initial syllables (Oxford 2015)

4.4 Historical Phonology: Proto-Algonquian Reflexes

4.4.1 PA Vowel System

Proto-Algonquian is reconstructed with an eight-vowel system: four qualities with a length contrast (Bloomfield 1946).

4.4.2 The Y-Dialect Classification

Mohegan-Pequot is classified as a **Y-dialect** of Southern New England Algonquian (SNEA). In the traditional classification based on the reflex of PA *r (sometimes reconstructed as *l):

This means that where Wôpanâak has *n*, Mohegan typically has *y*. Stephanie Fielding noted this correspondence operating even in modern Connecticut English pronunciation (e.g., “Sullivan” → “Suh-yuh-van”). Granberry (2003:13) explicitly documents the core correspondences: Pequot *p, t, k, ch, s* → Mohegan *b, d, g, j, z* form-initially, and final Pequot *p, t, k, ch, s* → Mohegan form-final *b, d, g, j, z*.

4.4.3 Consonant Reflexes

4.4.4 Cluster Reflexes

4.5 Comparison with Prince & Speck (1904)

4.5.1 The Prince-Speck Glossary

J. Dyneley Prince (Columbia University) and Frank G. Speck published “Glossary of the Mohegan-Pequot Language” in *American Anthropologist* (1904, vol. 6, no. 1, pp. 18–44). This glossary contains **446 words and forms** collected by Speck from Fidelia Fielding between 1902 and 1903, submitted to Prince in Fielding’s own spelling.

Prince and Speck deliberately preserved Fielding’s “imperfect” English-orthography system for “sentimental reasons” (1904:19), while providing phonetic transcriptions in parentheses.

4.5.2 Orthographic Differences

4.5.3 The Brothertown Material

A subset of Prince-Speck’s glossary consists of “Brothertown words” — items collected from an old Mohegan man who had lived at Brothertown, near Green Bay, Wisconsin, where relocated New England tribes (Tunxis, Wampanoag, Mohegans, Montauks) had settled. Prince and Speck note these words are

“merely corruptions of Ojibwe forms” (1904:19) — relevant since our corpus bears the name “Brother-town.”

4.5.4 Convergence and Divergence

Most Fielding (2006) entries with FF attestation can be mapped to cognates in Prince-Speck (1904). The primary differences are systematic orthographic conventions: Prince-Speck’s *g* initial corresponds to Fielding’s *k* (voicing neutralization); Prince-Speck’s *ch* corresponds to Fielding’s *c* (just a spelling difference); Prince-Speck’s vowel notation is inconsistent (reflecting English spelling conventions) while Fielding’s is phonemic; and Prince-Speck uses ' for schwa while Fielding omits schwa or encodes it as *u*, *a*, or *i*.

4.6 Current Orthographic Conventions

4.6.1 The Modern Practical Orthography

The orthography used by Fielding (2006) and adopted by the Mohegan Language Project is a **Latin-based phonemic alphabet** with the following conventions:

Alphabet (12 consonant symbols, 6 vowel symbols): a â c e ê h i k m n ô p q s sh t u
w y '

Key design principles:

1. **1:1 Phoneme-to-grapheme correspondence** — each symbol maps to exactly one phoneme
2. **Circumflex for length** — â, ê, ô are long; a, e, i, u are short
3. **c for /t/** — avoids the English *ch* digraph; never represents /k/
4. **q for /k/** — distinguishes from *kw* cluster; occurs word-finally
5. **' (okina) for glottal stop** — follows Hawaiian and other indigenous orthography conventions
6. **sh for /ʃ/** — standard English digraph
7. **No silent letters** — every written symbol is pronounced
8. **Bound morphemes marked with -** — prefixes: *nu-*, *ku-*; suffixes: *-côq*
9. **Part of speech tags** — VAI, VII, VTA, VTI, NI, NA following Algonquianist conventions
10. **Alternate spellings in parentheses** — showing earlier English-based forms

4.6.2 Comparison with Granberry's Orthography

Julian Granberry (2003) uses a somewhat different practical orthography, differing primarily in representing lenis-fortis distinctions:

Granberry's orthography explicitly notes the vowel /ə/ (schwa) as *u* — e.g., *wutche* vs. Fielding's *wicki*. He also uses *x* for /h/ in some environments. However, the Fielding orthography has become the standard through the Mohegan Language Project's adoption.

4.7 Problems Encountered

4.7.1 Small Corpus

Only 145 records with explicit Mohegan-to-phoneme field mappings. This is the smallest corpus among the three SNEA sources, limiting statistical approaches.

4.7.2 Alternate Spellings

The primary conversion challenge is English-based alternate spellings → Fielding orthography, not Fielding → IPA. The alternate spellings provide direct training data for converting Prince-Speck (1904) English-orthography forms.

4.7.3 Circularity Risk

Some Fielding dictionary entries are reconstructed FROM Prince-Speck (1904) — using Fielding to verify Prince-Speck is circular for those entries. Careful provenance tracking is required to distinguish entries with independent FF attestation from those reconstructed via comparative methods.

4.7.4 The Pequot Extinction and the Treaty of Hartford (1638)

The Pequot War (1636–1638) culminated in the **Mystic Massacre** (May 26, 1637), where English forces and allies set fire to a Pequot fortified village, blocking the exits and shooting anyone attempting to escape. Approximately 700 Pequots were killed or captured; survivors were sold into slavery in Bermuda and the West Indies, while others were dispersed as captives to the victorious tribes.

The **Treaty of Hartford** (September 21, 1638) explicitly sought to eradicate Pequot cultural identity:

- **Name prohibition:** Survivors could no longer call themselves Pequots — they were to be referred to as Mohegans or Narragansetts “for ever”

- **Land prohibition:** Pequots were forbidden from returning to their native country
- **Dispersion:** Surviving Pequots were divided: 80 to Mohegan Sachem Uncas, 80 to Narragansett Sachem Miantonomo, 20 to Niantic Sachem Ninigret
- **English control of Pequot country:** The Pequot lands went to the Connecticut River towns

The treaty effectively declared the Pequot nation extinct as a sovereign polity. A 1743 copy of the treaty survives and is housed at the Connecticut State Archives. Daragh Grant (2015, *William and Mary Quarterly*) reconstructed the original 13-provision text from multiple surviving copies, including a newly rediscovered 1665 copy in the British Library’s Lansdowne Manuscripts.

The Treaty of Hartford directly caused the language’s near-extinction. By outlawing Pequot speech and dispersing the population, the colonial government ensured the language would cease to be transmitted to children. Mohegan survived longer (until 1908) precisely because the Mohegan tribe was not subject to the same dispersal — their territory west of the Thames River remained intact.

4.8 Research Approach

4.8.1 Recommended Methodology

Direct 1:1 phoneme mapping. Fielding’s orthography is already phonemic — conversion is a direct lookup. The primary challenge is mapping alternate English-based spellings to the standard Fielding form.

For the 145 records in our corpus, the practical orthography’s transparency means:

- Phonemic conversion from the orthography is essentially 1:1 (98%+ regular)
- The primary challenge is not decoding Fielding orthography → phonemes, but mapping alternate spellings → Fielding orthography
- The alternate spellings provide direct training data for converting Prince-Speck (1904) English-orthography forms
- Vowel quality for /e/ (plain *e*) is the single remaining ambiguity — the phoneme ranges from /i/ to /e/ and its exact quality may need contextual disambiguation

4.9 Conversion Workflow

[Pending implementation --- see Issue #1364]

4.10 Linguist Feedback

[Template --- content pending review]

4.11 Related Work: Colonial Recorder Perception Problem

The core challenge of this research card — English-speaking recorders misperceiving and mis-transcribing Algonquian phonemes — is a known phenomenon documented across Eastern North America. Two researchers' work is directly relevant:

Dr. Keith Cunningham (Linguist, Naticoke Indian Tribe). His doctoral dissertation *A Phonological Analysis of Naticoke With Practical Applications for Language Revitalization* (Georgetown University) analyzes three 18th-century Naticoke word lists (Heckewelder 1785) using the same methodology: historical phonology from colonial recorders, with the same challenges of multilingual informants and orthographic complexity. His 2024 Algonquian Conference presentation *A Revised Phonological Analysis of the Heckewelder Vocabulary of Naticoke* demonstrates that the colonial recorder perception problem extends beyond SNEA into the broader Eastern Algonquian family. Cunningham's work on Naticoke revitalization (co-authoring a Naticoke textbook) parallels the practical application of the Fielding dictionary for Mohegan language reclamation.

Dr. Craig Kopriv (Independent scholar). His paper *Les sons wyandots perçus par des oreilles étrangères* (Wyandot sounds perceived by foreign ears, *Recherches amérindiennes au Québec*, 2014) is the exact framing of the Fielding/Prince-Speck problem: how European recorders systematically misperceived and mis-transcribed indigenous phonemes. His *Wyandot Phonology: Recovering the Sound System of an Extinct Language* applies the same recovery methodology to an unrelated language family (Iroquoian), demonstrating that the foreign-ear perception problem is universal across colonial documentation of North American languages. His Dictionary of Wyandot project (NEH award FN-255576-17) shows the long-term application of these methods to language revitalization.

The Mohegan-Pequot corpus is unique among the SNEA sources in having a last-speaker corpus (Fidelia Fielding's diaries) alongside a modern practical orthography (Fielding 2013). This positions it as the most phonologically reliable source — but the Prince-Speck (1904) glossary, with its German-influenced transcription, reintroduces the foreign-ear perception problem that Kopriv documents for Wyandot.

4.12 Mohegan Revitalization

4.12.1 The Mohegan Language Project

Following the death of Fidelia Fielding in 1908, Mohegan had no fluent speakers for nearly a century. In April 1998, the Council of Mohegan Tribal Elders resolved to use Jits Bodunaxa's speech as the basis for a restored Modern Mohegan (Granberry 2003:6). A Mohegan Language Committee was formed, and Julian Granberry was appointed Tribal Linguist from 1998 to 2001.

In the 21st century, the **Mohegan Language Project** was established, creating lessons, a dictionary, and online learning materials. The project developed a practical orthography using the Latin alphabet, based on the four phonetic diaries left by Fielding.

4.12.2 Current Revitalization Efforts (as of 2024)

The Mohegan Tribe operates an active **Language Restoration Project** that ramped up significantly in 2017. Key initiatives include:

- **Master-Apprentice Program:** Immersion-style learning with passive speaking → circumlocution → active speaking
- **Immersion Nests:** Yearlong cohort programs scaffolded to seasonal activities
- **Workbook Development:** First Mohegan language workbooks and grammar curricula
- **Two fluent teachers** (the most since Fidelia Fielding's time) leading classes
- **Online classes** open to all Mohegan tribal members and Mohegan household members

4.12.3 Collaboration with Jessie Little Doe Baird

MacArthur Fellow **Jessie Little Doe Baird** (Mashpee Wampanoag), renowned for her Wôpanâak language reclamation, has served as a guide for the Mohegan project. Using Wôpanâak (an N-dialect) as a sister language, the Mohegan teachers cross-reference sound changes (N→Y correspondence) to build vocabulary and grammar.

4.12.4 The Samson Occom Papers

In 2022, Dartmouth College repatriated the papers of **Samson Occom** (1723–1792), a Mohegan minister and scholar who wrote extensively in his native language. These 18th-century documents provide another

primary source layer for the Language Restoration Project, offering examples of the language written by a fluent speaker during the period when Mohegan was still actively spoken.

4.12.5 Terminology: Reclamation vs. Revitalization

Mohegan language teachers **Allyson Corbin** and **Autumn Cholewa**, who created much of the current curriculum, emphasize the term **reclamation** over revitalization or revival:

“Reclaim is much more empowering for a community. The connotation is more hands-on. More like, ‘This is up to you to do.’ To reclaim it is to bring it home. It’s a prideful thing instead of something people felt pressure not to do.” (CT Insider, Nov 2024)

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5 Wampanoag: Multi-Source (Wood, Winslow, Shepard, Eliot)

5.1 Source Description

This section covers all historical Wampanoag/Massachusetts source vocabularies beyond the Trumbull compilation (covered in §3). Four primary sources span the critical transition from raw English-hearer perception through systematic missionary orthography:

William Wood (1634) *New Englands Prospect*, ~262 words. Pre-Eliot English ad-hoc orthography.

Wood was an English colonist living in the Massachusetts Bay Colony with no linguistic training.

His transcriptions reflect the earliest English-hearer perception of Wampanoag sounds, before any regularization system existed.

Edward Winslow (1624) *Good Newes from New England*, 22 scattered words and phrases. The earliest published Wampanoag vocabulary. Winslow was a Mayflower passenger and Plymouth Colony governor who described the language as “very copious, large, and difficult” and was unusually honest about his linguistic limitations.

Thomas Shepard (1647) *The Day-Breaking, if not the sun-rising of the Gospell with the Indians in New-England*, 11 native words plus two one-sentence prayers. Predates Eliot’s grammar by 19 years and his Indian Primer by 22 years. The prayers represent Eliot’s earliest work in the language, before he had fully analyzed its grammar.

John Eliot (1666) *The Indian Grammar Begun*, the first published grammatical analysis of any Algonquian language. Eliot’s orthography is a deliberately constructed system, not a casual English-based spelling. He distinguishes aspirated vs. unaspirated stops using English voiced/voiceless pairs (p~b, t~d, k~g).

5.2 Orthography Systems

5.2.1 William Wood (1634)

Wood’s vocabulary is the largest pre-Eliot Massachusetts word list. Its orthography follows English sound-spelling conventions of the 1630s:

- **Initial /w/ retained:** *Wigwam, Web, Wompey*
- **⟨ea⟩ for /i /:** *Kean* (cf. Eliot’s *keen*)
- **⟨s⟩ for / / inconsistently:** *Sawup* where later sources might write *sh-*
- **⟨k⟩ preferred over ⟨c⟩:** Initial /k/ written *K-* even before front vowels (*Kean, Ksitta*)
- **No diacritics:** Wood uses plain English letters only
- **Vowel length not marked:** Short vs. long vowel distinction invisible
- **Pidginization noted:** Goddard (1977) observes a “good deal of pidginization” in Wood’s data (Salvucci 2002)

5.2.2 Edward Winslow (1624)

Winslow's orthography is purely English-hearer perception with zero regularization. He was recording an entirely unfamiliar language with no reference system. The vocabulary overwhelmingly represents the Pokanoket dialect of Wampanoag (Massasoit's people), with some possible influence from the Patuxet band through Tisquantum.

Key characteristics:

- **Pre-Eliot:** No systematic orthography; words are ad-hoc English approximations
- **⟨h⟩ written reliably:** Winslow writes /h/ more consistently than any other early Wampanoag source except Edwards (1787 Mahican). This is one of the most valuable aspects of his small corpus.
- **⟨l⟩ → ⟨n⟩:** *Winsnow* for “Winslow” — no /l/ in Wampanoag
- **Pidginization:** Goddard (1977) identifies clear evidence of pidginization, including use of transitive inanimate *namen* where transitive animate *nunaw* would be grammatically correct

5.2.3 Thomas Shepard (1647)

The Wampanoag words in this pamphlet show unstandardized English orthography with the following characteristics:

- **⟨ch⟩ for /č/:** *Chechainuppan*, *Chepian*, *Ceehesom*
- **⟨sh⟩ and ⟨s⟩ for /š/:** *Evangenesch*, *Iehovah* (given English biblical name spelling)
- **⟨w⟩ for /w/:** *Wigwam*, *Wowein*, *wicke*
- **Final -an for animate nouns:** *Chepian*, *Chechainuppan*
- **⟨oo⟩ for /u /:** *Noonatomen*
- **⟨p⟩ for /b/:** *Chepian* (cf. Eliot's *chepiân*)
- **Preaspiration:** *cke* in *wicke* may indicate preaspirated stop

The pamphlet contains the earliest recorded Wampanoag Christian prayers:

Amanaomen Iehovah tahassen metagh — “Take away Lord my Stony heart”

Ceehesom Iehovah kekowhogkew — “Wash Lord my soule”

5.2.4 John Eliot (1666)

Eliot's grammar presents a deliberately constructed orthography. His key innovation is distinguishing aspirated vs. unaspirated stops using English voiced/voiceless pairs (p~b, t~d, k~g). This is the most important orthographic feature for phonological reconstruction.

Eliot describes 5 primary vowels with length distinctions:

- **a** (short) vs. **â** (long)
- **e** /i / (always long)
- **i** (short) vs. **ee/ie** (long — inconsistently marked)
- **o** (short) vs. **ô** (long)
- **u** (short schwa) vs. **oo** (long /u /)

5.3 Vowel and Consonant Tables

5.4 Source-Specific Orthographic Patterns

Each source has characteristic “errors” or systematic deviations from a standardized phonemic representation:

Wood (1634) No vowel length; English digraphs predominate; treats unaspirated stops as voiced; pidginized forms; word-initial *a*- prefix frequently attached to roots.

Shepard (1647) No consistent system; words are ad-hoc English approximations; preaspiration inconsistently shown; ⟨ck⟩ for /k/.

Eliot (1666+) Most consistent system; treats unaspirated stops as voiced English letters (b, d, g) but means unaspirated; vowel length inconsistently marked on some vowels; macron on long vowels (â, ô); distinctive ⟨ ⟩ (oo ligature) for /u /.

Cotton (1708) Extends Eliot's system; adds breve for ultra-short vowels; shows late shift in some vowels; more consistent vowel marking.

Trumbull (1903) Standardized historical reconstruction; marks length and aspiration consistently; normalizes variation.

5.5 Problems Encountered

- **Source provenance correction:** The “Anonymous 1647” vocabulary in ALR Vol. 27 is actually Thomas Shepard’s *The Day-Breaking*. Furthermore, the sample words in the original ALR listing (*kean, kenomptom, wean, nukus, kodtomp, ompmom, sawop, acheta co*) do NOT match the words actually found in the 1647 text. These words originate from William Wood’s 1634 vocabulary. The correct Shepard word list contains 11 items including the two prayers.
- **Pre-Eliot vs. post-Eliot:** Sources span the transition from raw English-hearer perception (Wood, Winslow) through systematic missionary orthography (Eliot). The same letter sequence may have different phonetic values in different sources.
- **Eliot’s stop aspiration system:** Eliot is the only colonial source who systematically distinguishes aspirated from unaspirated stops using orthographic pairs (b/p, d/t, k/g). His ⟨b, d, g⟩ represent unaspirated lenis stops (not voiced), while ⟨p, t, k⟩ represent aspirated fortis stops.
- **Pidginization:** Goddard (1977) identifies pidginization in both Wood and Winslow. If the English were learning a simplified contact variety, word order may not reflect full Wampanoag syntax, verb forms may be bare stems rather than inflected forms, and vocabulary may be mixed from multiple Algonquian languages.
- **Intermediary filter:** Winslow’s data passed through Tisquantum (Squanto) and Hobbamock as interpreters. Tisquantum’s English was learned in London, not from the Pilgrims, and his Wampanoag was Patuxet, a sub-dialect closely related to but not identical with Pokanoket.
- **Dialect uncertainty:** Pokanoket (Winslow/Wood) vs. Natick (Eliot/Trumbull) differences are unresolved. The distinction may represent real dialect differences, register differences (pidgin vs. full language), chronological differences (1620s vs. 1660s), or orthographic convergence.
- **Vowel instability:** Wood’s triple spelling *abbomocho/abomocho/abamacho* for “devil” is the strongest evidence in the entire SNEA corpus for colonial vowel instability.
- **Preaspiration /h/ unreported:** Like all pre-Eliot sources, Wood does not write preaspirated /h/ before stops. Winslow is the exception — he writes /h/ more consistently than any other early source.

5.6 Research Approach

The multi-source parsing strategy follows a four-stage pipeline:

1. **Source detection:** Identify which source's orthographic fingerprint is present. Each source has a characteristic orthographic texture that should be identifiable from spelling patterns.
2. **Source normalizer:** Convert to Eliot-consistent form. Apply source-specific decoding rules — the same letter sequence may have different phonetic values in different sources.
3. **Eliot parser:** Apply Grammar rules to produce a phonemic representation. Eliot's orthography is the most systematic and serves as the intermediate representation.
4. **Unified output:** Produce standard modern transcription.

5.6.1 Source-Adaptive Parsing Strategy

Wood English sound-spelling → phoneme via lookup table. Key mappings: ea→/i /, oo→/u /, k→/k/, s→/s~š/.

Shepard Ad-hoc English → phoneme via context. Same as Wood but with higher ambiguity.

Eliot Eliot orthography → phoneme via Grammar rules. p/b→/p/ vs. /p /, t/d→/t/ vs. /t /, k/g→/k/ vs. /k /, â→/a /, ô→/o ~u /.

Cotton Same as Eliot + breve → extra-short vowel. Vowel quantity more reliable.

Trumbull Modern normalized → phoneme directly. The cleanest mapping.

5.6.2 Resolving Ambiguity

Where a spelling could represent multiple phonemes (e.g., Wood's ⟨ea⟩ = /i / or / /), the parser uses source context and Algonquian comparative evidence. The Proto-Algonquian reconstructions in Aubin (1975) provide the external check needed to disambiguate competing analyses.

5.7 Conversion Workflow

[Pending implementation --- see Issue #1365]

The conversion pipeline will be implemented as a Python module in `src/commons/parsing/`. The module will:

- Detect source fingerprint from orthographic patterns
- Apply source-specific normalizer functions
- Route through the unified Eliot parser
- Output standard modern transcription
- Handle the pre-Eliot vs. post-Eliot transition
- Resolve ambiguity using PA comparative evidence

5.8 Linguist Feedback Template

[Template --- content pending review]

The following questions are open for linguist review:

- Are the source-specific vowel and consonant mappings correct?
- Does the proposed Eliot-consistent intermediate representation capture the phonemic distinctions correctly?
- Are there additional sources that should be included?
- Is the pidginization analysis (Goddard 1977) accepted, or are there alternative interpretations?
- How should the Pokanoket/Natick dialect distinction be handled in the unified output?

5.9 Related Work: Colonial Recorder Perception Problem

The core challenge of this research card — English-speaking recorders misperceiving and mis-transcribing Algonquian phonemes — is a known phenomenon documented across Eastern North America. Two researchers' work is directly relevant:

Dr. Keith Cunningham (Linguist, Nanticoke Indian Tribe). His doctoral dissertation *A Phonological Analysis of Nanticoke With Practical Applications for Language Revitalization* (Georgetown University) analyzes three 18th-century Nanticoke word lists (Heckewelder 1785) using the same methodology: historical phonology from colonial recorders, with the same challenges of multilingual informants and orthographic complexity. His 2024 Algonquian Conference presentation *A Revised Phonological Analysis of the Heckewelder Vocabulary of Nanticoke* directly parallels the multi-source Wampanoag analysis — both involve multiple colonial recorders (Wood, Winslow, Shepard, Eliot for Wampanoag; Heckewelder,

Du Ponceau for Nanticoke) with different orthographic conventions that must be cross-calibrated. Cunningham’s work demonstrates that the colonial recorder perception problem extends beyond SNEA into the broader Eastern Algonquian family.

Dr. Craig Koprís (Independent scholar). His paper *Les sons wyandots perçus par des oreilles étrangères* (Wyandot sounds perceived by foreign ears, *Recherches amérindiennes au Québec*, 2014) is the exact framing of the multi-source Wampanoag problem: how European recorders with different linguistic backgrounds (17th c. English, 18th c. English, 19th c. German-influenced American English) systematically misperceived and mis-transcribed indigenous phonemes. His *Wyandot Phonology: Recovering the Sound System of an Extinct Language* applies the same recovery methodology to an unrelated language family (Iroquoian), demonstrating that the foreign-ear perception problem is universal across colonial documentation of North American languages.

The multi-source Wampanoag corpus (Wood 1634, Winslow 1624, Shepard 1647, Eliot 1666) spans the full range of colonial recorder competence — from raw English-hearer perception (Wood, Winslow) through systematic missionary orthography (Eliot). This range makes it the best case study for the recorder calibration problem that both Cunningham and Koprís address in their respective language families.

5.10 Key References

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6 Wampanoag: Winslow 1624

6.1 Source Description

Recorder: Edward Winslow (1595–1655), Mayflower passenger, Plymouth Colony governor for three terms (1633–1634, 1636–1637, 1644–1645), and the colony’s primary negotiator and diplomat with the Native peoples of New England.

Source: *Good Newes from New England* (1624), printed in London by I.D. for William Bladen and John Bellamie. The earliest published source of Wampanoag vocabulary in the SNEA corpus.

Language: Wampanoag, Pokanoket dialect [wam].

Corpus: 24 records (the smallest in the SNEA collection).

Orthography: English-hearer perception with zero regularization. Winslow was unusually honest about his linguistic limitations, describing the Massachusetts language as “very copious, large, and difficult. As yet we cannot attain to any great measure thereof.” This acknowledgment distinguishes Winslow from many other colonial recorders who pretended to greater competence. His transcriptions are therefore likely more reliable as raw phonetic impressions — he was not normalizing or regularizing what he heard.

6.2 Recorder Biography

Edward Winslow was among the 41 Mayflower passengers who signed the Mayflower Compact in November 1620. Key events relevant to his linguistic data:

- **March 1621:** First contact with Samoset (Abenaki sachem speaking English) and Massasoit.
- **March–July 1621:** Tisquantum (Squanto) serves as interpreter, teaching the Pilgrims Wampanoag vocabulary.
- **March 1623:** Winslow cures Massasoit of a life-threatening illness — the encounter that produced his most extensive Wampanoag dialogue.
- **August 1623:** Hobbamock warns Plymouth of Massasoit's alleged conspiracy with the Massachusetts; Standish's Wessagussett raid follows.
- **1624:** *Good News from New England* published in London.

After returning to England in 1646, Winslow served as Oliver Cromwell's chief agent in foreign missions. He died of yellow fever in 1655 en route to a diplomatic mission in Spanish Hispaniola.

6.3 Linguistic Intermediaries

Winslow did not learn Wampanoag directly — he worked through interpreters. The identity and linguistic background of these intermediaries is crucial for understanding potential biases in his transcriptions.

6.3.1 Tisquantum (Squanto, c. 1585–1622)

Tisquantum was a Patuxet Wampanoag with an extraordinary linguistic history:

- **1614:** Kidnaped by Captain Thomas Hunt, along with 20+ other Patuxet men.
- **1614–1618:** Sold as a slave in Málaga, Spain; escaped with the help of Spanish friars.
- **1618–1619:** Lived in London with John Slany, a merchant of the Newfoundland Company; learned English.
- **1619:** Returned to Patuxet (modern Plymouth) — only to find his entire village depopulated by the 1616–1619 epidemic.
- **1621:** Approached Plymouth Colony in March; served as interpreter and guide.

- **1622:** Died of “Indian fever” at Monomoit (Chatham, Cape Cod) while accompanying Winslow on a diplomatic mission.

Linguistic implications of Tisquantum’s mediation:

- His English was learned in London, not from the Pilgrims — it was 16th-century English, not colonial dialect.
- His Wampanoag was Patuxet, a sub-dialect of Wampanoag closely related to Pokanoket but not identical.
- His time in Spain and England (8+ years) may have affected his native language production (attrition).

6.3.2 Hobbamock (Hobomok, d. 1642)

Hobbamock was a Pokanoket *pniese* — a military champion and elite warrior — who served as Massasoit’s trusted advisor and was assigned to live among the English:

- **March 1621:** Assigned by Massasoit to reside at Plymouth Colony as a liaison.
- **March 1623:** Revealed Massasoit’s alleged conspiracy with the Massachusett, leading to Standish’s raid.
- **Accompanied Winslow** on the mission to cure Massasoit — his lament “Neen womasu sagimus” (“My loving sachem”) upon hearing of Massasoit’s apparent death is one of Winslow’s most famous transcriptions.
- **Died of European disease** in 1642.

Linguistic significance: Hobbamock likely spoke Pokanoket dialect natively. He was not a language teacher but may have provided a more natural sample of Wampanoag speech than Tisquantum. His English was likely limited — Winslow probably communicated with him through a mixture of Wampanoag and signs.

6.4 Complete Vocabulary

According to Salvucci (InteractiveALR Vol. 27) and confirmed by primary source reading, Winslow’s text contains 22 distinct words and phrases in Wampanoag:

6.5 Syntactic Complexity

Despite the small corpus, Winslow records several sentences showing Wampanoag syntax:

Keen Winsnow? Copular question: *Keen* = 2sg pronoun “thur”, predicate nominal.

Matta neen wonckanet namen Winsnow Negative declarative: *Matta* (neg) + *neen* (1sg) + *wonckanet* (again/future) + *namen* (see) + vocative.

Hinnaim namen, hinnaim michen, matta cuts Serial verb construction: three coordinate clauses with *hinnaim* (by-and-by) + *matta* (neg).

Neen womasu sagimus Possessive + noun: *neen* (I/my) + *womasu* (loving?) + *sagimus* (sachem + 1sg possessive).

6.6 Orthography System

Winslow wrote in Early Modern English orthography (pre-1625), before English spelling standardization. Understanding what Winslow’s spellings meant for English is essential before mapping to Wampanoag phonemes.

6.6.1 Vowel Grapheme-to-Phoneme Mapping

6.6.2 Consonant Grapheme-to-Phoneme Mapping

6.6.3 Preaspiration /h/ in Winslow

Winslow writes /h/ more consistently than any other early Wampanoag source except Edwards (1787 Mahican). This is one of the most valuable aspects of his small corpus.

Interpretation: Winslow’s relatively good /h/ transcription may reflect:

- His intermediaries (Tisquantum, Hobbamock) had clearer /h/ articulation
- The Pokanoket dialect had more salient /h/ than Natick
- Winslow was a more careful listener — his “ignorance” of the language made him more phonetically accurate (less normalization bias)

6.7 Pidginization Analysis (Goddard 1977)

Goddard's landmark analysis of early contact Algonquian data identified Winslow's vocabulary as showing clear evidence of pidginization — the English and Wampanoag were communicating via a simplified contact variety, not the full grammatical language.

6.7.1 Key Evidence

In the sentence *Matta neen wonckanet namen Winsnow*:

1. ***namen* (transitive inanimate “see it”)** is used instead of the grammatically correct transitive animate form (expected *nunaw* “I see him”). The verb agrees with the object for gender — *namen* treats “Winslow” as if he were an inanimate object.
2. **Pro-drop is lost:** Full independent pronouns (*neen* “I”, *keen* “thou”) are used where verbal inflection alone would carry person in the full language. This is a universal pidgin feature — morphology simplifies, syntax becomes analytic.
3. **No verbal inflection:** The expected inflected verb (*n-nam-en* = 1sg-see-TA) is replaced with what appears to be the citation form or a nominalized form.

6.7.2 The Massachusetts Pidgin Hypothesis

Goddard hypothesized that a Massachusetts Pidgin (similar in function to Delaware Pidgin or Mobilian Jargon) existed before English colonization, used for inter-tribal trade. If so, what Winslow recorded may not have been “pure” Wampanoag at all, but rather a trade pidgin. This would explain:

- The consistency of certain simplifications (pidgin features shared across sources)
- The spread of certain forms (e.g., *matta* for negation appears in all colonial sources)
- The divergence between “missionary” Massachusetts (Eliot 1663) and “contact” Massachusetts (Winslow, Wood)

6.8 Cross-Source Cognate Table

6.9 Dialect Affiliation: Pokanoket Wampanoag

Winslow's data was collected primarily at Plymouth (Patuxet) and Pokanoket (modern Bristol, RI — Massasoit's capital). The Wampanoag Confederacy in 1620 consisted of several bands: Pokanoket (Mas-

sasoit's people, primary), Patuxet (Plymouth, Tisquantum's people, depopulated by 1620), Nauset (Cape Cod), Monomoi (Chatham), Manomet (Sandwich/Bourne), and Mashpee (Cape Cod).

The key dialectal division within Massachusetts/Wampanoag:

Per Salvucci's assessment: "Winslow's data has not been systematically analyzed for dialect affiliation." The Pokanoket sub-corpus is so small (22 items) that confident dialect assignment is impossible without external evidence.

6.10 Problems Encountered

- **Tiny corpus:** 24 records is insufficient for any statistical approach. Every entry must be treated as a unique case. Winslow is a corroborative source only.
- **Pidginization:** Goddard (1977) identifies "a good deal of pidginization" in Winslow's data. This affects whether the data can be used for phonological reconstruction at all. If the English were learning a pidgin variety, word order may not reflect full Wampanoag syntax, verb forms may be bare stems, and vocabulary may be mixed from multiple Algonquian languages.
- **Intermediary filter:** Tisquantum (Squanto) and Hobbamock served as interpreters — their own linguistic backgrounds may have affected the transcriptions. Tisquantum's English was learned in London, not from the Pilgrims, and his Wampanoag was Patuxet, a sub-dialect closely related to but not identical with Pokanoket.
- **Dialect uncertainty:** Pokanoket vs. Natick differences are unresolved. The distinction may represent real dialect differences, register differences (pidgin vs. full language), chronological differences (1620s vs. 1660s), or orthographic convergence.
- **Proper name contamination:** Some entries include English names (Winsnow = Winslow), which follow English orthography and must be excluded from phonemic analysis.
- **Multi-word entries:** Several entries are entire sentences requiring syntactic segmentation before lexical analysis.
- **Uncertain dialect:** Pokanoket dialect vs. Patuxet (via Tisquantum) vs. generic pidgin. Not resolvable from Winslow's data alone.

6.11 Research Approach

Five-step corroboration pipeline using Trumbull and Wood as primary sources. Winslow data is treated as corroborative, not primary. Each entry is verified against expected Eliot/Trumbull forms.

6.11.1 Step 1: Corroboration-Only

Treat Winslow as corroborative evidence for rules derived from larger sources. For each Winslow entry, find the cognate in larger Wampanoag sources (Trumbull 4,491 records, Wood 339 records). Use the cross-source consensus for the phonemic mapping. Flag Winslow-only words for manual review.

6.11.2 Step 2: Key Cognate Mappings

6.11.3 Step 3: h-Transcription as Secondary Evidence

Winslow's relatively consistent /h/ transcriptions provide secondary evidence for where preaspiration occurs:

- When Winslow writes *h* and other sources also confirm it → well-attested
- When Winslow writes *h* and other sources omit it → possible but needs external confirmation
- When Winslow omits *h* and Edwards (Mahican) writes it → probable preaspiration unrecorded

6.11.4 Step 4: Segment k- Prefix

The 2nd person prefix *k-* is seen in *keen* (thou, 2sg pronoun) and should be segmented before conversion. Note that *k-* with vowel-initial stems would produce different surface forms than *k-* with consonant-initial stems.

6.11.5 Step 5: Verify Against Trumbull

For each converted Winslow entry, verify against the expected form in the Eliot/Trumbull orthography. Flag mismatches for manual review — especially:

- Where Winslow writes /h/ and Trumbull does not (possible preaspiration evidence)
- Where Winslow and Trumbull use different lexical roots (possible dialect divergence)
- Where Winslow's form contradicts known PA correspondences

6.12 Conversion Workflow

[Pending implementation --- see Issue #1366]

The conversion pipeline will be implemented as a Python module in `src/commons/parsing/`. The module will:

- Treat Winslow as corroborative only — no regex-based conversion
- For each Winslow entry, find the cognate in larger Wampanoag sources
- Use cross-source consensus for phonemic mapping
- Flag Winslow-only words for manual review
- Segment *k*- prefix before conversion
- Verify each converted entry against expected Eliot/Trumbull forms

6.13 Linguist Feedback Template

[Template --- content pending review]

The following questions are open for linguist review:

1. **Pidgin or genuine dialect?** Are the simplifications in Winslow’s data (e.g., *namen* for “see” used as transitive animate) evidence of a pre-existing Massachusetts Pidgin (Goddard 1977) or simply a learner’s imperfect recording? This affects whether Winslow’s data can be used for phonological reconstruction at all.
2. ***witeo* vs. *wetu*:** Is the *witeo* (Winslow) / *wetu* (Eliot) split evidence of Pokanoket vs. Natick dialect difference, or are both from PA *wi·kiwa·mi with different English ear-perceptions? Williams records Narragansett *wetu* (agreeing with Eliot), suggesting *witeo* may be uniquely Pokanoket.
3. ***hobbamock* volcanism:** Is *hobbamock* a genuine Wampanoag lexical item or a portmanteau of Wampanoag + English “hobgoblin”? The initial *h* and the *-ock* suffix appear in both the proper name and the common noun “devil”, suggesting folk etymology contamination is possible.
4. **Phonological reliability of pidgin data:** If pidgin phonology approximates the substrate, then Winslow’s transcriptions are phonologically useful. If the pidgin introduced phonological simplifications (cluster reduction, final devoicing, vowel merger), then even the phonology is unreliable.

5. **pniese etymology:** Is this the same root as PA *-epye-wa (“to be powerful, to be a man”)? The initial cluster /pn-/ is unusual for Massachusetts and may reflect a specific morphotactic condition or epenthesis.

6.14 Related Work: Colonial Recorder Perception Problem

The core challenge of this research card — English-speaking recorders misperceiving and mis-transcribing Algonquian phonemes — is a known phenomenon documented across Eastern North America. Two researchers’ work is directly relevant:

Dr. Keith Cunningham (Linguist, Naticoke Indian Tribe). His doctoral dissertation *A Phonological Analysis of Naticoke With Practical Applications for Language Revitalization* (Georgetown University) analyzes three 18th-century Naticoke word lists (Heckewelder 1785) using the same methodology: historical phonology from colonial recorders, with the same challenges of multilingual informants and orthographic complexity. His 2024 Algonquian Conference presentation *A Revised Phonological Analysis of the Heckewelder Vocabulary of Naticoke* directly parallels the Winslow analysis — both involve tiny corpora (Winslow 24 records, Heckewelder ~200) recorded by English speakers with no linguistic training, both require PA comparative calibration, and both must account for intermediary effects (Tisquantum/Hobbamock for Winslow, multilingual informants for Heckewelder). Cunningham’s work demonstrates that the colonial recorder perception problem extends beyond SNEA into the broader Eastern Algonquian family.

Dr. Craig Kopris (Independent scholar). His paper *Les sons wyandots perçus par des oreilles étrangères* (Wyandot sounds perceived by foreign ears, *Recherches amérindiennes au Québec*, 2014) is the exact framing of the Winslow problem: how European recorders systematically misperceived and mis-transcribed indigenous phonemes. His *Wyandot Phonology: Recovering the Sound System of an Extinct Language* applies the same recovery methodology to an unrelated language family (Iroquoian), demonstrating that the foreign-ear perception problem is universal across colonial documentation of North American languages.

The Winslow corpus (24 records, the smallest in the SNEA collection) represents the most extreme case of the tiny-corpus problem. The methodological question — whether such a small dataset can yield reliable phonological data — is directly addressed by both Cunningham’s work on small Naticoke word lists and Kopris’s work on recovering phonology from limited documentary sources.

6.15 Key References

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7 Wampanoag: Wood 1634

7.1 Source Description

Recorder: William Wood, English colonist living in the Massachusetts Bay Colony (1629–1633). He returned to London to publish his observations. Wood had no linguistic training; his orthography is purely English-based ad-hoc spelling.

Source: *New Englands Prospect: A true, lively, and experimentall description of that part of America, commonly called Nevv England* (1634), printed by Tho. Cotes for John Bellamie, London. Small 4to, contains a “small Nomenclator” of ~262 Massachusetts words organized by semantic category (body parts, animals, trade goods, place names, month names, day-name classifiers).

Language: Wampanoag [wam].

Corpus: 339 records (the second-largest pre-Eliot Wampanoag source).

Orthography: Pre-Eliot English ad-hoc, no linguistic training. The Eliot Massachusetts writing system was still decades away. This makes Wood’s transcriptions reflect raw English-hearer perception of Wampanoag sounds, without the regularization that Eliot later introduced.

7.2 Recorder Biography

William Wood was an English colonist who lived in the Massachusetts Bay Colony from 1629 to 1633. Little is known of his life beyond his publication. He returned to London and published *New Englands Prospect* in 1634, which went through multiple editions:

The 1865 Prince Society edition is the most accessible for modern researchers. Wood’s Wampanoag vocabulary appears in all editions (pp. 123–128 in the 1764 edition, pp. 94–100 in the 1865 reprint), suggesting the word list was stable across editions with no significant revision.

7.3 Orthography System

Wood’s orthography follows English sound-spelling conventions of the 1630s. It is a pre-Eliot system — the Eliot Massachusetts writing system was still decades away. Key characteristics:

- **Initial /w/ retained:** *Wigwam, Web, Wompey*
- **⟨ea⟩ for /i /:** *Kean* (cf. Eliot’s *keen*)
- **⟨s⟩ for / / inconsistently:** *Sawup* where later sources might write *sh-*
- **⟨k⟩ preferred over ⟨c⟩:** Initial /k/ written *K-* even before front vowels (*Kean, Ksitta*)
- **No diacritics:** Wood uses plain English letters only
- **Vowel length not marked:** Short vs. long vowel distinction invisible
- **Pidginization noted:** Goddard (1977) observes a “good deal of pidginization” in Wood’s data (Salvucci 2002)

7.4 Vowel Tables

7.5 Consonant Tables

7.6 Key Orthographic Observations

7.6.1 b vs. p Alternation

Wood writes *abbona* (five) and *abamocho* (devil) with *b* where the Eliot/Trumbull orthography would use *p*. This shows Wood perceived the unaspirated Wampanoag /p/ closer to English /b/ — direct evidence of the voicing neutralization.

7.6.2 Geminate Consonants

Wood frequently doubles consonants: *bb*, *pp*, *nn*, *ck*, *gg*. This probably indicates the preceding vowel is short/checked rather than true gemination.

7.6.3 qu Digraph

The *qu* digraph consistently represents /k/ — seen in *sucqunnocquock* (sleeps), *asquoquin*.

7.6.4 Word-Initial a- Prefix

Word-initial *a-* is extremely common: *abbona*, *abamocho*, *aberginian*, *abbomocho*, *abonetta*. This may be a demonstrative/indefinite prefix that Wood transcribed as part of the root. Roughly 25–30% of nouns and adjectives in Wood’s 339-entry word list begin with *a-*.

7.6.5 Preaspiration /h/

Wood does **not** write preaspirated /h/ before stops. Like Williams, English-trained ears miss this feature. Example: *matta* (no) — modern Wampanoag /maht-a/ or similar; *sucqunnocquock* — no *h* written despite probable /h/ in the medial cluster.

7.7 Number System

Wood has unusually good coverage of Wampanoag numbers (10+ distinct number entries). This is valuable because numerals are relatively easy to verify cross-linguistically:

7.8 Comparison: Wood vs. Eliot Orthography

Key difference: Eliot uses a systematic vowel-length distinction (â, ô vs. a, o) which Wood entirely lacks. Eliot also notes preaspiration (h) where Wood never writes it. Eliot never uses *b* or *g*, always *p* and *k*, confirming the voicing-neutralization analysis.

7.9 Triple-Spelling Deep Dive: “Devil”

The devil word is a case study in Wood’s orthographic instability:

Proposed reconstruction: /apoma čə/ or /apoma čo/.

- *a-* prefix (demonstrative/article) → segmentable
- *bb* → /p/ (geminate or short-vowel marker)
- *o* ~ *a* in second syllable → /a/ or /ə/ (vowel reduction)
- *m* → /m/ (stable across all variants)
- *ch* → /č/ (English ⟨ch⟩ = /t /)
- final *o* → /o/ or /ə/

The fact that Wood uses the same word for “devil” that appears in other contexts (without Christian gloss) suggests he is mapping the Christian concept onto an indigenous spirit-being category. This triple spelling is also useful diagnostically: if the same word appears in Wood with multiple spellings, the stable consonants (m, ch) are reliable while the unstable vowels (o/a, o/a) indicate reduced/unstressed vowels that English ears struggled to categorize.

7.10 A-Prefix Analysis

The *a-* prefix is one of the most distinctive features of Wood’s orthography. In the 339-entry word list, roughly 25–30% of nouns and adjectives begin with *a-*. Wood treats *a-* as part of the root (no word-boundary space), suggesting either:

1. **English hearer error:** Wood heard the prefixed form as a single phonological word and wrote it as such
2. **Demonstrative prefix:** Wampanoag had a demonstrative/determiner prefix *a-* (cf. Massachusetts *a-* “that, yonder”) that was frequently attached to nouns in connected speech

3. **Indefinite prefix:** Prefixed to indefinite/unspecified referents (a man, not the man)

4. **Nominalizing prefix:** The *a-* may be a nominalizer on verb stems

Cross-source comparison: The *a-* is absent in Eliot — suggesting either Eliot was more systematic about morphological segmentation, the *a-* was a casual-speech feature that Eliot deliberately excluded from his written standard, or Wood heard and recorded a discourse feature (article-like prefix) that Eliot's educated linguistic eye removed.

7.11 Pidginization Theory

Several scholars have suggested that Wood's word list may represent not fluent Wampanoag but a pidginized or foreigner-talk register.

7.11.1 Goddard's Analysis (1977, 1981)

Ives Goddard noted that early colonial word lists often show features of reduced morphology — the characteristic of pidgin/foreigner-talk registers:

- **Missing inflectional endings:** Wood's entries notably lack the inflectional suffixes that Eliot's grammar requires
- **Reduced pronominal marking:** Missing the complex subject/object marking that Eliot documents
- **Lexical rather than grammatical:** Wood records content words (nouns, verbs in citation form) rather than inflected forms
- **Simplified phonology:** Some consonant clusters that Eliot writes are simplified in Wood's transcriptions

7.11.2 Evidence in Wood's List

7.11.3 Counter-Argument

The numeral system argues against full pidginization: Wood's number series *nawhut-nees-quin-yoâw-abbona* is a close match to the attested Wampanoag number system. Pidgins typically simplify or restructure numeral systems; Wood's closely matches the standard language.

7.12 Problems Encountered

- **b/p alternation:** Wood writes *b* where Eliot/Trumbull use *p* — direct evidence of voicing neutralization. The same Wampanoag phoneme (unaspirated /p/) is variously written as *b* or *p* with no consistent rule.
- **Triple spelling:** *abbomocho/labomocho/ abamacho* for “devil” — strongest evidence of colonial vowel instability in the corpus. The stable consonants (m, ch) are reliable while the unstable vowels (o/a, o/a) indicate reduced/unstressed vowels.
- **Word-initial a- prefix:** Many entries begin with *a-* that may be a demonstrative/indefinite prefix — must be segmented before conversion. Roughly 25–30% of nouns and adjectives are affected.
- **Geminate consonants:** Frequent doubling (*bb*, *pp*, *nn*, *ck*, *gg*) likely marks preceding vowel as short rather than true gemination.
- **Preaspiration /h/ unreported:** Like all pre-Eliot sources, Wood does not write preaspirated /h/ before stops.
- **Vowel quality instability:** Multiple spellings for the same word show Wood had trouble distinguishing Wampanoag vowels.
- **Multi-word entries:** Many entries are phrases, not single lexemes (e.g., *abonetta ta sucqunnoc-quock* = “five sleeps”, *appepes naw aug* = “when I see it I will tell you”).
- **No phonetic guide:** No *\ph* entries for validation.

7.13 Research Approach

Treat Wood as unreliable but corroborative. When Wood agrees with Trumbull/Eliot, it confirms the reading. When disagreeing, prefer Trumbull. Normalize *b* to /p/ universally. FSM with vowel averaging across Wood’s multiple spellings.

7.13.1 Conversion Pipeline Recommendations

Priority 1: Normalize Consonants (High Confidence)

- $b \rightarrow p$ (always)
- $g \rightarrow k$ (always, but check if voiced /g/ in context)

- $ck \rightarrow k$ (always)
- $qu \rightarrow k$ (keep as digraph or convert to q for phonemic)

Priority 2: Normalize Vowels (Moderate Confidence)

- All short-vowel candidates treated as underspecified /ə/ ~ /a/ unless confirmed by Eliot
- Use Eliot/Trumbull/Winslow cross-reference as tiebreaker
- Double vowels (oo, ee) are more stable and map to /u /, /i /

Priority 3: Morphological Segmentation

- Segment *a*- prefix where cross-reference supports it
- Do not segment otherwise
- Flag ambiguous cases

Priority 4: Preaspiration /h/

- Insert /h/ between vowels and following stops where Eliot confirms
- Default rule: when a short vowel is followed by a stop consonant in a stressed syllable, check Eliot for /h/
- Without Eliot confirmation, do NOT insert /h/ speculatively

Priority 5: Multi-Word Entries

- Tokenize by whitespace
- Convert each token independently
- Reconstruct the phrase from converted tokens

7.14 Conversion Workflow

[Pending implementation --- see Issue #1367]

The conversion pipeline will be implemented as a Python module in `src/commons/parsing/`. The module will:

- Treat Wood as an unreliable but corroborative source
- Normalize $b \rightarrow /p/$ universally across all Wampanoag entries
- Use FSM with vowel averaging across Wood’s multiple spellings
- Segment a - prefix on a word-by-word basis
- Handle multi-word entries by whitespace tokenization
- Insert preaspiration $/h/$ only where Eliot confirms it
- Verify each converted entry against expected Eliot/Trumbull forms

7.15 Linguist Feedback Template

[Template --- content pending review]

The following questions are open for linguist review:

1. **b/p normalization:** Is the universal $b \rightarrow /p/$ normalization correct, or are there contexts where Wood’s b represents a different phoneme?
2. **a- prefix segmentation:** Is the a - prefix a genuine Wampanoag morpheme (demonstrative/indefinite) or an English hearer error? Should it be segmented universally or on a case-by-case basis?
3. **Pidginization:** Does the numeral system’s accuracy argue against full pidginization, or can a language have accurate numerals alongside reduced morphology?
4. **Preaspiration insertion:** Is the conservative approach (insert $/h/$ only where Eliot confirms) correct, or are there systematic contexts where $/h/$ can be reliably reconstructed?
5. **“Devil” reconstruction:** Is $/apoma \check{c}ə/$ a plausible reconstruction, or does the root derive from a different PA source?

7.16 Related Work: Colonial Recorder Perception Problem

The core challenge of this research card — English-speaking recorders misperceiving and mis-transcribing Algonquian phonemes — is a known phenomenon documented across Eastern North America. Two researchers’ work is directly relevant:

Dr. Keith Cunningham (Linguist, Nanticoke Indian Tribe). His doctoral dissertation *A Phonological Analysis of Nanticoke With Practical Applications for Language Revitalization* (Georgetown University) analyzes three 18th-century Nanticoke word lists (Heckewelder 1785) using the same methodology: historical phonology from colonial recorders, with the same challenges of multilingual informants and orthographic complexity. His 2024 Algonquian Conference presentation *A Revised Phonological Analysis of the Heckewelder Vocabulary of Nanticoke* directly parallels the Wood analysis — both involve pre-missionary English recorders with no linguistic training, both show b/p alternation as evidence of voicing neutralization, and both require PA comparative calibration. Cunningham’s work demonstrates that the colonial recorder perception problem extends beyond SNEA into the broader Eastern Algonquian family.

Dr. Craig Koprís (Independent scholar). His paper *Les sons wyandots perçus par des oreilles étrangères* (Wyandot sounds perceived by foreign ears, *Recherches amérindiennes au Québec*, 2014) is the exact framing of the Wood problem: how European recorders systematically misperceived and mis-transcribed indigenous phonemes. His *Wyandot Phonology: Recovering the Sound System of an Extinct Language* applies the same recovery methodology to an unrelated language family (Iroquoian), demonstrating that the foreign-ear perception problem is universal across colonial documentation of North American languages.

The Wood corpus (339 records, the second-largest pre-Eliot Wampanoag source) is the best case study for the pre-orthographic recorder problem. Wood’s triple spelling for a single word (*abbomocholabomocholabamacho*) is the strongest evidence in the entire SNEA corpus for colonial vowel instability — a phenomenon that both Cunningham (Nanticoke) and Koprís (Wyandot) document in their respective language families.

7.17 Key References

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8 Mohegan-Pequot: Prince-Speck 1904

8.1 Source Description

8.1.1 Publication History and Citation Clarification

The glossary was published twice in slightly different venues, which causes persistent citation confusion. The first printing appeared in *American Anthropologist* n.s. 6(1): 18–45 (January–March 1904). A typographically identical reprint appeared in the *Boas Anniversary Volume* (G. E. Stechert, New York, 1906), pp. 18–45. Because the glossary appeared first in *American Anthropologist* in 1904, that is the correct year for scholarly citation. The 1906 reprint is typographically identical—same page numbering, same text—which is why later references often cite the Boas volume without noticing the earlier AA publication.

The exact title is “Glossary of the Mohegan-Pequot Language” (not “The Mohegan-Pequot Language” as sometimes mis-cited). The work is a glossary with brief ethnographic introduction, not a grammar.

8.1.2 J. Dyneley Prince (1868–1945): The Transcriber

John Dyneley Prince was a remarkable polymath: diplomat, politician, and academic linguist. Born in New York City in 1868, he earned his A.B. (1888) and Ph.D. (1892) from Columbia University and also studied in Europe. He served as Professor of Germanic Languages at Columbia (later Dean), as a New Jersey State Senator, and as U.S. Minister to Denmark (1921–1926) and Yugoslavia (1926–1932). He was reported to know approximately 30 languages including Estonian, Turkish, Arabic, Slavic languages, and Hungarian.

Prince was trained in the **Neogrammarian (Junggrammatiker)** tradition, which dominated Germanic linguistics in the late 19th century. This background has specific consequences for his transcription:

- **Phonetic impressionism over phonemic analysis:** Neogrammarians believed in recording “exact” phonetic detail rather than phonemic contrasts. Prince’s transcription therefore varies between *a*, *ah*, *au*, *o* for what is likely a single phoneme /a/ or / /, writes vowel length inconsistently, and records allophonic variation as if it were phonemic.
- **Germanic spelling conventions:** Prince unconsciously applied German orthographic habits even when intending to use English spelling (see §8.3).
- **No IPA training:** The glossary predates the International Phonetic Alphabet (1888/1900). Prince had no standardized phonetic alphabet—he used a blended English-German ad-hoc system.

Critical distinction: Prince was NOT a fieldworker. He transcribed from Speck’s field notes. He never heard the language spoken himself except possibly in brief demonstration. This means every transcription is two steps removed from the speaker: Fidelia Fielding → Speck’s ears → Speck’s field notation → Prince’s desk transcription → published form.

8.1.3 Frank G. Speck (1881–1950): The Fieldworker

Frank G. Speck was a Boasian anthropologist whose fieldwork methodology was ethnographic rather than strictly linguistic. Born in New York in 1881, he studied at Columbia University under Franz Boas (Ph.D. 1905) and later taught at the University of Pennsylvania. He was a prolific ethnographer of Eastern Algonquian peoples (Penobscot, Mohegan, Nanticoke, Tutelo, etc.).

Speck conducted fieldwork with **Fidelia Fielding (Aiji, 1827–1908)**, the last fluent speaker of Mohegan-Pequot. Fielding was born in 1827 on the Mohegan reservation in Uncasville, Connecticut, a direct de-

scendant of the Mohegan chief Uncas. She was the keeper of Mohegan traditions, maintained a diary in Mohegan using her own spelling system, was a monolingual Mohegan speaker as a child (learning English later in life), and was known for her fierce preservation of the language—she refused to speak English with other Mohegans.

Fielding was literate in English and developed her own orthography for recording Mohegan, primarily in her personal diaries. Her spelling system was based on English conventions as she understood them, with considerable idiosyncrasy.

8.1.4 The Speck–Prince Handoff

The exact chain of custody for the data is:

1. Fidelia Fielding produces utterances in Mohegan
2. Frank Speck hears and records them in field notation
3. Speck provides his field notes to J. Dyneley Prince
4. Prince converts Speck’s notation into the published orthography
5. Prince writes the introductory notes and publishes under both names

This two-stage recording (fieldworker → transcriber) introduces compounding transcription errors: Speck’s hearing of Fielding’s speech (accent, speed, dialect), Speck’s field notation choices (ad-hoc English-based spelling), Prince’s interpretation of Speck’s notation (he never heard the original), Prince’s application of German orthographic conventions, and typesetting errors in the 1904 publication.

8.1.5 Fidelia Fielding’s Orthography and Its Influence

Prince and Speck explicitly state that they sometimes deferred to Fielding’s spelling for “sentimental reasons”—i.e., they kept her English-based spelling even when a more phonetic transcription would have been technically better. From analyses of her diaries, Fielding’s orthography shows:

- English vowel letters for approximate sounds (*a, e, i, o, u* as in English orthography) with no consistent length marking
- *gh* for /k/ or /x/, following English convention, which may reflect actual velar fricative
- *w* for /w/ and as glide, consistent with English

- Doubled consonants for occasional geminate marking, which may reflect actual length or stress
- No systematic nasalization marking—nasal vowels were likely present but unwritten

The Prince-Speck preface indicates they intentionally preserved some of Fielding’s spellings because she was the last speaker and her written form had cultural value. This means some words in the glossary may be Fielding’s **written** spelling rather than Prince’s **hearing-based** transcription, and the distinction between “written by Fielding” and “recorded by Speck/Prince” is not marked in the glossary.

8.1.6 Orthography System

The glossary presents a three-way orthographic competition:

8.2 Phonemic Reconstruction

8.2.1 Consonant Inventory (Reconstructed)

Notes: *b*, *d*, *g* are likely English loan phenomena or allophones of /p, t, k/. The velar fricative /x/ may exist but is not clearly distinguished from /h/ or /k/ in the transcription. Preaspiration / p, t, k/ is probably a suprasegmental feature of the syllable coda, not a phoneme sequence.

8.2.2 Vowel Inventory (Reconstructed)

Diphthongs: /aj/ (spelled *ai*), /aw/ (spelled *au*), /oj/ (if present).

8.2.3 Preaspiration Phonology

Preaspiration in SNEA languages is a debated feature. The Prince-Speck evidence suggests that preaspiration occurred before fortis stops in coda position: *ahk*, *ahp*, *aht*. It may have been a syllable-level prosodic feature (fortis/lenis distinction) rather than a segment. Prince-Speck writes it as *hC* sequences, but these may represent single preaspirated segments / C/. The fact that it appears in Prince-Speck and not in earlier colonial sources may mean: (a) it developed later in Mohegan-Pequot, (b) earlier recorders missed it, or (c) Prince’s Germanic ear was better tuned to it.

8.2.4 Nasalization

Mohegan-Pequot had nasal vowels (inherited from Proto-Algonquian *ē, *ā, etc.). Prince-Speck does not systematically mark nasalization, may occasionally indicate it through vowel quality choices (e.g., *o*

for nasalized /ã/), and may miss it entirely, mapping nasal vowels to oral vowels. Fielding’s Mohegan has lost nasal vowels, so the modern source is no help for this feature.

8.3 German Orthographic Influence

Prince’s Germanistic training is visible throughout the transcription.

8.3.1 Affricate Representations

The use of *tsh* for the /t / affricate is a distinctly Germanic choice. English colonial recorders typically used *ch* or *tch*. The *ts* prefix represents a German-trained linguist’s desire to show the affricate’s stop onset.

8.3.2 Sibilant Representations

- *sch* (if present): German *sch* = / / (English *sh* would be more typical of American recorders)
- *s* before vowels: German *s* = /z/ in some positions (English reads this as /s/)

8.3.3 Vowel Length Marking

German linguistics of the period marked vowel length with vowel doubling (*ee*, *oo*)—shared with English, so the source is ambiguous—and *h* after vowel (*ah*, *eh*)—German uses *Dehnungs-h* (lengthening h), which is NOT the same as English “ah” for /a /. Prince’s *ah* spelling could derive from either English “ah” (indicating quality as much as length) or German *Dehnungs-h* (indicating length primarily). The ambiguity cannot be resolved without acoustic evidence.

8.3.4 Consonant Fortition

German-influenced recorders sometimes write initial /b, d, g/ as *b*, *d*, *g* but intend fortis articulation, apply final devoicing even when hearing voiced finals—writing *d* but hearing /t/, and use unaspirated *p*, *t*, *k* where English recorders would write *b*, *d*, *g*.

8.3.5 Umlaut-Like Notation

German linguists sometimes used *ü*, *ö*, *ä* when hearing sounds not in English or German. Whether Prince does this in the glossary requires a full search, but it is a known pattern in German-trained linguist transcriptions.

8.4 Preaspiration Evidence

8.4.1 Explicit /h/ in Prince-Speck

One of Prince-Speck's most valuable features for SNEA historical phonology is its treatment of /h/:

- **Word-initial h:** *ahupanun* (come here)—English recorders often drop word-initial h
- **Intervocalic h:** *bahduntah* (rising)—often missed by colonial recorders
- **Cluster hC:** *bahkeder* (bitter?)—PREASPIRATION, critical for SNEA reconstruction

8.4.2 Preaspiration (/hk/, /hp/, /ht/)

The spelling *hk* in *bahkeder* is significant because it represents a phonetic feature that English-colonial recorders systematically miss. Roger Williams (Narragansett) writes plain *k* for both /k/ and /hk/. John Eliot (Wampanoag) uses doubled vowels to indicate syllable weight but does not write preaspiration. Ezra Stiles (Mohegan-Pequot) sometimes writes *hk* but inconsistently. **Prince writes *hk*, *hp*, *ht* clusters explicitly**—this is unique among the pre-modern SNEA sources.

8.4.3 Why Prince-Speck Catches Preaspiration

Prince's Germanic training gave him an ear for aspiration contrasts. German distinguishes aspirated /p , t , k / in certain positions from unaspirated /p, t, k/ in clusters, and the *Dehnungs-h* convention made him conscious of h-like elements. Additionally, Mohegan-Pequot may have preserved preaspiration longer than other SNEA languages, or Speck's field notes may have captured it from Fidelia Fielding's careful speech.

8.4.4 Comparison with Other Sources

This makes Prince-Speck a **key witness for preaspiration** in SNEA historical phonology, on par with Mayhew and Bourne (Wampanoag) for the same feature.

8.5 English Loanword Typology

The Prince-Speck glossary contains a very high proportion of English loanwords, reflecting the contact situation of early 20th-century Mohegan.

8.5.1 Loanword Categories

Category 1: Cultural/Technological (new concepts)

- *beed* (bed), *beeddunk* (bedstead), *beesh* (peas), *beetkuz* (lady's dress), *manshetshe* (matches), *tchiken* (chicken), *breeches* (breeches), *cock* (rooster), *nute* (nut), *rug* (rug), *soyler* (soldier), *spuner* (spoon), *shtoo* (stew), *tappe* (tap/faucet), *tool* (tool), *wamp* (wampum, earlier borrowing)

Category 2: Extended/Replaced Domains

- *lif* (leaf—English “leaf” replacing native word), *appece* (apple—meaning extended to any fruit), *cheese* (cheese), *beksees* (pig—English “pigs” with Mohegan suffix?), *square* (square)

Category 3: Calendric/Temporal

- *beitar* (Friday—possibly from English “Friday” via Dutch “Vrijdag”?)

Category 4: Phonological Nativization

- *manshetshe* (matches): /t/ → *tsh*; final vowel epenthesis
- *soyler* (soldier): r → /l/ (no native /r/); final /ər/ → /er/
- *spuner* (spoon): no initial cluster simplification; final /n/ → /nər/
- *appece* (apple): /l/ → ?; final schwa → *e*

8.5.2 Identifying Loanwords

Loanwords in the glossary can be identified by phonotactic cues (presence of /r/, /l/, initial /b,d,g/ without alternation, consonant clusters not typical of Algonquian), semantic cues (European cultural artifacts, introduced concepts, calendric terms), and orthographic cues (English spelling patterns that diverge from Mohegan-Pequot patterns).

8.5.3 Handling in Conversion Pipeline

Loanwords should be kept with identity mapping (loanword → same word) rather than pushed through Mohegan-Pequot phoneme conversion. The detection FSM should flag suspected loanwords at runtime based on phonotactic patterns and semantic context.

8.6 Comparison with Other SNEA Recording Traditions

8.6.1 Williams 1643 (Narragansett)

- /h/ recording: Williams poor, Prince-Speck good—Prince better for preaspiration
- Vowel system: Williams English short/long, Prince-Speck more detailed—both need normalization
- Loanword handling: Williams marked as loans, Prince-Speck unmarked—Prince needs loanword detection
- Transcription consistency: Williams moderate, Prince-Speck low (two-stage)—Prince less consistent
- Grammar coverage: Williams extensive, Prince-Speck minimal—Prince is glossary only

8.6.2 Eliot 1660s (Wampanoag Mass Bible)

- Orthography: Eliot semi-standardized, Prince-Speck ad-hoc
- Corpus size: Eliot massive (1000pp), Prince-Speck small (446 words)
- Phonemic value: Eliot low (English-colonial), Prince-Speck potentially high (German training)
- Preaspiration: Eliot not written, Prince-Speck written

8.6.3 Stiles 1760s–90s (Mohegan-Pequot)

Ezra Stiles (1727–1795), president of Yale, collected Mohegan-Pequot vocabulary from several speakers (including Occom family members) over decades.

- Time depth: Stiles 140 years earlier, Prince-Speck later, closer to extinction
- Number of speakers: Stiles multiple (including Occom), Prince-Speck single (Fidelia Fielding)
- Transcription: Stiles ad-hoc English, Prince-Speck Germanic-influenced
- /h/ and preaspiration: Stiles inconsistent, Prince-Speck explicit
- Dialect: both Mohegan-Pequot proper
- Loanword rate: Stiles lower, Prince-Speck higher (more contact)

Stiles and Prince-Speck together provide diachronic depth—comparing the same language approximately 140 years apart.

8.6.4 Comparison with Stephanie Fielding (2013) *Modern Mohegan*

Stephanie Fielding (Yôpôwi Yoht, “Morning Fire”), a Mohegan linguist educated at MIT, published the *Modern Mohegan Dictionary* in 2013 (revised 2015). This is the only modern comprehensive dictionary of Mohegan.

The pipeline of Prince-Speck → Fielding → Proto-SNEA is powerful for reconstruction. Fielding gives the modern phonemic form (what the language sounds like now); Prince-Speck gives the older phonetic form (what the language sounded like 110 years ago); differences between them indicate sound changes in progress. However, Fielding’s dictionary is reconstructive—many entries are not from fluent speakers but from archival sources (including Prince-Speck). This introduces circularity: Fielding may have reconstructed a word FROM Prince-Speck, so using Fielding to “verify” Prince-Speck is invalid for that word.

8.7 Problems Encountered

- **German orthographic influence:** Prince’s Neogrammarian training imposed German conventions: *tsh* for /t /, *z* for /ts/, *Dehnungs-h* for vowel length, final devoicing, and possible umlaut-like notation for non-English vowels.
- **English loanword density:** Approximately 20–30% of entries are English loanwords (bed, matches, chicken, spoon, soldier). These must be detected and excluded from phoneme conversion.
- **Fielding dictionary circularity:** Many Fielding (2013) entries are reconstructed FROM Prince-Speck—using Fielding to verify Prince-Speck is circular for those entries. A cross-reference is needed to identify which Fielding entries derive from Prince-Speck (self-source) vs. from other sources (independent verification).
- **Record count correction:** 446 entries, not 465 as previously stated in some references. The actual published glossary contains 446 numbered entries (some are phrases rather than single words).
- **Two-stage recording chain:** Speck collected, Prince transcribed—Prince never heard the language spoken. This introduces compounding transcription errors at each stage.

- **“Sentimental reasons” spellings:** Prince and Speck intentionally preserved some of Fielding’s English-based spellings for cultural value, making it impossible to distinguish Fielding’s written spelling from Prince’s hearing-based transcription.
- **Inconsistent vowel length marking:** Prince writes vowel length sometimes with *h*, sometimes with doubling, sometimes not at all—the source of the length marking (English “ah” vs. German *Dehnungs-h*) is ambiguous.
- **Multi-word entries:** The glossary includes multi-word phrases, which complicates conversion because phrases span different syntactic contexts.

8.8 Research Approach

The conversion pipeline follows this sequence:

1. **Loanword detection FSM:** Automated phonotactic scan of all 446 entries to identify and catalog English loanwords. Loanwords are flagged for identity mapping (excluded from phoneme conversion).
2. **German orthography normalizer:** Identify all words where German conventions (*tsh*, *z*, *Dehnungs-h*) appear and apply the correct phoneme mapping.
3. **Fielding cross-reference with circularity flag:** Map each Prince-Speck entry against Fielding (2013) with source tracing—identify which Fielding entries derive from Prince-Speck (circular) vs. independent sources.
4. **Preaspiration capture:** Extract all *hC* clusters from the glossary and classify by position (initial, medial, final). Prince-Speck is one of the few sources that writes preaspiration (*/hC/* clusters) explicitly—this must be preserved.
5. **Stiles comparison:** Cross-reference Prince-Speck entries against Stiles’s Mohegan-Pequot vocabulary for diachronic comparison.
6. **Phoneme-to-orthography mapping table:** Full conversion table from Prince-Speck spelling to phonemes, covering all 446 entries.

8.9 Conversion Workflow

[Pending implementation --- see Issue #1368]

The conversion workflow will implement the research approach described above as a multi-stage pipeline. Each stage will be implemented as a separate module with its own test suite.

8.10 Linguist Feedback

[Template --- content pending review]

The following aspects require linguist review:

- German orthography pattern extraction: confirmation of detected German conventions
- Preaspiration audit: verification of /hC/ cluster classification
- Loanword catalog: review of flagged loanwords for accuracy
- Fielding cross-reference: validation of circularity detection
- Phoneme mapping: review of the conversion table for accuracy

8.11 Related Work: Colonial Recorder Perception Problem

The core challenge of this research card—English-speaking recorders misperceiving and mis-transcribing Algonquian phonemes—is a known phenomenon documented across Eastern North America. Two researchers’ work is directly relevant.

Dr. Keith Cunningham (Linguist, Nanticoke Indian Tribe). His doctoral dissertation *A Phonological Analysis of Nanticoke With Practical Applications for Language Revitalization* (Georgetown University) analyzes three 18th-century Nanticoke word lists (Heckewelder 1785) using the same methodology: historical phonology from colonial recorders, with the same challenges of multilingual informants and orthographic complexity. His 2024 Algonquian Conference presentation *A Revised Phonological Analysis of the Heckewelder Vocabulary of Nanticoke* directly parallels the Prince-Speck analysis—both involve a two-stage recorder chain (fieldworker → desk transcriber) that introduces compounding transcription errors, and both require identifying the transcriber’s native-language orthographic conventions (German for Prince, English for Heckewelder). Cunningham’s work demonstrates that the colonial recorder perception problem extends beyond SNEA into the broader Eastern Algonquian family.

Dr. Craig Koprís (Independent scholar). His paper *Les sons wyandots perçus par des oreilles étrangères* (Wyandot sounds perceived by foreign ears, *Recherches amérindiennes au Québec*, 2014) is the exact framing of the Prince-Speck problem: how European recorders systematically misperceived and mis-transcribed indigenous phonemes. His *Wyandot Phonology: Recovering the Sound System of an*

Extinct Language applies the same recovery methodology to an unrelated language family (Iroquoian), demonstrating that the foreign-ear perception problem is universal across colonial documentation of North American languages. His Dictionary of Wyandot project (NEH award FN-255576-17) shows the long-term application of these methods to language revitalization.

The Prince-Speck glossary (446 entries) is unique among the SNEA sources in having a documented two-stage fieldworker-transcriber chain (Speck collected, Prince transcribed) with known linguistic biases (Prince's Neogrammarian training). This makes it the most directly comparable case to the recorder-chain problems that both Cunningham and Kopriv address in their respective language families.

8.12 References

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9 Mahican: Edwards 1787

9.1 Source Description

9.1.1 Jonathan Edwards Jr. — Biographical Context

Jonathan Edwards Jr. (May 26, 1745 – August 1, 1801) was a Congregationalist minister, theologian, college president, and linguist. He was the son of the theologian Jonathan Edwards Sr. (Princeton’s 3rd president). He graduated from Princeton University in 1765 and served as president of Union College in Schenectady, New York from 1799 until his death in office.

Edwards grew up in Stockbridge, Massachusetts, a “praying town” where Mahican people were the majority population (approximately 150 Mahican families versus approximately 12 white families). He learned Mahican as a child from playmates, becoming fluent. At age 10 (1755), his father sent him to live in the Iroquois settlement of Onaquaga (Onohoquaga) to learn Oneida, giving him first-hand knowledge of both Algonquian and Iroquoian language families.

Unlike his father, who refused to learn Mahican (calling it “barbarous”), Edwards Jr. defended the language’s complexity, demonstrated that it had abstract terms (love, hatred, malice, religion), and translated the Lord’s Prayer into Mahican. He consulted a Mahican elder, “Captain Yohum,” for revisions before publication. George Washington, James Madison, and Thomas Jefferson all received copies of his treatise.

9.1.2 The 1788 Publication

Full title: *Observations on the Language of the Muhhekaneew Indians; in which the Extent of that Language in North-America is shewn; its Genius Grammatically Traced; some of its Peculiarities, and some Instances of Analogy between that and the Hebrew are Pointed Out.*

Communicated to the Connecticut Society of Arts and Sciences on October 23, 1787. Published 1788 by Josiah Meigs, New Haven. A second edition with notes by John Pickering appeared in 1823 (in Massachusetts Historical Collections).

The treatise includes:

- A grammatical sketch of Mahican (parts of speech, person marking, tense)
- A comparative glossary: English–Mahican–Shawnee
- A comparative glossary: English–Mahican–Chippeway/Ojibwa

- Numerals 1–10 in Mahican and Mohawk
- The Lord’s Prayer in Mahican and in the “language of the Six Nations”
- Word lists organized by semantic domain
- Observations on the geographic extent of Algonquian languages
- Speculative comparison with Hebrew (a common 18th-century trope)

Edwards uses the term “Mohegan” interchangeably with “Muhhekaneew” for the language, though he distinguishes it clearly from Mohegan-Pequot (Southern New England). His dialect is specifically Stockbridge Mahican.

9.1.3 Orthography System

Edwards uses an English-based ad-hoc orthography typical of 18th-century colonial sources. He provides a pronunciation key indicating his intended sound values for vowel letters.

Consonant values (relatively clear):

Vowel values (more ambiguous):

The key insight (Goddard 2008, Masthay 1991, Warne) is that Edwards uses *e* to represent /i / in many cases. This is confirmed by his pronunciation key and by cognate comparison.

9.2 Phonemic Reconstruction

9.2.1 Key Phonological Correspondences (PA → Mahican)

From Warne’s analysis of the Schmick manuscript, confirmed across sources:

9.2.2 The /i / vs. /e / Issue

Recent work (documented in *ImportantMahicanVowel2.pdf* from munseedelaware.com, with consultation by Goddard and Masthay) has established that Edwards’ *e* frequently represents /i /, NOT /e /. Goddard confirmed (c. 2023 email, cited in the Mahican vowel paper) that the second vowel of the tribal name was absolutely not /e / and is properly the “lower high” version of the vowel /i /, produced by setting the throat and tongue in a position to say the English word “hit” and then producing the sound.

Evidence for this mapping:

- Edwards writes *Muhhekaneew*—the second vowel is /i /

- *Neesoh* ‘two’—cf. PA *ni šwi, Munsee /ni šə/
- *Seepoo* ‘river’—cf. PA *si po wi, Munsee /si pə w/
- Goddard’s 2008 paper transcribes Dick’s /kīsōx/ ‘sun’ and /θīpə w/ ‘river’ with /i /—even where the English spelling suggests “e”
- Goddard’s 2010 phoneme inventory for Mahican lists NO /e / phoneme

Impact on orthography conversion: Edwards’ *e* should NOT be mapped to /e/ or / / by default. The default mapping should be /i()/ unless cognate evidence specifically indicates a different value.

9.3 The *gh* Grapheme Mystery

Edwards uses *gh* in several words where the phonological value is uncertain. Cross-linguistic comparison suggests multiple possible values:

Recommended resolution: Use comparative evidence. If a cognate from Schmick, Munsee, or Wampanoag shows /x/, the mapping is /x/. If it shows /h/, the mapping is /h/. If no cognate exists, flag as ambiguous.

9.4 Phonological Processes Affecting Edwards’ Orthography

9.4.1 Vowel Reduction and *uh*

Edwards frequently writes *u* and *uh* in unstressed syllables. The PA *e regularly became schwa /ə/ in Proto-Eastern Algonquian. In Edwards’ orthography, unstressed *u* most likely = /ə/; *uh* may represent /ə h/ or breathy schwa in a checked syllable. Munsee shows parallel reduction: PA *e → PEA *ə → Munsee /ə/ (raised to /i/ or /u/ before certain clusters).

9.4.2 Nasalization of *a

PA *a developed a nasal quality in Mahican, marked by Schmick with a tilde (ã). Warne considers this a “secondary feature” that was optional in Old Mahican but generalized to ALL instances of /a / by the Modern Mahican period. Edwards does NOT mark nasalization, so Edwards’ *a* may represent either oral /a / or nasal /ã / depending on cognate evidence.

9.4.3 Word-Final Vowel Deletion

PA word-final vowels were deleted in Mahican (a pan-Eastern Algonquian process). Edwards sometimes writes these (reflecting an older/more careful pronunciation) and sometimes omits them. For example, PA *-i (plural marker) sometimes appears as *-eh* or *-uh* in Edwards, sometimes absent.

9.4.4 Schwa Lowering Before /h, x/

Before /h/ and /x/, schwa /ə/ is lowered to /a/ in Mahican (Warne). This accounts for alternations where Edwards writes *a* where PA reconstruction predicts *e. Example: *ntah* ‘my heart’ < PA *-te h- (with *e → *a and schwa-lowering in certain environments).

9.4.5 Rounding Assimilation

PA *e rounds to Mahican /o/ when adjacent to *w. This explains forms like the word for ‘tree’ and ‘beaver’ where Edwards’ *o* corresponds to PA *e. (Warne’s rule ordering: rounding before laxing.)

9.5 Morphological Patterns in Edwards’ Data

Edwards’ data reveals several regular morphological patterns that can aid word segmentation before orthography conversion.

9.5.1 Person Prefixes

Edwards himself noted these correspondences and compared them to Hebrew prefixes, which was one of his major observations.

9.5.2 Verbal Suffixes

Edwards records frequent verbal suffixes: *-uh*, *-eh*, *-seh* (likely aspect/mood markers), *-oo* (3rd person singular independent indicative, e.g., *pumiffoo*), and *-nuh* (1st person plural, e.g., *ncmeetfchnuh* ‘we eat’).

9.5.3 Noun Inflection

Possessive prefixes precede nouns (e.g., *noghkuh* ‘my father’). Plural formation occurs via suffixation (e.g., *penumpaufoo* ‘boy’ → *penumpaufoouk* ‘boys’).

9.6 Comparative Sources for Orthography Interpretation

9.6.1 The Schmick Manuscript (Primary Internal Source)

- **Author:** Johann Jacob Schmick, Moravian missionary
- **Period:** c. 1753–1767 (before Edwards)
- **Dialect:** Moravian/Western Mahican (Ohio/PA branch)
- **Format:** Two volumes (322 pp.), German-based orthography
- **Publication:** Masthay (ed.), *Schmick's Mahican Dictionary*, APS Memoir 197 (1991)
- **Contains:** Mahican historical phonology by David Pentland

Schmick's manuscript uses a German-based orthography which is in some ways more precise than Edwards' English-based system: German *ch* for /x/ (vs. Edwards' ambiguous *gh*), more consistent vowel representation, and extensive vocabulary (much larger than Edwards' 144 records). However, Schmick shows inconsistency with vowels—especially the reflex of PA *a which he writes variously as *a*, *aa*, *ã*, *ā*, *a* (Warne). The tilde likely indicates a secondary nasalization feature.

Key finding: Schmick is much more consistent with consonants than vowels. Identical forms appear written with different vowel symbols or sequences (Warne, *Time-Depth in Mahican Diachronic Phonology*).

Pentland's historical phonology (in Masthay 1991) provides the connecting framework between PA reconstructions and Mahican forms. Goddard (2007–2008) provided important corrections to Pentland's phonology, particularly in the treatment of vowel length and the PA *e reflexes.

9.6.2 Zeisberger's Delaware Materials

David Zeisberger (1721–1808), Moravian missionary, produced extensive documentation of Munsee Delaware (a close relative of Mahican within the Delawarean subgroup): *Zeisberger's Delaware Vocabulary of 1772* (earliest extensive Munsee vocabulary), *Grammar of the Language of the Lenni Lenape or Delaware Indians* (1827), and the *Diary of David Zeisberger* (several volumes). These are important for comparative cognate work because Munsee and Mahican share a common ancestor (Delawarean, per Goddard) and often show parallel developments.

9.6.3 Later Mahican Sources

- **Hendrick Aupaumut writings** (c. 1790s–1820s): Stockbridge Mahican letters and documents by a fluent Mahican speaker
- **Wm. Dick (Michelson notes)** (c. 1910s): Modern Mahican (Wisconsin) phonetic transcriptions
- **Bernice Robinson recordings** (mid-20th century): Audio recordings of the last speakers

9.7 Problems Encountered

- **The *gh* grapheme mystery:** *ghusoo*h (eight) contains *gh*—the most unusual grapheme across all eight sources. Candidates include /x/ (voiceless velar fricative), /h/, or /k/. Must be resolved per cognate on a case-by-case basis. This is the single biggest ambiguity.
- ***uh* reduction:** Edwards’ orthography is dominated by *u* and *uh* sequences representing reduced /ə/. May represent /ə/, /ə h/, or /uh/ depending on context.
- **Word-initial consonant clusters:** Highly unusual by English standards (*wnissoo*, *hkeesque*, *hpoon*, *kmeetseh*)—indicate rich Mahican morphology. Need morphological analysis to determine whether initial CC sequences are prefix+root or root-internal.
- **Sentence-level entries:** Many entries are full sentences, not lexemes—require morphological segmentation before conversion.
- **Vowel quality ambiguity:** Edwards’ English-based vowel letters are ambiguous. The /i/ = *e* mapping (Goddard) is the most important correction. *a* could be /a/, /a /, or /ã /.
- **No phonetic transcriptions:** Edwards did not use a systematic phonetic alphabet. No phonetic entries exist.
- **Small corpus:** Approximately 144 records limits statistical approaches.
- **Dialect differences:** Stockbridge Mahican (Edwards) vs. Moravian Mahican (Schmick) may have phonological differences.
- **18th-century conventions:** Edwards’ spellings reflect his own English pronunciation as much as Mahican sounds; e.g., *oo* may be /u/ or /o/.

9.8 Research Approach

The conversion pipeline follows this sequence:

1. **Preprocessing:** Split multi-word entries into individual tokens. Flag entries with *gh* for comparative cognate resolution. Normalize Edwards' typographic conventions (long s, etc.).
2. **Phoneme mapping with cognate-based ambiguity resolution:** Apply the default mapping table with overrides per cognate. The *e* = /i / default (Goddard) is the most important rule. *gh* is resolved per cognate (Schmick /x/ or /h/).
3. **Morphological segmentation:** Identify and strip person prefixes (n-, k-, w-). Segment verbal suffixes (-uh, -eh, -seh, -oo, -nuh). Produce root forms for lexicon building.
4. **Cross-validation:** Compare converted forms against Schmick's phonemicized forms, against Munsee cognates (Zeisberger, Goddard), and against PA reconstructions (Pentland, Siebert). Flag entries where converted phonemes mismatch expected reflexes.

9.8.1 Reference Cognate Set for Validation

The following Edwards forms have well-understood cognates and should be used as the validation baseline for the conversion:

9.8.2 Software Architecture

Given the ambiguities and the need for comparative-evidence-based resolution, the conversion system should be implemented as:

- **FSM core** for the unambiguous mappings (consonants, clear vowels)
- **Cognate database** providing priors for ambiguous mappings (*gh*, *e* values, vowel length)
- **Resolver module** that queries the cognate database before applying ambiguous rules; falls back to a default with a confidence flag if no cognate found
- **Output format:** phonemic transcription with confidence markers: [high] = confirmed by cognate, [med] = default rule no contradicting cognate, [low] = guess no cognate available, [FLAG] = contradictory evidence manual review needed

9.9 Conversion Workflow

[Pending implementation --- see Issue #1369]

The conversion workflow will implement the research approach described above as a multi-stage pipeline. Each stage will be implemented as a separate module with its own test suite.

9.10 Linguist Feedback

[Template --- content pending review]

The following aspects require linguist review:

- *gh* grapheme resolution: confirmation of /x/ vs. /h/ mapping per cognate
- *e* = /i / default: verification of the Goddard finding across all entries
- Morphological segmentation: review of prefix/suffix identification
- Cognate database: validation of reference cognate set
- Confidence markers: review of [low] and [FLAG] entries

9.11 Related Work: Colonial Recorder Perception Problem

The core challenge of this research card—English-speaking recorders misperceiving and mis-transcribing Algonquian phonemes—is a known phenomenon documented across Eastern North America. Two researchers’ work is directly relevant.

Dr. Keith Cunningham (Linguist, Nanticoke Indian Tribe). His doctoral dissertation *A Phonological Analysis of Nanticoke With Practical Applications for Language Revitalization* (Georgetown University) analyzes three 18th-century Nanticoke word lists (Heckewelder 1785) using the same methodology: historical phonology from colonial recorders, with the same challenges of multilingual informants and orthographic complexity. His 2024 Algonquian Conference presentation *A Revised Phonological Analysis of the Heckewelder Vocabulary of Nanticoke* directly parallels the Edwards analysis—both involve 18th-century recorders with greater linguistic awareness than their 17th-century predecessors, both document preaspiration more reliably than earlier sources, and both require PA comparative calibration. Cunningham’s work demonstrates that the colonial recorder perception problem extends beyond SNEA into the broader Eastern Algonquian family.

Dr. Craig Kopris (Independent scholar). His paper *Les sons wyandots perçus par des oreilles étrangères* (Wyandot sounds perceived by foreign ears, *Recherches amérindiennes au Québec*, 2014) is the exact framing of the Edwards problem: how European recorders systematically misperceived and mis-transcribed indigenous phonemes. His *Wyandot Phonology: Recovering the Sound System of an Extinct Language* applies the same recovery methodology to an unrelated language family (Iroquoian), demonstrating that the foreign-ear perception problem is universal across colonial documentation of North American languages. His Dictionary of Wyandot project (NEH award FN-255576-17) shows the long-term application of these methods to language revitalization.

The Edwards Mahican corpus (144 records) is unique among the SNEA sources in being recorded by a fluent second-language speaker of the language (Edwards learned Mahican as a child from playmates). This makes Edwards the most phonetically reliable colonial recorder in the corpus—but the *gh* grapheme mystery and *uh* reduction patterns documented in this card show that even a fluent L2 speaker’s orthography is filtered through English phonological categories, the same problem that both Cunningham and Kopris address in their respective language families.

9.12 References

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Part II

Thematic Synthesis

10 Preaspiration Across Sources

Preaspiration — the occurrence of /h/ before stops (/hp/, /ht/, /hk/) — is one of the most debated features in SNEA historical phonology. The eight sources in this study show dramatically different rates of preaspiration reporting, which tells us more about the recorders than about the language. English has no phonemic preaspiration, and English-speaking recorders routinely failed to hear this feature. The result is a systematic gap in the documentary record that must be filled through comparative and morphotactic evidence.

10.1 Cross-Source Comparison of Preaspiration Reporting

Table 61 summarizes how each source treats preaspiration. The contrast between Edwards (who writes it consistently) and Williams (who almost never writes it) is the most striking.

10.1.1 Edwards (1787) — The Gold Standard

Jonathan Edwards Jr. is the only colonial source who can be considered reliable for /h/ detection. He was a fluent L2 speaker of Mahican, having learned the language as a child from playmates in Stockbridge,

Massachusetts. His transcriptions consistently write /h/ in positions where PA reconstruction predicts it: *hkeesque* 'eye', *hpoon* 'winter', *keesogh* 'sun' (with /h/ in the final fricative). Edwards' training in Hebrew and Greek, combined with his childhood fluency, gave him phonological categories that included /h/ as a salient segment.

10.1.2 Prince-Speck (1904) — The Trained Linguist

J. Dyneley Prince, trained in the Neogrammarian tradition, explicitly writes /hC/ clusters where all earlier SNEA sources miss them. The spelling *bahkeder* 'bitter' shows /hk/ written; Williams or Eliot would write only *k*. Prince's Germanic training gave him an ear for aspiration contrasts: German distinguishes aspirated /p^h, t^h, k^h/ in certain positions from unaspirated /p, t, k/ in clusters, and the *Dehnungs-h* (lengthening h) convention made him conscious of h-like elements. Additionally, Mohegan-Pequot may have preserved preaspiration longer than other SNEA languages, or Speck's field notes may have captured it from Fidelia Fielding's careful speech.

10.1.3 Eliot (1666) — Partial and Inconsistent

John Eliot writes *hk*, *hp*, *ht* clusters in approximately 15% of expected positions. His *Indian Grammar Begun* (1666) distinguishes aspirated from unaspirated stops using English voiced/voiceless pairs (b/p, d/t, k/g), but he does not apply this consistently in the Bible translation. The apostrophe in Eliot's orthography may also encode preaspiration or glottal stop — the function is ambiguous without comparative evidence.

10.1.4 Winslow (1624) — Partial but Notable

Edward Winslow writes /h/ more consistently than any other early Wampanoag source except Edwards. His *kiehtan* (the supreme God) shows the /ht/ cluster, and *hobbamock* (devil) shows word-initial /h/ where Wood's cognate *abamocho* omits it. This relatively good /h/ transcription may reflect: (1) his intermediaries (Tisquantum, Hobbamock) had clearer /h/ articulation, (2) the Pokanoket dialect had more salient /h/ than Natick, or (3) Winslow's "ignorance" of the language made him more phonetically accurate (less normalization bias).

10.1.5 Williams (1643) — The Invisible /h/

Roger Williams almost never wrote preaspirated /h/ before stops. This is the single most important finding for the Narragansett source. In Narragansett (and other SNEA languages), /h/ commonly precedes

/p/, /t/, /k/ in word-medial position. English speakers perceive these as single stops /k/, /p/, /t/ because preaspiration is not a phonemic distinction in English. Williams wrote *k* where the actual pronunciation was /hk/ (preaspirated *k*).

The database contains only 3 records with phonetic (\ph) fields out of 2,134 total. All three show the same pattern:

10.2 Cross-Linguistic Evidence for Preaspiration

The newly acquired reference dictionaries provide independent evidence for the preaspiration patterns observed in the SNEA sources:

- **Arapaho register variation:** The Arapaho Dictionary (Cowell et al. 2012) documents a formal/careful vs. everyday pronunciation distinction where preaspiration and glottal stops are more salient in careful speech (e.g., *Nih'oo3oo* → *Nihoo3oo*). This suggests that Fidelia Fielding's careful diary speech may have preserved preaspiration that Prince-Speck captured, while everyday Mohegan had already lost it — explaining the discrepancy between Prince-Speck's /hC/ notation and Fielding's modern orthography.
- **Watkins Cree dialect chain:** Watkins (1865) documents the /l/n/th/y/ dialect chain across Cree, where the same PA segment surfaces differently depending on dialect. This provides a model for understanding why Edwards (fluent L2 Mahican) writes /h/ consistently while Williams (English L1 Narragansett) does not — the difference may be as much about the recorder's phonological categories as about the language itself.
- **Brinton Lenape terminal -g:** Brinton (1888) describes terminal -g in Lenape as “a strong, suddenly-checked expiration” — essentially a glottal stop or preaspiration. This confirms that preaspiration-like phenomena were present across Eastern Algonquian and were not unique to SNEA.

These comparative data strengthen the conclusion that preaspiration was a genuine phonological feature of SNEA languages, not a recording artifact. The variation in reporting reflects recorder competence and attention, not absence of the feature in the language.

All three show Williams writing *k* where the reconstruction has /hk/. The /h/ is invisible to English ears. If this pattern generalizes — and comparative evidence from Massachusetts and Mohegan-Pequot suggests it does — then more than 50% of Williams' stops may be missing a preceding /h/.

10.3 Cowan (1973): Indirect Evidence for Preaspiration

William Cowan’s landmark study “The Development of PA *-k- and *-hk- in Narragansett” (1973) demonstrated that PA *-k- (simple velar) and *-hk- (preaspirated velar) merged phonetically in Narragansett as /hk/ medially, but can sometimes be distinguished through the behavior of preceding vowels. PA *-k- shortens a preceding long vowel; PA *-hk- does not. This provides an indirect method for reconstructing preaspiration from vowel length patterns — even though Williams never writes the /h/.

Cowan’s finding means that vowel length in Williams’ orthography is not just a vowel quality marker; it carries information about the following consonant’s preaspiration status. A long vowel before a stop suggests the stop was simple (PA *-k-); a short vowel before a stop suggests the stop was preaspirated (PA *-hk-). This interaction between vowel length and preaspiration is a critical input to the conversion workflow.

10.4 Implications for Conversion Workflows

The pattern is clear: preaspiration was a genuine phonological feature of SNEA languages, but English-speaking recorders routinely failed to hear it. The recorders who wrote it most consistently (Edwards, Prince-Speck) were either fluent L2 speakers or trained linguists with non-English phonological categories. This has direct implications for the conversion workflows:

1. **Preaspiration must be inferred, not read.** For Williams, Wood, and most of Eliot, preaspiration /h/ must be inferred from morphotactic and comparative evidence, not from the orthography.
2. **Morphotactic position is the primary cue.** Preaspiration is most common before stops in the coda of stressed syllables. A stop preceded by a short vowel in a stressed syllable is a candidate for preaspiration.
3. **Comparative evidence from Edwards is the best guide.** Edwards’ consistent /h/ marking in Mahican provides a model for predicting preaspiration in sources where it is unwritten. Where Edwards writes /h/ and Williams omits it, the preaspiration is probable.
4. **Cowan’s vowel-length diagnostic.** For Narragansett specifically, vowel length patterns can serve as an indirect cue: a short vowel before a stop suggests preaspiration (PA *-hk-), while a long vowel suggests a simple stop (PA *-k-).
5. **PA reconstructions provide the baseline.** Where PA reconstruction shows */h/ in a corresponding

position, preaspiration is expected in the daughter language unless a specific sound change removed it.

6. **Prince-Speck as a key witness.** The Prince-Speck glossary (1904) is unique among the pre-modern SNEA sources in writing /hC/ clusters explicitly. This makes it a key witness for preaspiration in SNEA historical phonology, on par with Mayhew and Bourne (Wampanoag) for the same feature.

Without these inference strategies, the conversion pipeline would silently drop a phonemic /h/ from every word where it was present but unwritten — producing a systematic underdifferentiation that would propagate through all downstream phonological analysis.

11 Colonial English Phonology as a Filter

Each recorder’s English orthography is a lens through which Algonquian phonemes were filtered. Understanding the recorder’s native phonological system is essential to recovering the target language’s sound system. The colonial recorders did not write what they heard — they wrote what their English phonological categories allowed them to perceive. This section establishes the Early Modern English baseline for each recorder and documents how their native systems systematically distorted their transcriptions.

11.1 Early Modern English Phonology Baseline (c. 1600–1620)

The Great Vowel Shift (GVS) was complete by approximately 1600–1620 for most English dialects. This means the colonial recorders’ vowel system was significantly different from both Middle English and Modern English. Table 63 shows the vowel values that the earliest recorders (Winslow, Wood, Williams) would have used.

The critical point for Algonquian transcription is that the *ea* digraph in 1620–1643 represented / / or /e /, NOT the modern /i /. When Williams writes *wean* for a Narragansett word, the *ea* represents a mid front vowel, not a high front vowel. Similarly, *oo* in 1620 represented /o / (still raising toward /u /), not the modern /u /. These differences matter for mapping colonial spellings to Algonquian phonemes.

11.2 Per-Recorder Linguistic Background

Each recorder brought a different linguistic background to the task of transcribing an unfamiliar language. Table 64 summarizes their training and its impact.

11.2.1 Winslow (1624) — The Honest Amateur

Edward Winslow was unusually honest about his linguistic limitations, describing the Massachusett language as “very copious, large, and difficult.” His transcriptions are raw phonetic impressions — he was not normalizing or regularizing what he heard, unlike Eliot (who developed a systematic orthography) or Williams (who, though linguistically gifted, worked through an English lens). Winslow’s data passed through two intermediaries: Tisquantum (Squanto), a Patuxet Wampanoag who learned English in London and may have experienced language attrition, and Hobbamock, a Pokanoket *pniese* (military champion) with limited English. This intermediary filter means every transcription is at least two steps removed from the speaker.

11.2.2 Wood (1634) — The Pre-Orthographic Recorder

William Wood’s orthography is pre-Eliot — the Massachusett writing system was still decades away. His transcriptions reflect raw English-hearer perception without regularization. The most diagnostic feature is his b/p alternation: Wood writes *abbona* (five) and *abamocho* (devil) with *b* where the Eliot/Trumbull orthography uses *p*. This shows Wood perceived the unaspirated Wampanoag /p/ as closer to English /b/ — direct evidence of the voicing neutralization that characterizes SNEA stop systems. Wood’s triple spelling for a single word (*abbomocholabomocholabamacho*) is the strongest evidence in the entire SNEA corpus for colonial vowel instability.

11.2.3 Williams (1643) — The Linguistically Gifted Amateur

Roger Williams was a theologian with no formal linguistic training, but his orthography is the most phonetically sophisticated of the 17th-century sources. His unique contribution is the systematic use of diacritics for stress: acute (á) for primary stress, grave (à) for low/falling tone, and circumflex (â) for rising-falling tone. This is a major advantage over Eliot’s Massachusett Bible (1663), which uses no accent marks, making stress reconstruction for Massachusett far more difficult. However, Williams’ English phonological filter still produced systematic gaps: he almost never wrote preaspirated /h/, and his five-vowel system (a, e, i, o, u) must represent 8–10 Narragansett vowel phonemes.

11.2.4 Eliot (1666) — The Systematic Missionary

John Eliot developed the first standardized writing system for an Algonquian language, working with Massachusett collaborators including Job Nesutan and John Sassamon. His orthography was phonemi-

cally oriented for its time — he attempted to represent the sounds of Massachusetts systematically, not just through English ear-approximations. His key innovation was distinguishing aspirated from unaspirated stops using English voiced/voiceless pairs (b/p, d/t, k/g). However, this system was still filtered through English phonological categories: Eliot’s *b*, *d*, *g* represent unaspirated lenis stops (not voiced), and his *p*, *t*, *k* represent aspirated fortis stops. The distinction is phonetically real in Massachusetts but mapped onto an English voicing contrast that does not align perfectly.

11.2.5 Edwards (1787) — The Fluent L2 Speaker

Jonathan Edwards Jr. is unique among the colonial recorders in being a fluent L2 speaker of Mahican, having learned the language as a child in Stockbridge, Massachusetts, where Mahican families were the majority population. His training in Hebrew and Greek gave him explicit phonological categories, and his childhood fluency gave him native-like perception. The result is the most reliable colonial transcription in the entire SNEA corpus. Edwards consistently writes /h/ where PA reconstruction predicts it, and his vowel transcriptions are more stable than any other colonial source. His *Observations on the Language of the Muhhekaneew Indians* (1788) includes comparative glossaries with Shawnee and Ojibwa, demonstrating a comparative method that was far ahead of its time.

11.2.6 Prince-Speck (1904) — The Two-Stage Chain

J. Dyneley Prince and Frank Speck represent a two-stage recording chain (fieldworker → desk transcriber) that introduces compounding transcription errors. Speck, a Boasian anthropologist with minimal linguistic training, collected field notes from Fidelia Fielding, the last fluent speaker of Mohegan-Pequot. Prince, a Neogrammarian linguist who never heard the language spoken, converted Speck’s notes into the published orthography. Prince’s Germanic training introduced specific biases: *tsh* for /t/ (following German *tsch*), *z* for /ts/ (German *z*), and *Dehnungs-h* (lengthening h) for vowel length. The glossary also preserves some of Fielding’s own English-based spellings for “sentimental reasons” (Prince & Speck 1904:19), creating a three-way orthographic competition between English colonial conventions, Fielding’s vernacular spelling, and Prince’s German linguistic conventions.

11.3 The Koprís (2014) Framework

Craig Koprís’ paper “Les sons wyandots perçus par des oreilles étrangères” (Wyandot sounds perceived by foreign ears, 2014) provides the exact theoretical framework for the colonial recorder problem. Koprís demonstrates that European recorders systematically misperceived and mis-transcribed indigenous

phonemes because their native phonological systems lacked the necessary categories. His key insight is that the recorder’s L1 is not a source of random error but a systematic filter: the errors are predictable given the recorder’s native phoneme inventory.

Applied to the SNEA corpus, the Koprís framework predicts:

1. **English recorders will merge Algonquian stop series.** English has a two-way laryngeal contrast (voiced vs. voiceless aspirated); Algonquian languages have a different contrast (plain vs. preaspirated, or fortis vs. lenis). The mapping is systematically distorted.
2. **Vowel quality will be underdifferentiated.** English has more vowel qualities than the five-letter Latin alphabet can represent, but colonial recorders still used only five vowel letters. The mapping from Algonquian vowels to English letters is systematically ambiguous.
3. **Stress and length will be confused.** English stress is dynamic (loudness-based); Algonquian stress may be tonal or length-based. Recorders who mark stress (Williams) may be indirectly marking length, and vice versa.
4. **Preaspiration will be invisible.** English has no phonemic preaspiration, so English-speaking recorders will systematically fail to hear it — unless they have training (Prince) or L2 fluency (Edwards) that provides the necessary category.

The Koprís framework transforms the colonial recorder problem from a liability into an asset: once we know the recorder’s L1 system, we can predict the systematic distortions and reverse them. This is the theoretical foundation for the cross-source calibration methodology described in Section 13.

12 Vowel Ambiguity Patterns

Every colonial source in this study exhibits vowel underdetermination, but the patterns differ systematically. Understanding what is systematic vs. random in the vowel ambiguity is essential for designing conversion workflows. The core problem is that each recorder used a five-letter Latin alphabet (a, e, i, o, u) to represent an Algonquian vowel system of 8–10 phonemes, including length distinctions and nasalization that the Latin alphabet was not designed to encode.

12.1 Per-Source Vowel Ambiguity

Table 65 shows how many Algonquian phonemes each grapheme maps to in each source. The number of possible mappings per grapheme ranges from 1 (for the most consistent digraphs) to 4 (for the most

ambiguous single letters).

The table reveals a clear hierarchy of ambiguity. The most reliable transcriptions are the English digraphs *ee* and *oo*, which consistently map to /i / and /u / respectively across all sources. The most ambiguous are Williams' *e* (4 possible values) and *o* (4 possible values), reflecting the fact that Narragansett had a larger vowel inventory than the five Latin letters could represent.

12.1.1 Williams' Vowel System in Detail

Williams' orthography is the most ambiguous because Narragansett had the largest vowel inventory among the SNEA languages documented. His simple vowel mappings (from O'Brien's Chart B) illustrate the problem:

- *a* → /ə/, / /, /a / — three-way ambiguity covering schwa, open-o, and long a
- *ah* → /a /, / / — long variants
- *e* → /i /, / /, /ə/, — four-way ambiguity; silent word-final
- *ê* → /i / — circumflex = long i (consistent)
- *i* → /ə/, /i /, / / — three-way; often = schwa in unstressed
- *o* → /ə/, /a /, /u /; after *w*: /a /, / / — broad range
- *oo*, *ô* → /u / — consistent long u
- *ie*, *ee* → /i / — consistent
- *ei* → /i /, /ə/ — two-way
- *ea* → /i /, / ə/, /a / — three-way

12.1.2 Eliot's Vowel System

Eliot's orthography is more systematic but still underdetermined. His five vowel letters (*a*, *e*, *i*, *o*, *u*) must represent 8–10 Massachusetts phonemes. The key disambiguating feature is that Eliot sometimes uses vowel doubling to mark length: *ee* = /i /, *oo* = /u /, *aa* = /a /. However, this doubling is not applied consistently across the Bible translation. Eliot's distinctive (oo ligature, often mis-encoded as ∞) represents long /u /, contrasting with *oo* which may represent a different vowel quality or length.

12.1.3 Wood's Vowel Instability

Wood's vowel system is the least stable. His triple spelling for 'devil' (*abbomocho/abomocho/abamacho*) shows that the same recorder, recording the same word, produced three different vowel transcriptions. The stable consonants (*m, ch*) are reliable, while the unstable vowels (*o/a*) indicate reduced or unstressed vowels that English ears struggled to categorize. This pattern is diagnostic: when a word appears with multiple spellings in Wood, the stable consonants are reliable while the unstable vowels indicate schwa or unstressed reduction.

12.1.4 Edwards' Vowel System — The /i/ Correction

Recent work (Goddard 2007–2008, confirmed by Masthay 2024) has established that Edwards' *e* frequently represents /i/, NOT /e/ or /ɪ/. This is the most important correction to the interpretation of Edwards' orthography. Evidence includes:

- *Muhhekaneew* — the second vowel is /i/, not /e/
- *Neesoh* 'two' — cf. PA *ni šwi, Munsee /ni šə/
- *Seepoo* 'river' — cf. PA *si po wi, Munsee /si pəw/
- Goddard's 2010 phoneme inventory for Mahican lists NO /e/ phoneme

The default mapping for Edwards' *e* should be /i()/ unless cognate evidence specifically indicates a different value.

12.2 Systematic vs. Random Variation

A critical distinction for the conversion workflow is whether vowel variation is systematic (reflecting real phonological or dialect differences) or random (reflecting the recorder's inability to consistently categorize the same sound).

12.2.1 Random Variation: Wood's Triple Spelling

Wood's *abbomocho/abomocho/abamacho* for 'devil' is the clearest example of random variation. The same recorder, in the same publication, produces three different transcriptions of the same word. The vowel in the second syllable alternates between *o* and *a*, suggesting a reduced vowel that Wood could not consistently categorize. This type of variation can be averaged out: the stable reading across multiple attestations is likely the correct vowel quality.

12.2.2 Systematic Variation: Williams vs. Winslow

The consistent difference between Williams' *wétu* and Winslow's *witeo* for 'house' represents systematic variation that may reflect real dialect differences. Williams records Narragansett (Rhode Island), while Winslow records Pokanoket Wampanoag (Plymouth area). The *e/i* alternation in the first syllable and the final *u/o* alternation are consistent across multiple attestations, suggesting a real phonological difference between the dialects rather than random transcription error.

12.2.3 Diagnostic Criteria

To distinguish systematic from random variation, we apply the following criteria:

1. **Consistency across attestations:** If the same source consistently writes the same vowel for the same word across multiple occurrences, the variation is likely systematic. If the vowel varies unpredictably, it is likely random.
2. **Cross-source agreement:** If two independent sources agree on a vowel value, the variation is likely systematic. If they disagree, the disagreement may reflect dialect differences (systematic) or recorder error (random).
3. **PA correspondence:** If the PA reconstruction predicts a specific vowel reflex, the variation can be evaluated against the prediction. Variation that matches the prediction is systematic; variation that does not is likely random.
4. **Phonological environment:** Variation in unstressed syllables is more likely to be random (schwa reduction). Variation in stressed syllables is more likely to be systematic.

12.3 Schwa Reduction Patterns

Schwa (/ə/) is the most variably transcribed vowel across all sources. Colonial English recorders had no consistent way to represent schwa, so it appears as *a*, *e*, *i*, *o*, or *u* depending on context and recorder. The following patterns emerge:

- **Williams:** Schwa is most often written as *i* or *u* in unstressed syllables. Word-final schwa is often written as *e* (which may be silent).
- **Eliot:** Schwa is most often written as *u* in unstressed syllables. Eliot's grammar describes *u* as the "obscure" vowel, corresponding to schwa.

- **Wood:** Schwa alternates unpredictably between *a*, *e*, *i*, *o*, and *u*. The triple spelling for 'devil' shows *o* and *a* alternating for what is likely a schwa.
- **Winslow:** Schwa is most often written as *e* in final position (e.g., *ahhe*, *witeo*).
- **Edwards:** Schwa is most often written as *u* in unstressed syllables. The sequence *uh* may represent /əh/ or breathy schwa in a checked syllable.
- **Prince-Speck:** Schwa is written as ' (apostrophe) in their phonetic key, but in the orthographic forms it appears as *a*, *e*, or *u*.

The schwa reduction pattern is so consistent across sources that it can be used as a diagnostic: when a vowel in an unstressed syllable varies across attestations, the default interpretation should be schwa.

12.4 Nasalization Ambiguity

Nasalization is one of the most poorly documented features in the colonial sources. Proto-Algonquian *a and *o developed nasalized reflexes in SNEA languages before certain consonants, but no colonial source marks nasalization consistently.

- **Williams:** Nasalization is implied but not marked. He uses nasal coda consonant digraphs (*an*, *aun*, *om*, *on*) to imply nasal vowels, but these same sequences also represent oral /an/, /aun/, /om/, /on/. Disambiguating requires checking the Massachusetts cognate (Eliot marks nasal /ô/ explicitly) or the PA reconstruction.
- **Eliot:** The only colonial source that marks nasalization explicitly, using *ô*, *û*, or an overline. However, this marking is not applied consistently across the Bible translation.
- **Wood, Winslow:** Nasalization is entirely unmarked.
- **Edwards:** Nasalization is not marked. PA *a developed a nasal quality in Mahican (marked by Schmick with a tilde), but Edwards does not indicate this.
- **Prince-Speck:** Nasalization is not systematically marked. May occasionally indicate it through vowel quality choices (e.g., *o* for nasalized /ã/).
- **Fielding:** Modern Mohegan has lost nasal vowels, so the modern source provides no help for this feature.

The nasalization ambiguity is consequential because it is phonemic in SNEA languages. In Narragansett, *n now* (yes) and *núnow* (no) are distinguished only by nasality, and Williams collapses this distinction orthographically in related forms. The conversion workflow must infer nasalization from comparative evidence rather than reading it from the orthography.

12.5 Cross-Linguistic Vowel System Comparison

The newly acquired reference dictionaries provide independent models for understanding the PA vowel system that underlies the SNEA colonial transcriptions:

- **Sauk 4-vowel length system:** The Sauk Dictionary (Whittaker 2005) documents a clean 4-vowel length system: short *a/e/i/o* and long *â/ê/î/ô*. This provides a template for the underlying PA vowel system that the SNEA colonial sources were attempting to represent. The Sauk system suggests that Williams’ extreme vowel ambiguity (2–4 possible phonemes per grapheme) is not because Narragansett had an unusually complex vowel system, but because Williams lacked the orthographic tools to mark length distinctions that were phonemic.
- **Watkins Cree vowel notation:** Watkins (1865) describes a 3-way vowel length distinction for Cree (short, long, overlong) using diacritics (*a, ā, â*). This confirms that PA vowel length was a three-way contrast in some daughter languages, and that the SNEA colonial sources’ failure to mark length is a systematic recording gap, not evidence that length was not phonemic.
- **Arapaho triple vowels:** The Arapaho Dictionary (Cowell et al. 2012) documents triple vowels (*eee, iii, ooo, uuu*) forming two syllables. While Arapaho is the most divergent Algonquian language, its vowel system demonstrates that Algonquian languages can develop complex vowel length and quality distinctions that far exceed the five-letter Latin alphabet.

These comparative data confirm that the vowel ambiguity in SNEA colonial sources is primarily a recording artifact, not a reflection of the languages’ actual phonological systems. The conversion workflows must therefore reconstruct the underlying length and quality distinctions from comparative evidence rather than reading them from the orthography.

13 Cross-Source Calibration Methodology

One of the key findings of this project is that data from one source can resolve ambiguity in another. This cross-source calibration is the most powerful tool in the conversion toolkit. Because each recorder had

different strengths and weaknesses — Edwards was reliable for /h/ but not for vowel quality, Williams was reliable for stress but not for preaspiration, Eliot was systematic but English-filtered — the sources complement each other. By calibrating across sources, we can recover phonological information that no single source provides.

13.1 The Calibration Pipeline

The calibration pipeline proceeds from most reliable to least reliable, with each source used to calibrate the next. The ordering is determined by the recorder’s linguistic training, fluency, and orthographic system:

1. **Fielding (2013)** — Modern phonemic orthography with 1:1 phoneme-to-grapheme correspondence. The most phonologically reliable source, but limited to Mohegan-Pequot and representing the modern revived language.
2. **Trumbull/Eliot (1903/1666)** — Systematic missionary orthography with the largest corpus (4,491 records). Eliot’s grammar provides explicit phonological descriptions, and the large corpus enables statistical approaches.
3. **Edwards (1787)** — Fluent L2 speaker with the most reliable /h/ detection. Small corpus (144 records) but high per-record reliability.
4. **Prince-Speck (1904)** — Trained linguist with explicit /hC/ cluster notation. Medium corpus (446 records) but two-stage recording chain introduces compounding errors.
5. **Williams (1643)** — Linguistically gifted amateur with unique stress diacritics. Large corpus (2,134 records) but high vowel ambiguity and no /h/ detection.
6. **Wood/Winslow (1634/1624)** — Raw English perception with no linguistic training. Small corpora (339 and 24 records) but valuable as the earliest attestations.

13.2 Cross-Source Disambiguation: Specific Examples

13.2.1 Massachusetts Disambiguates Narragansett Vowels

Williams’ vowel ambiguity in Narragansett is partially resolved by comparing against Eliot’s Massachusetts cognates, which have a more systematic orthography. For example, Williams’ *a* can represent /ə/, / /, or /a /. When the Massachusetts cognate shows *â* (Eliot’s long *a* marker), the Narragansett vowel is likely

/a/. When the Massachusett cognate shows *u* (Eliot’s schwa marker), the Narragansett vowel is likely /ə/.

The cognate pair for ‘dog’ illustrates this: Williams writes *ayim*, Eliot writes *arum*. The PA reconstruction *a emwa confirms the vowel quality and shows the / / → /j/ (Narragansett) vs. / / → /r/ (Massachusett) reflex difference. The cross-source comparison resolves both the vowel quality (first vowel /a/) and the consonant reflex.

13.2.2 Edwards Disambiguates Preaspiration

Edwards’ consistent /h/ marking in Mahican provides a model for predicting preaspiration in Williams and Wood, where it is unwritten. When Edwards writes *hkeesque* ‘eye’ with initial /hk/, and the PA reconstruction shows */h/ in a corresponding position, we can infer that the cognate Narragansett form (which Williams writes without /h/) also had preaspiration. This inference is strengthened when the morphotactic position matches: a stop in the coda of a stressed syllable is a candidate for preaspiration.

13.2.3 Prince-Speck Confirms Fielding

Where Prince-Speck (1904) and Fielding (2013) agree on a Mohegan-Pequot form, the confidence is high. For example, both record *wetu* for ‘house’, showing remarkable lexical stability across 110 years. Where they disagree, the circularity risk must be assessed: Fielding’s dictionary is partially reconstructive, and some entries may derive from Prince-Speck itself. Using Fielding to “verify” Prince-Speck for such entries would be circular. The conversion workflow must track which Fielding entries derive from Prince-Speck (self-source) vs. from independent sources.

13.2.4 Wood and Winslow Corroborate Each Other

Wood (339 records) and Winslow (24 records) are the two earliest Wampanoag sources, separated by only 10 years. Where they agree on a form, the reading is likely correct. For example, both record *matta* for ‘no’, and both record *sachim* for ‘chief’. Where they disagree — as with Wood’s *abamocho* vs. Winslow’s *hobbamock* for ‘devil’ — the disagreement may reflect real dialect differences (Wood records Massachusetts Bay, Winslow records Pokanoket) or different lexical roots entirely.

13.3 Sister-Language Dictionaries as Calibration Anchors

Beyond the eight primary sources, a growing catalog of online Algonquian dictionaries provides independent calibration anchors for verifying SNEA phoneme mappings. These resources are organized by

language family group in the research cards at [.issues/research-cards/](#).

13.3.1 Eastern Algonquian (Closest Genetic Relatives)

Eastern Algonquian is the closest genetic relative of the SNEA languages. Key online dictionaries include:

- **Mi'kmaq Talking Dictionary** (<https://mikmaqonline.org>) — 6,000+ words with audio from 3+ speakers each. Preserves PA cluster distinctions collapsed in SNEA colonial sources.
- **Passamaquoddy-Maliseet Dictionary** (<https://pmportal.org>) — 18,000+ words with video archive. Preserves PA vowel length distinctions that are ambiguous in Williams and Wood.
- **Western Abenaki Dictionary** (Day 1994) — 2 volumes, 538 pp. Abenaki is the closest living relative of the SNEA languages and provides the best comparative data for resolving SNEA phonological questions.
- **Lenape Talking Dictionary** (<https://www.talk-lenape.org>) — 17,000+ words with audio. Lenape is transitional between SNEA and Central Algonquian, crucial for the Mahican analysis.

13.3.2 Central Algonquian (Most Conservative Phonologically)

Central Algonquian languages are the most conservative phonologically and provide the best evidence for PA reflex verification:

- **Cree dictionaries** (<https://dictionary.eastcree.org>, <https://itwewina.altlab.app>) — Cree preserves PA *e: and *i: distinction (collapsed in SNEA colonial sources) and PA cluster distinctions (*hp *ht *hk *šk) directly relevant to the SNEA preaspiration problem.
- **Ojibwe dictionaries** (<https://ojibwe.lib.umn.edu>, <https://dictionary.nishnaabemwin.atlas-ling.ca>) — Ojibwe provides the Y-dialect reference for PA *r → /y/ (vs. SNEA N-dialect /n/).
- **Menominee** (Bloomfield 1962) — The most phonologically conservative Algonquian language. Preserves PA vowel length and *h preaspiration most faithfully of any living Algonquian language.

13.3.3 Plains Algonquian (Divergent, Boundary Conditions)

Plains Algonquian languages (Blackfoot, Cheyenne, Arapaho) are the most divergent from PA and provide the outer bounds for PA reflex verification. Cheyenne (Petter 1915, <https://cheyennelanguage.org>) and Arapaho (Cowell et al. 2012) have undergone radical phonological changes that provide a stress-test for PA reconstruction methods used in SNEA calibration.

13.4 The Cognate Set Approach

The most powerful calibration tool is the cognate set: a word with the same meaning attested in multiple SNEA languages, with PA reconstruction as the anchor. Table 66 shows a representative cognate set for basic vocabulary items.

The cognate set reveals several patterns:

1. **Lexical stability:** Basic vocabulary items like 'house', 'water', 'I', and 'two' are nearly identical across all SNEA languages and across 260+ years (Williams 1643 to Fielding 2013). This confirms the reliability of the core vocabulary for cross-source calibration.
2. **Regular sound correspondences:** The PA *r reflex distinguishes N-dialects (Massachusett *neen* from *ni ra) from Y-dialects (Mohegan *neen* with /n/ from a different PA source). The PA * reflex distinguishes Narragansett /j/ (*ayim*) from Massachusett /r/ (*arum*).
3. **Cross-source vowel confirmation:** The vowel in 'house' is consistently *e* across Williams, Eliot, and Fielding, confirming the /i/ quality (from PA *i). The vowel in 'water' is consistently *i* across all sources, confirming the / / quality (from PA *e).
4. **Edwards' Mahican as an outlier:** Edwards' *neesoh* 'two' shows the Mahican final vowel (-*oh*) that is absent in the other SNEA languages, reflecting Mahican's transitional status between SNEA and Delaware.

13.5 How PA Reconstructions Anchor Vowel Quality

Proto-Algonquian reconstructions (Bloomfield 1925, 1946; Aubin 1975; Pentland 1979) provide the comparative evidence needed to choose among ambiguous vowel mappings. The logic is:

1. For a given colonial spelling, identify the set of possible phonemic interpretations (from the grapheme-to-phoneme mapping table).

2. Find the PA reconstruction for the cognate set.
3. Apply the known sound correspondences (PA → daughter language) to predict the expected reflex.
4. Select the phonemic interpretation that matches the expected reflex.

For example, Williams’ *wétu* ‘house’ has *e* in the first syllable. The possible phonemic interpretations of Williams’ *e* are /i /, / /, /ə/, or . The PA reconstruction is *wi kiwa mi. The known correspondence is PA *i → Narragansett /i /. Therefore, the *e* in *wétu* represents /i /, not / / or /ə/. This is confirmed by the consistent *ee* spelling in Eliot’s Massachusett cognate *wetu* (where *ee* = /i /).

This logic applies to every ambiguous vowel in the corpus. The PA reconstruction is the anchor that resolves the ambiguity.

13.6 Calibration Confidence Levels

Each converted form is assigned a confidence level based on the strength of the calibration evidence:

- **[high]** — Confirmed by cognate in a more reliable source or by PA reconstruction. No contradictory evidence.
- **[med]** — Default rule applies, no contradicting cognate, but no confirming cognate either.
- **[low]** — Guess; no cognate available, default rule may not apply.
- **[FLAG]** — Contradictory evidence; manual review needed.

The calibration pipeline produces a confidence-tagged output that allows downstream users to assess the reliability of each converted form. This transparency is essential for both scholarly analysis and revitalization applications, where the confidence level determines how the form can be used.

14 Dialect Classification Evidence

The SNEA languages belong to different dialect groups within the Eastern Algonquian family. The corpus data provides evidence for these classifications, and understanding the dialect relationships is essential for cross-source calibration. A form that looks like a transcription error in one dialect may be a regular phonological reflex in another.

14.1 The Four-Way Dialect Classification

Southern New England Algonquian languages are traditionally classified by the reflex of Proto-Algonquian *r (sometimes reconstructed as *l or *θ). Four dialect types are recognized:

The PA *r reflex is the most diagnostic isogloss for SNEA dialect classification, but it is not the only one. Costa (2007) identifies a network of isoglosses that collectively define the SNEA dialect continuum.

14.2 Costa (2007) Isoglosses

David Costa's definitive study "The Dialectology of Southern New England Algonquian" (2007) identifies the following key isoglosses that distinguish the SNEA languages:

1. **PA *r reflex:** /n/ (N-dialect) vs. /j/ (Y-dialect) vs. /r/ (R-dialect) vs. /l/ (L-dialect). This is the primary classification criterion.
2. **PA *θ reflex:** In N-dialects, PA *θ merges with /n/ (e.g., Massachusett *neen* 'I' from PA *ni ra, where *r → /n/ and *θ → /n/). In Y-dialects, PA *θ → /j/ in some environments.
3. **PA *š reflex:** In Mahican, PA *š merges with /s/ (a distinguishing feature from the other SNEA languages, where /š/ is retained as a separate phoneme).
4. **Word-final vowel deletion:** Mahican shows more advanced word-final vowel deletion than the other SNEA languages, reflecting its transitional status toward Delaware.
5. **Preaspiration retention:** Mohegan-Pequot (as recorded by Prince-Speck) appears to have retained preaspiration longer than Narragansett or Massachusett, though this may be a recording artifact.
6. **Stress patterns:** Narragansett stress patterns (as recorded by Williams' diacritics) appear similar to Loup A and Massachusett, with a preference for stress on long syllables and alternating rhythm on short syllables.

14.3 Mohegan-Pequot as a Y-Dialect: The Evidence

Mohegan-Pequot is classified as a Y-dialect based on the reflex of PA *r → /j/. The evidence from the corpus includes:

- **PA *r → /j/ in cognate sets:** Where Massachusett (N-dialect) has /n/, Mohegan-Pequot has /j/. Stephanie Fielding noted this correspondence operating even in modern Connecticut English pronunciation (e.g., "Sullivan" → "Suh-yuh-van").

- **The Y-dialect isogloss bundle:** Costa (2007) shows that the Y-dialect area (Mohegan-Pequot, Montauk) forms a coherent isogloss bundle that separates it from the N-dialect area (Massachusett, Narragansett, Nipmuc).
- **Prince-Speck evidence:** The Prince-Speck glossary (1904) confirms the Y-dialect classification for early 20th-century Mohegan-Pequot, showing /j/ reflexes where N-dialects have /n/.
- **Fielding evidence:** The modern Mohegan dictionary (2013) confirms the Y-dialect classification for the revived language.

Granberry (2003:13) explicitly documents the core correspondences: Pequot *p, t, k, ch, s* → Mohegan *b, d, g, j, z*

14.4 Central Algonquian Reference Points

The dialect classification can be further verified against Central Algonquian languages, which preserve the PA *r reflex distinctions more clearly than the SNEA colonial sources:

- **Cree (Y-dialect):** Plains Cree (<https://itwewina.altlab.app>) is a Y-dialect (PA *r → /y/), providing a reference for the Mohegan-Pequot Y-dialect classification. Cree also preserves the PA vowel system more faithfully than any SNEA source.
- **Ojibwe (N-dialect):** The Ojibwe People’s Dictionary (<https://ojibwe.lib.umn.edu>) represents an N-dialect (PA *r → /n/), providing a reference for the Wampanoag and Narragansett N-dialect classification.
- **Munsee Delaware (L-dialect):** The Lenape Talking Dictionary (<https://www.talk-lenape.org>) represents an L-dialect (PA *r → /l/), providing a reference for the Mahican transitional status between SNEA and Delaware.

These Central Algonquian reference points confirm that the SNEA dialect classifications are consistent with the broader Algonquian family patterns. The Y/N/L/R classification is not unique to SNEA — it reflects a continent-wide pattern of PA *r reflexes across the Algonquian family. form-initially, and final Pequot *p, t, k, ch, s* → Mohegan form-final *b, d, g, j, z*. This lenis-fortis alternation is a distinctive feature of the Mohegan-Pequot dialect complex.

14.5 Mahican as Transitional

Mahican occupies a transitional position between the SNEA languages and the Delaware languages.

Evidence for this transitional status includes:

- **PA *š → /s/:** Mahican merges PA *š with /s/, a feature shared with Delaware but not with the other SNEA languages (which retain /š/ as a separate phoneme).
- **PA *θ → /n/:** Mahican merges PA *θ with /n/, like the N-dialects of SNEA, but unlike Delaware (which retains /θ/ or shifts it to /s/).
- **Word-final vowel deletion:** Mahican shows more advanced word-final vowel deletion than the other SNEA languages, approaching the Delaware pattern.
- **PA *e → /a/:** Mahican retracts/lowers PA *e to /a/, a development shared with some Delaware dialects but not with the other SNEA languages (where PA *e → /i/).
- **Edwards' data:** Edwards' Mahican forms show features of both groups. His *neesoh* 'two' shows the SNEA /n/ reflex of PA *r, but the final *-oh* suffix is more characteristic of Delaware.

This transitional status means that Mahican cannot be used as a direct model for the other SNEA languages without accounting for the differences. The conversion workflow must apply different sound correspondences for Mahican than for the N-dialect or Y-dialect languages.

14.6 Cross-Linguistic Cognate Evidence Table

Table 68 shows how the same PA root is reflected across the four dialect types, illustrating the regular sound correspondences that define the classification.

The table reveals several important patterns:

1. **The PA *r reflex is the primary classifier.** Massachusetts (N-dialect) and Mohegan (Y-dialect) both show /n/ in 'I' and 'thou', but the underlying PA source differs: Massachusetts /n/ comes from PA *r, while Mohegan /n/ comes from a different source (PA *n). The Y-dialect reflex of PA *r is /j/, which appears in other lexical items.
2. **Mahican patterns with N-dialects for *r.** Edwards' Mahican shows /n/ for PA *r, aligning it with the N-dialect group for this isogloss.

3. **Munsee (L-dialect) is clearly distinct.** The L-dialect reflex /l/ for PA *r separates Munsee from all the SNEA languages.
4. **The PA * reflex distinguishes Narragansett from Massachusetts.** The 'dog' cognate shows Narragansett /j/ (*ayim*) vs. Massachusetts /r/ (*arum*), both from PA *a emwa. This is a secondary isogloss within the N-dialect group.
5. **Lexical stability across dialect boundaries.** Basic vocabulary items like 'house', 'water', and 'three' are remarkably similar across all four dialect types, confirming the reliability of these items for cross-source calibration.

14.7 Implications for the Conversion Workflow

The dialect classification has direct implications for the conversion workflow:

1. **Source-specific sound correspondences.** Each source language requires its own set of PA → daughter sound correspondences. Applying Massachusetts correspondences to Mohegan-Pequot data would produce systematic errors.
2. **Cross-dialect calibration is valid only for shared isoglosses.** When using Massachusetts to disambiguate Narragansett (both N-dialects), the calibration is valid for shared features. When using Massachusetts to disambiguate Mohegan-Pequot (N-dialect vs. Y-dialect), the calibration must account for the isogloss differences.
3. **Mahican requires separate treatment.** Mahican's transitional status means it cannot be grouped with either the N-dialect or Y-dialect languages for conversion purposes. A separate set of sound correspondences is required.
4. **The cognate set approach is the most reliable calibration method.** By anchoring each conversion in a known PA reconstruction and applying the correct dialect-specific sound correspondences, the conversion workflow can produce reliable phonemic transcriptions even from highly ambiguous colonial orthographies.

14.8 Cross-Linguistic Reassessment of Dialect Classification

The newly acquired reference dictionaries provide independent evidence that both confirms and complicates the traditional four-way dialect classification:

- **Watkins Cree dialect chain model:** Watkins (1865) documents a /l/n/th/y/ dialect chain across Cree, where the same PA segment surfaces as /l/, /n/, /θ/, or /j/ depending on dialect. This is a continuum, not a discrete classification. The SNEA Y/N/R/L classification may similarly represent a continuum rather than discrete groups — the apparent binary between Mohegan (Y-dialect) and Wampanoag (N-dialect) may be an artifact of limited documentation rather than a genuine linguistic boundary. The Cree model suggests that intermediate dialects (R-dialect Loup A, L-dialect Munsee) are not separate types but points on a continuum.
- **Brinton Lenape Unami/Minsi split:** Brinton (1888) documents a dialect split within Lenape (Unami vs. Minsi) that parallels the SNEA N-dialect/Y-dialect split. The Unami/Minsi distinction involves the same PA *r reflex (Unami /l/ vs. Minsi /n/), confirming that the Y/N/R/L classification is a recurring pattern across Eastern Algonquian, not unique to SNEA.
- **Sauk vowel system as PA model:** The Sauk 4-vowel length system (Whittaker 2005) provides a template for the PA vowel system that underlies all dialect classifications. The fact that Sauk (Central Algonquian) preserves a clean 4-way length distinction suggests that the SNEA languages' original vowel systems were similarly structured, and that the ambiguity in colonial transcriptions reflects recording limitations rather than phonological merger.
- **Arapaho divergence as boundary condition:** The Arapaho Dictionary (Cowell et al. 2012) represents the extreme end of Algonquian phonological divergence. Arapaho's radical changes (PA *p → /b/, PA *k → /g/ or /k/, development of /θ/, /x/, triple vowels) provide the outer bounds for what is possible in Algonquian sound change. The SNEA languages, by contrast, are relatively conservative — their sound changes are minor compared to Arapaho's transformations.

These comparative data suggest that the traditional four-way classification is essentially correct but should be understood as a continuum rather than discrete categories. The conversion workflows should treat the dialect classification as a gradient feature, with each source occupying a position on the continuum rather than belonging to a discrete type.

Part III

External Resources and Open Questions

15 Non-SNEA Algonquian Resources

This project draws on a range of Algonquian resources beyond the SNEA languages directly represented in the corpus. These are organized by language family group below. Full details for each resource are documented in the research cards at [.issues/research-cards/](#).

15.1 Proto-Algonquian Reconstructions

- **Bloomfield (1925, 1946):** Foundational PA reconstruction.
- **Aubin (1975):** *A Proto-Algonquian Dictionary*. Mercury Series, Canadian Ethnology Service Paper No. 39. ~2,300 reconstructions. Analyzed Narragansett as substantially accurate relative to PA.
- **Hewson (1993):** *A Computer-Generated Dictionary of Proto-Algonquian*. 4,066 entries. Online at <https://protoalgonquian.atlas-ling.ca>. Typos corrected 2015–2017.
- **Pentland (1979):** *Algonquian Historical Phonology*. Comprehensive PA-to-daughter phonological developments.

15.2 Eastern Algonquian (Closest Genetic Relatives)

Eastern Algonquian is the closest genetic relative of the SNEA languages. These resources provide the best comparative data for resolving SNEA phonological questions.

- **Mi'kmaq Talking Dictionary** (<https://mikmaqonline.org>) — 6,000+ words, each recorded by 3+ speakers. Preserves PA cluster distinctions collapsed in SNEA colonial sources.
- **Passamaquoddy-Maliseet Dictionary** (Francis & Leavitt, <https://pmportal.org>) — 18,000+ words with video archive. 50+ years of collaboration among speakers, educators, and linguists.
- **Western Abenaki Dictionary** (Day 1994, 2 vols, 538 pp) — Documents the language as spoken in the last half of the 20th century. Abenaki is the closest living relative of the SNEA languages.

- **Nulhegan Abenaki Dictionary** (2025, <https://abenakionline.com>) — First publicly shareable English-Abenaki dictionary, released with National Archives grant support.
- **Lenape Talking Dictionary** (<https://www.talk-lenape.org>) — 17,000+ words with audio, stories, usage examples. Lenape is transitional between SNEA and Central Algonquian.
- **A Lenâpé-English Dictionary** (Brinton, from Moravian MS, <https://archive.org>) — 19th-century dictionary based on Zeisberger, Heckewelder, and other Moravian sources.

15.3 Central Algonquian (Most Conservative Phonologically)

Central Algonquian languages are the most conservative phonologically and provide the best evidence for PA reflex verification.

Cree-Innu Group

- **Eastern James Bay Cree Dictionary** (<https://dictionary.eastcree.org>) — Northern and Southern dialects, English-Cree-French.
- **itwêwina Plains Cree Dictionary** (<https://itwewina.altlab.app>) — Searchable Plains Cree (Y-dialect).
- **Online Cree Dictionary** (<https://www.creedictionary.com>) — Aggregates 4 dictionaries: Alberta Elders', Earle Waugh, Wolvengrey, Maskwacis.
- **A Dictionary of the Cree Language** (Watkins 1865/1938, <https://www.creeliteracy.org>) — Historical dictionary for diachronic comparison.

Ojibwe-Potawatomi Group

- **Ojibwe People's Dictionary** (<https://ojibwe.lib.umn.edu>) — 7,000+ entries with audio. Southwestern Ojibwe (N-dialect).
- **Nishnaabemwin Online Dictionary** (<https://dictionary.nishnaabemwin.atlas-ling.ca>) — 12,000+ words. Odawa dialects of Lake Huron.
- **A Dictionary of the Ojibway Language** (Baraga 1853/1992, <https://archive.org>) — Historical dictionary by Bishop Frederic Baraga.

- **Potawatomi Language Dictionary** (<https://www.potawatomidictionary.com>) — Citizen Potawatomi Nation.

Sauk-Fox-Kickapoo Group

- **A Concise Dictionary of the Sauk Language** (<https://www.sacandfoxnation-nsn.gov>) — Based on fieldwork 1990s–2000s.
- **Kickapoo Vocabulary** (Voorhis 1988) — Algonquian and Iroquoian Linguistics, Memoir 6. 205 pp.

Miami-Illinois and Other Central

- **Myaamia ILDA Dictionary** (<https://aacimotaatiiyankwi.org>) — Miami-Illinois online dictionary with mobile app. Myaamia Center, Miami University.
- **The Menomini Language** (Bloomfield 1962, <https://archive.org>) — 515 pp. The most phonologically conservative Algonquian language. Preserves PA vowel length and *h preaspiration most faithfully.
- **English-Shawnee Dictionary** (Emerson) — Practical dictionary.

15.4 Plains Algonquian (Highly Divergent)

Plains Algonquian languages are the most divergent from PA and provide the outer bounds for PA reflex verification.

- **Blackfoot Online Dictionary** (<https://dictionary.blackfoot.atlas-ling.ca>) — Part of the Algonquian Dictionaries Project.
- **English-Cheyenne Dictionary** (Petter 1915, <https://cheyennelanguage.org>) — 1,126 pp with handwritten annotations. Cheyenne shows extreme PA consonant shifts.
- **Dictionary of the Arapaho Language** (Salzmann 1983, Cowell et al. 2012, <https://www.colorado.edu/project/arapaho>) — 243 pp (4th ed.). Arapaho shows extreme PA vowel reduction.

15.5 Comparative SNEA Dialectology

- **Costa (2007):** “The Dialectology of Southern New England Algonquian.” Definitive modern dialect classification of SNEA.
- **Goddard & Bragdon (1988):** *Native Writings in Massachusetts*. Comparative phonology/orthography.
- **Bragdon (2009):** Updated linguistic analysis of Massachusetts.
- **Cunningham (2024):** Nanticoke phonological analysis — directly parallel methodology.
- **Masthay (2001):** Mahican language word list.
- **Goddard (1978–2010):** Eastern Algonquian surveys, Munsee Delaware phonology, Mahican phoneme inventory.
- **Siebert (1975):** Methods for reconstructing from early documentary sources (Powhatan).

16 Open Problems

The following problems remain unresolved across the SNEA sources. These are drawn from the “Open Questions for Linguist Review” sections of each research card.

16.1 Phonological

1. **Preaspiration inference rules:** Can a reliable set of morphotactic rules predict when preaspiration /h/ was present but unwritten? Or does this require per-entry comparative verification?
2. **The *gh* grapheme:** Is Edwards’ *gh* /x/, /h/, or /k/? Cross-linguistic cognate evidence from Munsee Delaware is needed.
3. **Nasalization disambiguation:** When does *an/aun* represent /ã/ vs. /an/? Massachusetts cognate comparison is the primary method, but it is not always available.

16.2 Methodological

1. **Pidgin vs. genuine dialect:** Are the simplifications in Winslow’s data evidence of a pre-existing Massachusetts Pidgin (Goddard 1977) or simply a learner’s imperfect recording?

2. **Dialect stratification:** How much of the variation between Natick (Eliot) and Pokanoket (Winslow/Wood) is orthographic vs. phonological vs. dialectal?
3. **Fielding circularity:** Which Fielding (2013) entries derive from Prince-Speck vs. independent sources? A source-tracing audit is needed.

16.3 Corpus-Level

1. **Training data shortage:** Only 3 of 2,134 Narragansett records have phonetic fields. Hand-annotating ~50 entries is recommended before ML approaches are viable.
2. **Tiny corpus reliability:** Winslow's 24 records — can a dataset this small yield reliable phonological data?

17 Aggregated Linguist Feedback

This section will be populated as linguist review cycles are completed for each source workflow. The template below shows the structure.

17.1 Feedback Categories

Each source workflow's feedback section records:

- **Conversion accuracy:** Per-entry verification against linguist judgment.
- **Problem identification:** Issues the linguist flags that the research phase missed.
- **Methodology critique:** Linguist assessment of the FSM/pipeline approach.
- **Revitalization applicability:** Whether the output is usable for practical language teaching.
- **Cross-cutting patterns:** Feedback that applies across multiple sources.

17.2 Cross-Cutting Patterns

[To be populated after review cycles]

18 Future Work

The combined research across all eight sources suggests several directions for future work that may feed back into the dependent spec requirements.

18.1 Immediate Priorities

1. **Hand-annotated training data:** Create a gold-standard set of ~50 Narragansett entries with verified phoneme mappings, to enable ML classifier approaches.
2. **Preaspiration inference model:** Develop morphotactic rules for predicting unwritten /h/ across all colonial sources, using Edwards and Prince-Speck as training data.
3. **Fielding circularity audit:** Trace which Fielding (2013) entries derive from Prince-Speck vs. independent sources.

18.2 Longer-Term Directions

1. **Unified conversion pipeline:** Combine all eight source workflows into a single FSM with source-detection frontend.
2. **Cross-linguistic calibration database:** Build a database of confirmed cognate sets across all SNEA languages for automated disambiguation.
3. **Revitalization-ready output:** Generate pronunciation guides and teaching materials directly from the conversion pipeline.

Table 1: Technique comparison across SNEA sources

Technique	Sources	Expected Accuracy	Rationale
Regex-based conversion	None	Low	Rejected — ambiguities are fundamentally context-dependent; regex cannot distinguish /a/ from /ə/ from /a:/ in Williams’ orthography without comparative evidence.
FSM with weighted transitions	Narragansett, Wampanoag, Mahican	Moderate	Selected — handles context-dependent ambiguity via state transitions; weights encode probability from PA cognate frequency.
1:1 phoneme mapping	Mohegan-Pequot (Fielding)	High	Selected — Fielding’s modern practical orthography is already phonemic; conversion is a direct lookup.
ML classifier	Narragansett	Unknown	Selected but blocked — only 3 records have phonetic (ph) fields; training data insufficient without hand-annotating ~50 entries.
Comparative calibration	All colonial sources	High for confirmed cognates	Selected — PA reconstructions and sister-language evidence (Massachusetts, Mohegan) disambiguate vowel quality and preaspiration.
4-stage FSM pipeline	Wampanoag (Trumbull)	Moderate–High	Selected — preprocessing → phonemic conversion → comparative fallback → statistical verification.
Loanword detection FSM	Mohegan-Pequot (Prince-Speck)	High	Selected — ~20–30% English loanwords must be identified before phonemic conversion to avoid false cognate matches.
German orthography normalizer	Mohegan-Pequot (Prince-Speck)	High	Selected — Prince’s Neogrammarian training imposed German conventions (<i>tsh</i> for / /, <i>z</i> for /ts/, <i>Dehnungs-h</i>).
Morphological segmentation	Mahican (Edwards)	Moderate	Selected — sentence-level entries require prefix/suffix parsing (<i>k</i> - 2nd person, <i>n</i> - 1st person) before conversion.

Table 2: Source status and proto-KPI targets

Source	Research	Implementation	Expected Recall	Expected Accuracy
Narragansett (Williams 1643)	Complete	Pending	85–90%	70–80%
Wampanoag (Trumbull 1903)	Complete	Pending	90–95%	80–90%
Mohegan-Pequot (Fielding 2013)	Complete	Pending	95–98%	90–95%
Wampanoag multi-source	Complete	Pending	85–90%	75–85%
Wampanoag (Winslow 1624)	Complete	Pending	70–80%	60–70%
Wampanoag (Wood 1634)	Complete	Pending	80–85%	65–75%
Mohegan-Pequot (Prince-Speck 1904)	Complete	Pending	85–90%	75–85%
Mahican (Edwards 1787)	Complete	Pending	80–85%	70–80%

Mark	Meaning
á	Primary stress (or high pitch — debated)
à	Low/falling tone (secondary stress rarely written)
â	Rising-falling tone

Table 3: Stress marking in Williams’ orthography

Williams	IPA	Narragansett Phoneme	Notes
a	/ə/, / /, /a /	PA *a *e *a	Three-way ambiguity
ah	/a /, / /	PA *a	Long variants
e	/i /, / /, /ə/,	PA *i *e *a	Four-way ambiguity; silent word-final
ê	/i /	PA *i	Circumflex = long i
i	/ə/, /i /, / /	PA *a *i *i	Often = schwa in unstressed
o	/ə/, /a /, /u /; after w: /a /, / /	PA *a *a *o	Broad range
oo, ô	/u /	PA *o	Consistent long u
u	/ə/, /a/; word-initially possibly /w/	PA *a	Ambiguous schwa-ish
ie	/i /	PA *i	
ee	/i /	PA *i	Consistent
ei	/i /, /ə/	PA *i *a	
ih	/ə/	PA *a	
oh	/ə/, /o /	PA *a	
eu	/ju /	—	Rare, from English loan phonology
ea	/i /, / ə/, /a /	PA *i *a	

Table 4: Simple vowel mappings in Williams’ orthography

Williams	IPA
an, aum, aun	/ã/ (PA *a → ą in some environments)
om, on	/õ/ (PA *o → ȝ)
ook	/o k/, /o k / (may be nasalized in context)

Table 5: Nasalized vowel patterns

Williams	IPA	Notes
au	/ /, /a /	
aw	/ /, / /	
ai	/a /, /e /	
aü	/a .u /	Rare; hiatal /a / + /u /, not a true diphthong

Table 6: Diphthong-like sequences

Williams	IPA	Notes
p, pp	/b/ /p/ [p]	Unaspirated lenis stop; alternates with <i>b</i>
t	/d/ /t/ /t /	Before /i/ may palatalize to /t /
tt	/t/ /d/ /t /	Geminate, also palatalized in palatal contexts
k, kk	/k/ /k / /x/	Labialized before /o/ /u/; guttural before /a/
c, cc	/k/ /k /	Before a/o/u = /k/; before e/i = /s/ (English soft-c)

Table 7: Stop consonants

Williams	IPA	Notes
ch	/t /	As in English “chair” — may also represent /t/ palatalized
tch	/t /	As in English “itch”
ddt, dt	/d/ /t/ /t /	Complex ch-t sound; among the most ambiguous

Table 8: Affricate consonants

Williams	IPA	Notes
m, mm	/m/	
n, nn	/n/	Before consonant can be syllabic /n/ (→ /nə/ for English speakers)
n (syllabic)	/n/ → /nə/	Unwritten schwa before n in consonant clusters

Table 9: Nasal consonants

Williams	IPA	Notes
s, ss	/s/	Single <i>s</i> after vowel; geminate <i>ss</i> also /s/
sh	/ / word-initially; /s/ before consonant	Context-dependent ambiguity
z	/s/ (not /z/)	English-style voiced s for fortis /s/
sc, shk, sk, shq	/sk/	Cluster spellings
sq	/sk / /x/	Labialized or guttural depending on following vowel

Table 10: Fricative consonants

Williams	IPA	Notes
w, ww	/w/ /j/ before e/i	Before e/i can be /j/ (yes)
q (before vowel)	/k /	As in English “queen” — consistent
q (before w)	/k/ /k / /x/	Context-dependent; may represent guttural

Table 11: Glide consonants

ID	Williams	\ph (phonetic)	Pattern
3834	Aaunchemókaw	ą čimo hka w	$k \rightarrow hk$
4286	Anakáusichick	ano hka si či k	$k \rightarrow hk$
4285	Anakáusu	ano hka səw	$k \rightarrow hk$

Table 12: Evidence from the 3 \ph entries in the database

English	Williams Narragansett	Eliot Massachusett	PA reconstruction
dog	ayim	arum	*a emwa
man	weenawash	wénawaš	
water	nipp	nippe	*nepyi
fire	apeisuwau	apeisuwâ	
house	wétu	wétu	*wi kiwa mi
I (say)	neen	neen	*ni ra

Table 13: Cognate pairs for calibration

Eliot	Phoneme	IPA	Example	Notes
a	Short a / Schwa	/a/ /ə/	<i>wadchu</i> (mountain)	Unstressed $\rightarrow [ə]$
a	Long a	/a /	<i>mahche</i> (already)	Lengthened
e	Long e	/i /	<i>wetu</i> (house)	Two different underlying vowels
e	Short e	/ /	<i>penashimwog</i>	
e	Final schwa	/ə/	<i>nish</i> (two)	Often silent but written
i	Long i	/i /	<i>pish</i> (future)	
i	Short i	/ /	<i>nippe</i> (water)	
o	Long o	/u /	<i>noh</i> (father)	
o	Short o	/ /	<i>nuppa</i> (sleep)	
o	Schwa	/ə/	<i>onk</i> (and)	Unstressed reduction
u	Schwa	/ə/	<i>ummon</i> (breast)	Unstressed position
u	Short u	/ /	<i>musqui</i> (grass)	

Table 14: Simple vowel mappings in Eliot’s orthography

Eliot Digraph	IPA	Example	Notes
ee	/i /	<i>neen</i> (I), <i>keetompauog</i> (men)	Consistent — most reliable
oo	/u /	<i>woskot</i> (firewood), <i>sook</i> (pouring)	Consistent
/ ∞	/u / or complex	See §3.4	The debated symbol
aa	/a /	<i>waam</i> (long), <i>maangu</i> (wolf)	
ii	/i /	<i>niin</i> (my)	Less common
au	/ /	<i>mauche</i> (come)	Rare
ai	/a j/	<i>wai</i> (woe)	Diphthong
eu	/e w/	<i>seup</i> (river)	Possibly /əw/

Table 15: Vowel digraphs

Eliot	IPA	Example	Notes
p	/p/ /b/	<i>peche</i> (tomorrow), <i>nuppa</i> (sleep)	Unaspirated, lenis medial
t	/t/ /d/	<i>tummunk</i> (worm), <i>wetu</i> (house)	Unaspirated, medially lenis
k	/k/ / /	<i>kenug</i> (fish), <i>askkuhnk</i> (serpent)	Unaspirated
c (a/o/u)	/k/	<i>commaco</i> (feast), <i>cutsh</i> (speak)	Hard c — English convention
c (e/i)	/s/	<i>ce</i> (thou), <i>nuppacoh</i> (chaff)	Soft c — English convention
ch	/t /	<i>choh</i> (child), <i>mache</i> (go)	Affricate
sh	/ /	<i>asuhsh</i> (fine), <i>shuckquan</i> (rain)	
s	/s/	<i>simma</i> (various)	Word-initial s often written ss
ss	/s / /s/	<i>massa</i> (great)	Geminate
m	/m/	<i>matta</i> (no)	
n	/n/	<i>neen</i> (I)	
w	/w/	<i>wetu</i> (house), <i>waantam</i> (wise)	
y	/j/	<i>yeuyeu</i> (gradually)	
q	/k /	<i>qussuk</i> (stone)	Labialized k — from PA *kw
h	/h/	<i>ohke</i> (earth), <i>kehtan</i> (great)	Also marks preaspiration
ht	/ht/ /h/	<i>kehtan</i> (great), <i>kehte</i> (truly)	Preaspirated /t/
hp	/hp/ /h/	<i>wuhpom</i> (honey)	Preaspirated /p/
hk	/hk/ /h/	<i>ohke</i> (earth)	Preaspirated /k/
' (apostrophe)	/h/ / /	<i>m'tah</i> (be tired)	Elided syllable, often = /h/

Table 16: Consonant inventory

Pattern	Count (approx.)	Example
∞ (bare)	500	<i>achm∞wonk</i> (news)
∞ (with breve)	50	∞ <i>ppa</i> (various) — phonetically reduced
∞ in compounds	100	Various morpheme-internal occurrences
Total	654	14.6% of 4,491 records

Table 17: ∞ in the database corpus

Abbreviation	Source	Approx. %
C.	Eliot's Catechism (various editions)	40%
B.	Eliot's Bible (1663, 1685)	30%
P.	Psalms translated by Eliot	10%
G.	Eliot's Grammar (1666)	5%
Pr.	Eliot's Primer(s)	5%
Pl.	Plymouth/colonial records	3%
(unmarked)	Unknown or Trumbull's own reconstruction	7%

Table 18: Source abbreviations in Trumbull

Abbreviation	Full Form	Meaning
vbl. n.	verbal noun	Action nominalization
adj.	adjective	Modifier
pro.	pronoun	Personal/demonstrative pronoun
adv.	adverb	Modifier of verb/adjective
prep.	preposition	Relational word
conj.	conjunction	Connective
interj.	interjection	Exclamation
n.	noun	Substantive
v.	verb	Verb (unmarked for transitivity)
v. a.	verb active	Transitive verb
v. n.	verb neuter	Intransitive verb
part.	particle	Uninflected word
s.	substantive	Noun (alternative to n.)
pl.	plural	Plural form
comp.	compound	Compound word
in comp.	in compound	Used only in compounds
dim.	diminutive	Diminutive form

Table 19: Grammatical label abbreviations

Source	Word	Notes
Winslow (1624)	<i>hobbamock</i>	Initial h written; also a proper name
Wood (1634)	<i>abamocho, abomocho</i>	No initial h; o-final
Williams (1643)	<i>ábammocho</i>	Narragansett; consistent with Wood
Trumbull/Eliot	<i>achm[∞]wonk</i>	Different root; <i>ch</i> not <i>b/p</i>

Table 20: The devil problem across sources

PA	Massachusetts Reflex	Eliot Grapheme	Example
*a	/a/	a	*a lapy → <i>anupp</i> (bowstring)
e	/ə/	u, a	*e θkwe wa → <i>squa</i> (woman)
i	/i/	i, e	*i li- → <i>en-</i> (thus)
o	/ /	o, a	*o θa na → <i>-adchan</i>
a	/a /	aa, a	*a mo wa → <i>amaug</i> (bee)
e	/i /	ee	*e hšwa → <i>ash</i> (three)
i	/i /	ee	*i niwa → <i>neen</i> (I)
o	/u /	oo,	*o ta na → <i>adchan</i>
p	/p/	p	*pexpenya → <i>pepeng</i>
t	/t/	t	*ta nya → <i>ta</i> (there)
k	/k/	k, c	*ka kino → <i>kakine</i> (all)
m	/m/	m	*maškyekwi → <i>maskiht</i> (medicine)
n	/n/	n	*na pe wa → <i>nap</i> (man)
s	/s/	s, ss	*so kya → <i>sohki</i> (hard)
š	/ /	sh	*e špowa ki → <i>ashpowau</i> (above)
θ	/s/	s	*e θkwe wa → <i>squa</i> (woman)
l	/n/	n	*la ke wa → <i>noh</i> (father)
w	/w/	w	*wa poswa → <i>womp</i> (white)
y	/j/	y	*aya pe wa → <i>ayim</i> (man?)
h	/h/	h, (unwritten)	*a hšo wa → <i>ashsh</i> (?)

Table 21: PA to Massachusetts regular correspondences

Gloss	Trumbull/Eliot (expected)	Confidence	PA Source
‘no, not’	<i>matta</i>	High	PA *mati-
‘woman’	<i>squa</i>	High	PA *e θkwe wa
‘man’	<i>nap</i>	High	PA *na pe wa
‘house’	<i>wetu</i>	High	PA *wi kiwa mi
‘water’	<i>nippe</i>	High	PA *nepyi
‘stone’	<i>qussuk</i>	High	PA *a senya
‘fire’	<i>nuta</i> (?)	Moderate	PA *eškw-
‘three’	<i>ash</i>	High	PA *nšwa šika
‘white’	<i>womp(ish)</i>	High	PA *wa poswa (rabbit → white)
‘devil’	<i>achm[∞]wonk</i>	Moderate	See §20
‘snake’	<i>askkuhnk</i> (?)	Moderate	PA *ašk-?

Table 22: Confirmed cognate mappings

Grapheme	IPA	Length	Notes
a	/a/	short	Low central
â	/a /	long	
e	/i / /e/	short	Front, historically /i / lowered; can be /e/
ê	/e /	long	Rare; mid front
i	/i/	short	High front
o	(not used as plain vowel)	—	Only occurs in diphthongs
ô	/o /	long	Very common; mid back rounded
u	/u/	short	High back rounded
	/ə/	short	Schwa, not written consistently

Table 23: Mohegan vowel system

Grapheme	IPA	Notes
ay	/aj/	
aw	/aw/	
uy	/uj/	

Table 24: Mohegan diphthongs

Grapheme	IPA	Voicing	Place	Manner	Notes
p	/p/	Voiceless	Bilabial	Stop	Unaspirated lenis
t	/t/	Voiceless	Dental	Stop	Unaspirated lenis
k	/k/	Voiceless	Velar	Stop	Unaspirated lenis
c	/t /	Voiceless	Palatal	Affricate	Always /t /, never /k/
s	/s/	Voiceless	Alveolar	Fricative	Fortis/lenis not marked
sh	/ /	Voiceless	Palatal	Fricative	
h	/h/	Voiceless	Glottal	Fricative	Explicitly written
m	/m/	Voiced	Bilabial	Nasal	
n	/n/	Voiced	Alveolar	Nasal	
w	/w/	Voiced	Labiovelar	Approximant	
y	/j/	Voiced	Palatal	Approximant	
kw	/k /	Voiceless	Labiovelar	Stop	Labialized
q	/k /	Voiceless	Labiovelar	Stop	Word-final /k /
'	/ /	—	Glottal	Stop	Okina

Table 25: Mohegan consonant inventory

PA	Mohegan Reflex	Example
*a	a	
a	â	
e	i, a, ə (variable)	
e	ê	
i	i	
i	e	
o	u	
o	ô	

Table 26: PA to Mohegan vowel reflexes

Dialect Type	Reflex of PA *r	Example Language
N-dialect	/n/	Massachusetts (Wôpanâak), Narragansett, Nipmuc
Y-dialect	/y/	Mohegan-Pequot , Montauk
R-dialect	/r/	Western Connecticut (Loup area)
L-dialect	/l/	Munsee Delaware, Unami

Table 27: SNEA dialect classification

PA	Mohegan	Notes
*p	p	Unaspirated
t	t	Unaspirated; palatalized to c /t / before *i, *i , *y
č	c /t /	
k	k	Unaspirated
m	m	
n	n	
θ	n (N-dialects), y (Y-dialects)	Key PA distinction
r (controversial)	y	The Y-dialect defining reflex
s	s	
š	sh / /	
h	h	Explicit in modern orthography
w	w	

Table 28: PA to Mohegan consonant reflexes

PA Cluster	Mohegan Reflex	Notes
*ht	ht	Retained in some forms, simplified in others
hk	k	
hp	p	
šk	shk / k/	
θp	p	With devoicing
mp	p	Nasal cluster simplification
nt	t	
nk	k	
nč	c /t /	

Table 29: PA cluster reflexes in Mohegan

Feature	Fielding (2006)	Prince-Speck (1904)
Affricate /t /	<i>c</i> always	<i>ch</i> or contextual
Velar /k/	<i>k</i>	<i>g</i> initially, <i>k</i> , <i>c</i> elsewhere
Voicing	Unaspirated voiceless	Inconsistent (<i>g/k</i> , <i>b/p</i> variation)
Vowel length	Circumflex (â, ê, ô)	Colon or context
Glottal stop	' (okina)	Inverted comma
Schwa	Not written	' (apostrophe)
/ /	<i>sh</i>	<i>sh</i>
Stops	p, t, k (unaspirated)	b/g/p/t/k/d (inconsistent)

Table 30: Orthographic differences between Fielding and Prince-Speck

Fielding	Granberry	IPA
p	b, p	/p/
t	d, t	/t/
k	g, k	/k/
c	j, ch	/t /
s	z, s	/s/
sh	zh, sh	/ /

Table 31: Fielding vs. Granberry orthography

Table 32: Comparative Orthography Table

English	Wood (1634)	Shepard (1647)	Eliot (1666+)	Cotton (1708)	Trumbull (1903)
man	—	—	wosketomp	wosketomp	wosketomp
woman	—	—	mittamwossis	mittamwoʻssis	mittámwossis
house	Wigwam	Wigwam	wetu	wetu	wétu
devil	—	Chepian	chepiân	chepiân	chepeân
king/sachem	—	Sachem	sôchim	sôchim	sôchim
I	Kean	—	keen	keen	kēn
very angry	—	musquantum	—	—	—
good	Winnet	—	wunne	wunne	wunne
white	Wompey	—	wompi	wompi	wômpei
sleep (classifier)	Sawup	—	—	—	—

Table 33: Vowel Representations by Source

Phoneme	Wood	Shepard	Eliot	Cotton	Trumbull
/i /	ea, ee	e, ee	e, ee	e, ee	ē
/a/	a	a	a	a	a
/a /	a, aw	a	â	â	â
/u /	oo	o, oo	o, ô,	o, ô	ô, u
/ə/	—	a, e	u	u, uʻ	u
/o/	o	o	o	o	o

Table 34: Consonant Representations by Source

Phoneme	Wood	Shepard	Eliot	Cotton	Trumbull
/p/	p	p	p~b	p~b	p
/p /	—	—	p	p	p (with dot)
/t/	t	t	t~d	t~d	t
/t /	—	—	t	t	t (with dot)
/k/	k	k	k~g	k~g	k
/k /	—	—	k	k	k (with dot)
/č/	ch	ch	ch	ch	tc
/š/	sh	sh	sh	sh	sh
/w/	w	w	w	w	w
/j/	—	—	y	y	y

Table 35: Winslow’s Wampanoag Lexicon

Word/Phrase	English	Notes
Keen Winsnow?	“Art thou Winslow?”	Massasoit’s greeting; l→n
Ahhe	“Yes”	Winslow’s reply
Matta neen wonckanet namen Winsnow	“O Winslow, I shall never see thee again!”	Deathbed lament
Winsnow	“Winslow”	/l/ → /n/; no /l/ in Wampanoag
Neen womasu sagimus	“My loving sachem”	Hobbamock’s lament
Sachimo comaco	“The sachem’s house”	Sachem + house at Pokanoket
Witeo	“An ordinary house”	Contrasts with comaco
Squasachim	“The sachem’s wife”	Squa (woman) + sachim (sachem)
Maskiet	“Physic/Medicine”	Healing context
Kiehtan	(The supreme God)	Explains Wampanoag theism
Chise	“An old man”	Winslow’s etymology for Kiehtan
Kiehchise	“A very old man”	Comparative: chise → kiehchise
Hinnaim namen, hinnaim michen, matta cuts	“By and by see, eat, but not speak”	Complex sentence
Quatchet	“Walk abroad”	Imperative?
Sachim	“King/Chief”	General title
Pniese (also pinse, pinese)	“Chief champion / man of valor”	Military elite class
Powah	“Shaman / healer”	Calls upon the devil
Squaw	“Woman”	Occurring in compounds
Massasoit	“Great Chief”	Proper name / title
Canonickus	Massasoit’s older brother	Proper name with -us?
Wamsutta	Massasoit’s son	Proper name
Namasquemit	“Powah” or dreaming healer	One reference only

Table 36: Winslow Vowel Mapping

Grapheme	IPA	Example	Confidence
a (short)	/a/ ~ /ə/	matta	High
a (long)	/a /	ahhe (first a)	Moderate
e (short)	/ /	chise (final e)	Moderate
e (final)	/ə/	ahhe, witeo, maskiet	High
i	/ / ~ /i/	kiehchise, chise	Moderate
o	/ / ~ /ə/	hobbamock, commaco	Moderate
u	/ / ~ /ə/	cuts	Low
ie	/i /	kiehtan, kiehchise	High
oo	/u /	askooke	High
ah	/ah/	ahhe	Moderate
ee	/i /	keen (thou)	High

Table 37: Winslow Consonant Mapping

Grapheme	IPA	Example	Confidence
h	/h/	ahhe, hobbamock, kiehtan	High
k	/k/	kiehtan, askooke, keen	High
c (before a/o)	/k/	commaco, cuts, comaco	High
ch	/t /	chise, kiehchise	High
sh	/ /	Only in proper names	Low
m	/m/	commaco, matta, namen	High
n	/n/	neen, namen, Winsnow	High
s	/s/	sachim, squaw, sagimus	High
t	/t/	matta, kiehtan, quatchet	High
w	/w/	winsnow, witeo	High
p	/p/	pniese (initial pn-!)	Moderate
b	/b/	hobbamock (medial)	Moderate

Table 38: Cross-Source Comparison for /h/

Source	hobbamock initial h	kiehtan medial ht	Overall /h/ Reliability
Winslow (1624)	YES	YES	Highest among early sources
Wood (1634)	NO (abamocho)	N/A	Unreliable for /h/
Williams (1643)	Mixed (ábammocho)	N/A	Unreliable
Eliot (1663)	Different root	Likely kehtan	Partial
Edwards (1787)	N/A (Mahican)	YES	Most reliable overall

Table 39: Confirmed Cognates Across Sources

Gloss	Winslow (1624)	Wood (1634)	Trumbull/Eliot	Williams (1643)
‘no’	matta	matta	matta	matta
‘woman’	squa(achim)	squaw	squa	squa
‘devil’	hobbamock	abamocho	achm∞wonk	ábammocho
‘yes’	ahhe	—	—	—
‘house’	witeo	wetu/weetu	wetu	—
‘snake’	askooke	askook	askkuhnk	askug
‘king’	sachim	sachim	sachim	—
‘old man’	chise	—	—	—
‘speak’	cuts	—	cuts	ketes
‘walk’	quatchet	—	—	—
‘medicine’	maskiet	—	maskiht	—
‘shaman’	powah	powaw	—	powwáw

Table 40: Pokanoket vs. Natick Features

Feature	Pokanoket (Winslow/Wood)	Natick (Eliot/Trumbull)	Notes
‘house’	witeo (Winslow)	wetu (Eliot)	Real lexical difference?
‘devil’	hobbamock (Winslow)	achm∞wonk (Eliot)	Different roots entirely
‘man of valor’	pniese (Winslow)	Not attested	Nativized English?
Initial /h/	Written (Winslow)	Often omitted (Eliot)	Possibly orthographic
/l/ substitution	n only (Winslow)	n only (Eliot)	Consistent across dialects

Table 41: Key Cognate Mappings

Winslow	Wood	Trumbull/Eliot (expected)	PA Source
hobbamock (devil)	abamocho	Different root	See §11 — two different roots?
matta (no)	matta	matta	PA *mati-
kiehtan (God)	—	kehtan	PA *ki·ht-?
sachim (chief)	sachim	sachim	PA *sa·kima·wa
squa (woman)	squaw	squa	PA *e·θkwe·wa
askooke (snake)	askook	askkuhnk	PA *aθkw-?

Table 42: Publication History of *New Englands Prospect*

Edition	Year	Printer	Place
1st	1634	Tho. Cotes for John Bellamie	London
2nd	1635	—	London
3rd	1639	John Dawson for John Bellamie	London
4th (Boston)	1764	Thomas and John Fleet	Boston
Prince Society	1865	—	Boston

Table 43: Wood Vowel Rules

Wood Grapheme	Likely Phoneme	Example	Notes
a	/a/ ~ /ə/	abbona (five), matta (no), abamocho	Short a dominant
e	/ / ~ /i/	aberginian (an Indian), apponees (twelve)	
i	/ / ~ /i/	aberginian	
o	/ / ~ /o/ ~ /ə/	abbona, abamocho, abbomocho	Broad range
u	/ə/ ~ / /	sucqunnocquock (sleeps), nugge (sieve)	
a followed by a	/a / (implied)	abbona, abamacho	Double letters often for emphasis
oo	/u / ~ / /	askooke (in Winslow), week (in Wood)	
au	/ / ~ /aw/	pausawniscosu (half fathom), aug (I)	
ee	/i /	apponees (twelve), nees (two)	
ai	/e / ~ /aj/	pausawniscosu, appause (morning)	

Table 44: Wood Consonant Rules

Wood Grapheme	Likely Phoneme	Example	Notes
b	/p/ ~ /b/	abbona (five), abamocho (devil)	Wood uses b where later sources use p
p	/p/ ~ /b/	apponees, appause	
m	/m/	matta (no), abamocho	
n	/n/	abbona, an nu ocke (a bed)	
t	/t/ ~ /d/	matta, abbona quit	
s	/s/	apponees, pausawniscosu	
ch	/t /		
sh	/ /		
k	/k/ ~ /g/	nugge (sieve), asquoquin	
g	/g/ ~ /k/	nugge	
qu	/k /	sucqunnocquock, asquoquin	Labialized k
w	/w/	naw aug (I will tell), yoaw	
ck	/k/	asquoquin, an nu ocke	

Table 45: Wood's Number System

Wood	Meaning	Expected Wampanoag
abbona	five	(p)àpàna?
nawhut	maybe one	nequt
nees	two	nees
quín	three	qush, shwa
yoâw	four	yaw, yau
apponees	twelve	—
apponabonna	fifteen	—
apponaquina	sixteen	—
apponasquoquin	nineteen	—
apponenotta	seventeen	—

Table 46: Wood vs. Eliot Grapheme Mapping

Wood	Eliot	Phoneme	Reliability
a	a, â	/a/, /ã/	Low (Wood conflates length/nasal)
e	e	/ /	Moderate
i	i, u	/ /, /ə/	Low (often confused)
o	o, ô	/ /, /ã/	Low
u	u	/ə/	Low
oo	oo	/u /	High
ee	ee	/i /	High
au	au, âu	/aw/, / /	Moderate
b	p (never b)	/p/	Context-dependent (normalize to p)
p	p	/p/	High
t	t	/t/	High
k/c	k	/k/	High
ch	ch	/č/	High
sh	sh	/š/	High
qu	q	/k /	High
m	m	/m/	High
n	n	/n/	High
w	w	/w/	High
y	y	/y/	High
s	s	/s/	High
h	h	/h/ (including preaspiration)	Missing in Wood
g	k (never g)	/k/	Normalize to k

Table 47: Wood's Spellings for "Devil"

Spelling	Source
abbomocho	Primary Wood spelling
abomocho	Variant (single b)
abamacho	Variant (a for o in second position)
abamocho	Variant

Table 48: Pidginization Evidence in Wood

Feature	Example	Pidgin Character
Citation-form verbs	sucqunnocquock (sleeps) — appears as-is, not inflected	Foreigner-talk
Missing possessive prefixes	an nu ocke (a bed) — literal “a bed” not “my bed”	Reduced morph
Paratactic phrases	abonetta ta sucqunnocquock (six sleeps) — juxtaposed numerals+noun	No numeral ag
Unusual word order	Some entries show English word order with Wampanoag words	Code interfere

Source	Convention	Example	Interpretation
English colonial	<i>ch</i> for /t /, <i>sh</i> for / /	“cheese”	Familiar, predictable
English vernacular (Fielding)	Ad-hoc English-based	Variable spelling	Unpredictable
German linguistic (Prince)	<i>tsh</i> for /t /, <i>z</i> for /ts/, <i>Dehnungs-h</i>	Specific forms	Detectable pattern

Table 49: Orthographic confusion index

	Bilabial	Alveolar	Palatal	Velar	Glottal
Stops	p	t		k	
Voiced stops	b?	d?		g?	
Affricates			t		
Fricatives		s		(x?)	h
Nasals	m	n			
Approximants	w		j		

Table 50: Reconstructed consonant inventory for Mohegan-Pequot

	Front	Central	Back
High	i /i /		u /u /
Mid	e / /	ə /ə/	o /o/
Low		a /a/	/ /

Table 51: Reconstructed vowel inventory for Mohegan-Pequot

Prince-Speck	German convention	English equivalent	Phoneme
<i>tsh</i>	German <i>tsch</i> (as in <i>deutsch</i>)	Rare, used in “batsha”	/t /
<i>z</i>	German <i>z</i> = /ts/	English <i>z</i> = /z/	ambiguous

Table 52: Affricate representations

Source	Preaspiration /hk/	Intervocalic /h/	Word-initial /h/
Williams 1643	Not written	Sometimes	Rarely
Eliot 1660s	Not written	Rarely	Rarely
Stiles 1760s–90s	Inconsistent	Inconsistent	Inconsistent
Prince-Speck 1904	Written	Written	Written
Fielding 2013	Not phonemic	Not present	Not present

Table 53: Preaspiration and /h/ recording across SNEA sources

Feature	Fielding (2013)	Prince-Speck (1904)
Base	Phonemic (designed)	Phonetic (impressionistic)
Vowel length	Explicit	Inconsistent
/h/	Not phonemic	Orthographic
Preaspiration	Not marked	Marked
Nasalization	Not marked	Not marked
Loanwords	Identified	Unidentified
Consonant system	Simplified (modern)	More complex (archaic)

Table 54: Orthography comparison: Fielding vs. Prince-Speck

Grapheme	Likely value	Evidence
<i>p, t, k</i>	/p, t, k/ (unaspirated or lightly aspirated)	PA *p, *t, *k > Mahican /p, t, k/
<i>b, d, g</i>	Probably /p, t, k/ after nasals or in voiced contexts	Unclear; English-orthography convention
<i>m, n</i>	/m, n/	Straightforward
<i>s</i>	/s/	PA *s, *š merged to /s/ in Mahican by modern period
<i>sh</i>	/š/	English convention
<i>h</i>	/h/	Explicitly written; Edwards is unusually consistent
<i>w, y</i>	/w, j/	Glides
<i>ch</i>	/č/	English convention
<i>gh</i>	UNKNOWN	See §9.3

Table 55: Edwards' consonant graphemes and likely values

Edwards grapheme	Likely Mahican phoneme	PA source	Example
<i>ee</i>	/i /	PA *i	<i>neesoh</i> 'two' (PA *i)
<i>e</i>	/i / or /i/	PA *i()	<i>seepoo</i> 'river' (PA *i)
<i>a</i>	/a/ or /a /	PA *a, *e → PEA *a	<i>nannah</i> 'five' (PA *a)
<i>o, oo</i>	/o/ or /u/	PA *o()	
<i>u</i>	/ə/ (schwa) in unstressed; possibly /o/ elsewhere	PA *e → PEA *ə	<i>muhhekaneew</i> (with)
<i>au, aw</i>	/a/ + /w/ or /a /		
<i>uh</i>	/ə h/ or /ə/		Frequent in final syllables

Table 56: Edwards' vowel graphemes and likely values

PA	Old Mahican	Modern Mahican	Notes
*p	/p/	/p/	No change
t	/t/	/t/	No change
k	/k/	/k/	No change
č	/č/	/č/	No change
θ	/n/	/n/	Merged with *n by Old Mahican period
l	/n/	/n/	Merged with *n by Old Mahican period
n	/n/	/n/	
s	/s/	/s/	
š	/s/	/s/	Merged to /s/ by Modern Mahican
m	/m/	/m/	
h	/h/	/h/	
i	/i /	/i /	
i	/i/	/i/	
e	/a/ (lax low)	/a/	Retracted/lowered from *e to /a/ in PEA
e	/ə/ (schwa)	/ə/	Reduced to schwa (pan-EA development)
a	/a ã /	/ã /	Nasalized in modern Mahican
a	/a/	/a/	
o	/u /	/u /	
o	/o/	/o/	Marginal phoneme even in PA

Table 57: PA → Mahican sound correspondences

Candidate	Evidence For
/x/ (voiceless velar fricative)	Schmick uses German <i>ch</i> for /x/ in cognate forms; PA *h before consonant sometimes a
/h/	Simple /h/ is the default reflex of PA *h
/k/ (lenis/voiced allophone)	English <i>gh</i> occasionally used for /k/ in colonial sources

Table 58: Candidate values for the *gh* grapheme

Person	Prefix	Edwards example	Gloss
1st sg.	n-	<i>ncmeetseh, ncmeetfeh</i>	‘I eat’
2nd sg.	k-	<i>kcmeetseh, ka meetjeh</i>	‘you eat’
3rd sg.	w-/ -	<i>pumiffoo</i>	‘he walks’

Table 59: Person prefixes in Edwards’ data

Edwards form	Gloss	PA reconstruction	Expected Mahican
<i>neesoh</i>	‘two’	*ni šwi	/ni sə/
<i>seepoo</i>	‘river’	*si po wi	/si pə w/
<i>keesogh</i>	‘sun’	*ki šo θ a	/ki sox/?
<i>hpoon</i>	‘winter’	*pepoon-	/pə pu n/?
<i>hkeesque</i>	‘eye’		/ki skw/?
<i>nannah</i>	‘five’	*nya lanwi	/na nə w/?
<i>penumpaufoo</i>	‘boy’		
<i>pumiffoo</i>	‘he walks’	*pemihθ e wa	/pə mihse wa/?
<i>ncmeetseh</i>	‘I eat’	*mi či-	/nə mi čə/?
<i>ghusooh</i>	‘eight’	*nešwa šika	
<i>noghkah</i>	‘my father’		/nə xk-/
<i>muhhekaneew</i>	‘Mahican’		/mə hi kani w/

Table 60: Reference cognate set for validation

Table 61: Preaspiration reporting across SNEA sources

Source	Year	/h/ written?	Reliability
Edwards (Mahican)	1787	YES — consistent	Most reliable
Prince-Speck (Mohegan-Pequot)	1904	YES — explicit /hC/ clusters	High
Eliot (Wampanoag)	1666	Partial — 15% of expected positions	Moderate
Winslow (Wampanoag)	1624	Sometimes — <i>kiehtan</i> , <i>hobbamock</i>	Partial
Williams (Narragansett)	1643	Rarely — 0 of 2,134 records	Unreliable
Wood (Wampanoag)	1634	Never	Unreliable
Trumbull (Wampanoag)	1903	Inherits Eliot’s inconsistency	Moderate
Fielding (Mohegan)	2013	Not phonemic (modern loss)	N/A

Table 62: The three \ph entries in the Williams corpus

ID	Williams spelling	\ph (phonetic)	Pattern
3834	Aaunchemókaw	ą čimo hka w	<i>k</i> → /hk/
4286	Anakáusichick	ano hka si či k	<i>k</i> → /hk/
4285	Anakáusu	ano hka səw	<i>k</i> → /hk/

Table 63: Early Modern English vowel values (c. 1600–1620)

Spelling	ME value	c. 1600 value	Modern value
<i>a</i> (short)	/a/	/a/	/æ/
<i>e</i> (short)	//	//	//
<i>i</i> (short)	//	//	//
<i>o</i> (short)	//	//	///
<i>u</i> (short)	//	//	//
<i>a</i> (long)	/a/	/ / /e/	/e/
<i>ee</i> (long)	/e/	/i/	/i/
<i>ie</i> (long)	/ /	/i/	/i/
<i>oo</i> (long)	/o/	/u/	/u/
<i>ou</i> (long)	/u/	/a/	/a/
<i>ea</i> (digraph)	/ /	/ / /e/	/i/
<i>ai/ay</i>	/a/	/ / /e/	/e/

Table 64: Recorder linguistic background and transcription impact

Recorder	Training	Key bias	Impact
Winslow (1624)	None (colonist)	English-hearer perception	Raw phonetic im
Wood (1634)	None (colonist)	English-hearer perception	b/p alternation, u
Williams (1643)	Theologian, self-taught linguist	English categories + stress awareness	Diacritics for stre
Eliot (1666)	Missionary, Hebrew/Greek training	Systematic but English-filtered	Aspiration via vo
Edwards (1787)	Theologian, fluent L2 speaker	Hebrew/Greek + childhood fluency	Most reliable /h/
Prince (1904)	Neogrammarian linguist	German orthographic conventions	<i>tsh</i> for /t /, <i>Dehnu</i>
Speck (1904)	Boasian anthropologist	Ethnographic, minimal linguistic training	Two-stage record
Fielding (2013)	Modern linguist (MIT)	Phonemic orthography	1:1 phoneme-to-g

Table 65: Per-source vowel ambiguity: number of Algonquian phonemes per grapheme

Grapheme	Williams	Eliot	Wood	Winslow	Edwards	Fielding
<i>a</i>	3 (/ə, , a /)	2 (/a, ə/)	2 (/a, ə/)	2 (/a, ə/)	2 (/a, a /)	1 (/a/)
<i>e</i>	4 (/i , , ə, /)	3 (/i , , ə/)	2 (/ , i/)	2 (/ , ə/)	2 (/i , i/)	1 (/i /)
<i>i</i>	3 (/ə, i , /)	2 (/i , /)	2 (/ , i/)	2 (/ , i/)	1 (/i/)	1 (/i/)
<i>o</i>	4 (/ə, a , u , /)	3 (/u , , ə/)	3 (/ , o, ə/)	2 (/ , ə/)	2 (/o, u/)	—
<i>u</i>	2 (/ə, a/)	2 (/ə, /)	2 (/ə, /)	1 (/ə/)	2 (/ə, o/)	1 (/u/)
<i>ee</i>	1 (/i /)	1 (/i /)	1 (/i /)	1 (/i /)	1 (/i /)	—
<i>oo</i>	1 (/u /)	1 (/u /)	1 (/u /)	1 (/u /)	1 (/u /)	—
Total phonemes	8–10	8–10	6–8	6–7	7–8	6

Table 66: Cross-source cognate set for basic vocabulary

Gloss	Williams (NRG)	Eliot (WMP)	Edwards (MAH)	Fielding (MPQ)	PA
'house'	wétu	wetu	—	wetu	*wi kiwa mi
'water'	nipp	nippe	—	nip	*nepyi
'dog'	ayim	arum	—	ayum	*a emwa
'man'	weenawash	wénawaš	—	—	—
'fire'	apeisuwau	apeisuwâ	—	—	—
'I'	neen	neen	neen	neen	*ni ra
'two'	nees	nees	neesoh	nees	*ni šwi
'no'	matta	matta	—	matta	*mati-
'woman'	squa	squa	—	squa	*e θkwe wa
'five'	—	pàpàna	nannah	—	*nya lanwi

Table 67: SNEA dialect classification by PA *r reflex

Dialect type	Reflex of PA *r	Languages	Sources
N-dialect	/n/	Massachusetts (Wampanoag), Narragansett, Nipmuc	Eliot, Trumbull, Williams
Y-dialect	/j/	Mohegan-Pequot, Montauk	Prince-Speck, Fielding
R-dialect	/r/	Western Connecticut (Loup A)	(limited)
L-dialect	/l/	Munsee Delaware, Unami	Zeisberger

Table 68: Cross-dialect cognate evidence for PA *r reflex

Gloss	PA	N-dialect (Mass.)	Y-dialect (Mohegan)	Mahican (Edwards)	L-dialect (Munsee)
'I'	*ni ra	neen	neen	neen	ni l
'thou'	*ki ra	keen	keen	keen	ki l
'man'	*na pe wa	napp	nap	—	ná pe w
'house'	*wi kiwa mi	wetu	wetu	—	wi kiwa m
'water'	*nepyi	nippe	nip	—	mbí
'dog'	*a emwa	arum	ayum	—	—
'three'	*ne θwi	shwa	shwa	—	nəxá
'four'	*nye wi	yaw	yaw	—	né wi
'five'	*nya lanwi	pàpàna	—	nannah	ná lan
PA *r reflex		/n/	/j/	/n/	/l/